



DESIGN PRACTICES

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Revision Memo

12/00 General revision of all text sections, along with editorial changes to entire practice.

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SCOPE

This section describes vaporizing equipment, provides design guidelines, and presents simplified hand calculation procedures for the heat transfer and hydraulic balance of vaporizers. The calculation procedures are based on methods presented in the Heat Transfer Research, Inc., (HTRI) Design Manual, Volumes 2 and 3, and Design Practices [Section XIV, Fluid Flow](#). The recommended design procedure is to use the appropriate computer program as described in [Section IX-K](#). Tube selection, circulation rate, and general design criteria for steam generators are also covered in this section. [Sections IX-A, B, and C](#) describe basic vaporizer considerations such as tube size and pitch, fouling factors, etc. [Section IX-A](#) also includes a brief description of enhanced boiling tubes that can be used to increase the nucleate boiling coefficient.

REFERENCES

HTRI

Report BG1-1, *Nucleate Pool Boiling of Hydrocarbon Mixtures*, May 1983
 Report BT-1, *Boiling Heat Transfer in a Vertical Thermosiphon Reboiler*, May 1978
 Report BT-3, *Boiling of Mixtures inside Vertical Tubes*, Sept. 1978
 Report BT-4, *Two-Phase Flow Instability in Vertical Thermosiphon Reboilers*, Dec. 1978
 Report BK-1-1, *Thermal Design of Kettle Reboiler Bundles*, July 1969
 Design Manual, Volumes 2 and 3

EQUIPMENT TYPES AND APPLICATIONS

REBOILER TYPES

General

Distillation column reboilers can be divided into three groups according to their method of fluid circulation:

1. Thermosiphon or natural circulation reboilers - The driving force is provided by the liquid head from the bottom of the tower. The available head must be sufficient to overcome pressure drops in the inlet and outlet piping and in the exchanger. The boiling fluid can be placed on the tubeside, or on the shell side of a conventional shell-and-tube exchanger, or on the shell side of a kettle type exchanger.
2. Pump-through or forced circulation reboilers - A pump is used to circulate the boiling fluid through the exchanger and back to the tower. The boiling fluid is generally placed on the tubeside of a conventional shell-and-tube exchanger.
3. Internal or stab-in reboilers - A tube bundle is inserted into the tower eliminating the need to provide for circulation of the fluid.

Figure A1 shows sketches of different reboiler types and lists the advantages and disadvantages of each. The selection of the type of unit is based on the heat transfer performance, fouling tendency, and the economics of the design; hydrodynamic performance varies widely for the different types of exchangers, and is an important consideration. The economic factors include foundation and piping costs, exchanger shell and bundle costs, tower elevation, and operating costs. The performance factors are the rate of heat transfer and vaporization, pressure drop, and flow stability. The Reboiler Selection Chart below is a guide to equipment selection. This is a rough guide and the ultimate selection depends on many more specific criteria.

Kettle Reboilers

The components of a typical kettle reboiler are shown in **Figure A2**. The bundles are normally two-pass, and U tubes if a clean heating fluid such as steam is used, or a pull through floating head (T type) if the heating fluid is fouling. The bundle is inserted in an enlarged shell to provide vapor-liquid disengaging space above the bundle. An overflow weir is normally provided to maintain the liquid level above the top of the tubes.

Kettle reboilers may have 100% vaporization for clean fluids such as refrigerants. If the boiling fluid tends to foul then an intermittent blowdown connection is provided at the bottom of the shell to remove fouling material accumulating at the bottom of the shell.

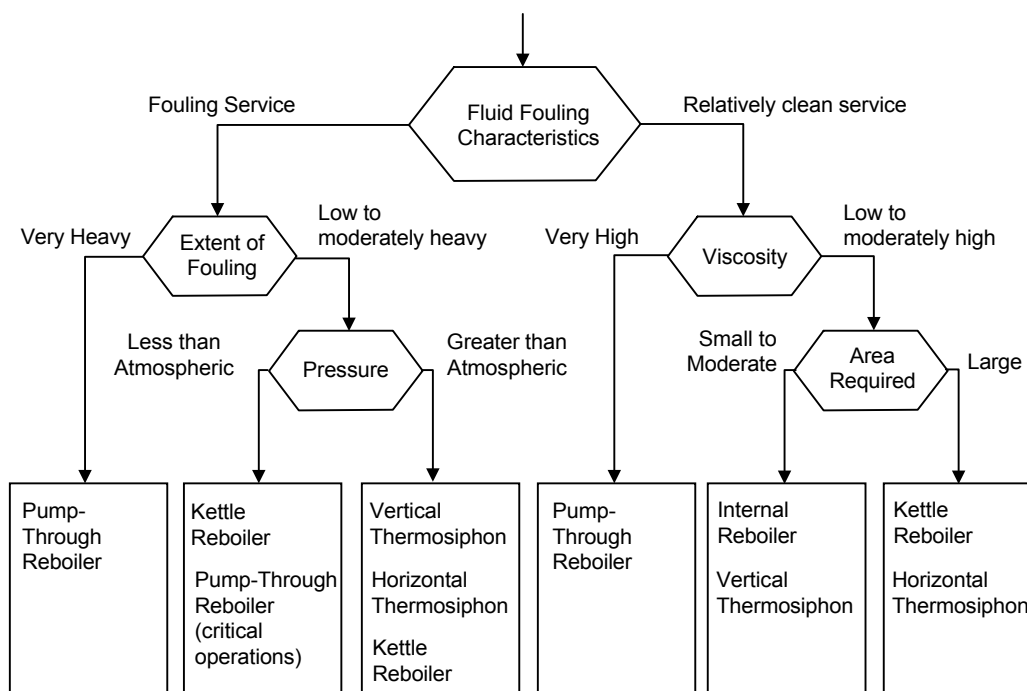
With kettle reboilers the bottom section of the column is simplified. Product hold-up or vapor-liquid disengaging space is not required in the tower, so internal baffling is minimized. If the necessary liquid hold-up beyond the weir in the exchanger requires more than 4 ft of shell length a thermosiphon should be considered. Kettles require the lowest tower elevation of the various reboiler types and so find application where the tower bottoms are not pumped away.

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EQUIPMENT TYPES AND APPLICATIONS (Cont)

Kettles are more expensive than the other types of reboiler units because of the enlarged shell. Their use is, therefore, limited to process conditions that find it more suitable, i.e. fluids with narrow boiling range, heat-sensitive materials, vacuum services, low temperature difference to the heating medium. Kettle reboilers are not well suited for fluids with wide boiling range and high pressure conditions.

REBOILER SELECTION CHART



DP9E10a

Vertical Thermosiphon Reboilers

These are usually single tube-pass units with vaporization in the tubes. Installation costs are low since the exchanger may be directly mounted on the tower minimizing foundation, structures, and plot area; on the other hand, if independently supported in the vertical orientation, these installation costs are probably the same as a horizontal unit or sometimes even higher. Bottoms product hold-up and vapor-liquid disengaging space are provided in the tower bottom. Piping must be short but of large diameter to reduce the pressure drop and ensure flow stability. The shell side fluid should be non-fouling so as to reduce the cost of having to remove the bundle vertically and to permit fixed tubesheet construction.

Vertical thermosiphons should be restricted to a maximum of 50% vaporization because at higher rates of vaporization hydrodynamic head losses may exceed the available driving force and cause unstable flow by creating temporary backflow of vapor. It is common industrial practice to limit vaporization to 30%.

Flow instability can be prevented by increasing tube diameter, especially for low pressure services. Tubes of 3/4 in. (19.05 mm) OD should be avoided if possible. Increasing tube length may promote instability by increasing pressure drop. Rules of thumb to obtain a stable design are: (a) the pressure drop in the exit piping should be less than 0.3 times the total driving head, (b) the flow area of the exit piping should be no less than the total flow area of the tubes, and (c) the flow area of the inlet piping should be in the range of 0.1 to 0.3 times the total flow area of the tubes.

Detailed information on operational characteristics and design suggestions are given in **Tables 1 and 2**. Thermosiphons should be avoided for viscous fluids and for operations where the available liquid head is variable.

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EQUIPMENT TYPES AND APPLICATIONS (Cont)

Horizontal Thermosiphon Reboilers

The boiling fluid is placed on the shell side and multiple inlets and outlets, and split horizontal baffles are often used to provide good distribution and higher velocities. Bottoms product hold-up and vapor-liquid disengaging are provided in the bottom of the tower. This type is limited to moderately fouling services and the use of a removable bundle with square tube pitch is recommended.

Horizontal thermosiphons require separate foundations, more plot area than vertical units, and more complex piping. Thus their overall cost is generally higher than vertical thermosiphons.

Vaporization should be restricted to 50% for hydrocarbons and 10% for aqueous solutions for clean services and 25% for relatively more fouling services. The advantage of horizontal thermosiphons is their lower pressure drop and larger liquid volume hold-up than the vertical thermosiphons, that allows for greater operational flexibility and their lower head requirements. Horizontal thermosiphons can handle fluid mixtures with wider boiling range; however, viscous services and those with variable liquid head should be avoided.

Pump-Through Reboilers

In a pump-through reboiler the vaporizing fluid is circulated by a pump accommodating a higher pressure drop and correspondingly higher heat transfer coefficients. Piping diameters may be smaller than for other reboiler types and the exchanger can be located away from the tower. Initial pump costs and operating costs are high, but they can be justified under the following circumstances:

1. The service is very viscous or highly fouling.
2. A high circulation rate, about 10:1, or a low increase in temperature is required to prevent thermal degradation of the bottoms product.
3. The liquid head available from the tower is limited. However, tower elevation must be high enough to provide the required NPSH.

In highly fouling services the vapor fraction at the outlet should be limited to about 5% by weight.

Internal Reboilers

An internal reboiler consists of a tube bundle inserted directly into the tower using a flange in the side of the tower. This eliminates the costs of a shell and piping. The flange can be relatively expensive, but on an installed cost per square foot basis internal reboilers are frequently the least expensive of all the types. A major limitation of this type is that the length of the bundle is restricted by the tower diameter, so that an uneconomically large bundle diameter may be required. As a rule of thumb, if the bundle diameter is 36 in. (914 mm) or greater, internal reboilers should not be considered. The major disadvantage of internal reboilers is that they cannot be serviced and cleaned without shutting the tower down. For that reason welded tube-to-tubesheet joints should be provided.

In sizing the bundle and the flange connection, sufficient clearance should be provided between the bundle and the flange neck inside diameter to allow vapor to escape into the tower and avoid vapor blanketing of that portion of the tubes. Fouling tendencies of internal reboilers are similar to kettles.

REBOILERS AND TOWER ELEVATION

The bottom of the shell of a reboiler is located at the minimum practical distance above grade. This distance is set by space requirements for piping, and is usually about 3 ft (1 m).

For natural-circulation reboilers a pressure balance through the reboiler circuit determines the exact tower elevation. The usual points of reference for the pressure balance are the bottom tangent line of the tower and the bottom of the reboiler shell or the bottom tubesheet for a vertical thermosiphon reboiler. As a rough guide, the distance between these points will normally be 6 to 10 ft (1.8 to 3 m) for kettle reboilers and 8 to 20 ft (2.4 to 6.1 m) for thermosiphon reboilers.

For forced-circulation reboilers, the tower is normally elevated 15 ft (4.6 m) above grade to provide an adequate pump NPSH. If necessary this elevation can be decreased by reducing the pump NPSH requirement to a minimum consistent with pump availability. However, this will usually cause a considerable increase in the cost of the pump.

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REBOILER STAGE EFFICIENCY

The stage efficiency of recirculating type reboilers (that are not baffled to prevent bottom tray liquid from bypassing the reboiler) may be estimated from the following equation.

$$S_F = (1 - N_L / (N_V K / A + B)) / (1 - N_L / (N_V K + B))$$

where: A = $1 + y(K - 1)$
 B = Bottoms product rate, moles/hr
 S_F = Fraction of theoretical stage
 K = Equilibrium constant of light key component
 N_L = Liquid rate from bottom tray, moles/hr
 N_V = Vapor rate from reboiler, moles/hr
 y = Mole fraction vaporized in the reboiler

CHILLERS

The operation of a chiller is like that of a reboiler. The fluid being cooled on the tube side vaporizes refrigerant on the shell side. A chiller is normally constructed like a kettle reboiler, but without a weir. It may also be made from a regular shell and tube exchanger by eliminating tubes in the top portion of the shell to provide a space for vapor-liquid disengaging. The calculation procedure for chillers is the same as for kettle reboilers.

PREHEATERS

A preheater, like a pump-through reboiler, consists of a conventional shell-and-tube exchanger with the liquid to be vaporized fed either by a pump or pressurized from upstream equipment. Most preheaters add both latent heat and sensible heat to the fluid being vaporized (one exception would be the vaporization of a pure component under very little back pressure). The fluid to be vaporized is generally placed in the tubes to avoid potential phase separation that would occur if fluid is placed on the shellside. Preheaters are most often used for feed streams to distillation columns. A lower level of heat can be used in tower feed preheating rather than in reboilers. Also, tower loadings can be balanced by trading off between feed preheat and heat input to the reboiler.

WASTE HEAT BOILERS

Applications

Waste heat boilers (steam generators) are commonly used to recover heat from catalytic cracker regenerator flue gas, thermal reactor and condensers in a sulfur plant, reformer effluent, gas turbine exhaust, catalyst-oil slurries, and fixed bed reactors.

Types of Units

There are two types of units used for waste heat boilers (WHB): firetube or water tube. In firetube units (the more common type) the heating fluid passes through the tubes and steam is generated outside the tubes (shell side).

Firetube Units

Firetube units usually come as thermosiphon or kettle type waste heat boilers. Generally, kettles are provided for such services as hot product coolers (Resid, FCC Bottoms), or for Sulfur Condensers. Maximum heat flux normally does not exceed 35,000 Btu/hr ft² (110,000 W/m²) and maximum process inlet temperature is 800°F (430°C). Where the maximum heat flux is greater than 40,000 Btu/hr ft² (125,000 W/m²) and where the process inlet temperature is greater than 900°F (480°C), a thermosiphon type WHB is generally recommended. In a thermosiphon type WHB, steam is generated on the shellside of a tube bundle but the steam disengagement occurs in a separate shell, called the Steam Drum. Flow between the WHB and the Steam Drum is maintained by natural circulation because of the large density difference between the water "leg", i.e., the downcomer, and the steam "leg", i.e., the riser. The circulation rate for thermosiphon-type WHB, is generally between 15 to 20:1, depending on the severity of the process and the design maximum heat flux. Reliable designs for the WHB would take into account the possibility of vapor blanketing at the hot end, where the maximum heat flux is encountered. WHB having a high inlet process temperature, > 1000°F (535°C), are supplied with a protective refractory lining inside the inlet channel and on the tubesheet; metallic or ceramic ferrules are supplied at each tube end.

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EQUIPMENT TYPES AND APPLICATIONS (Cont)

Water Tube Units

Water tube units which generate steam inside the tubes are commonly used in flue gas exhaust from gas turbines, CO boilers, FCC units, etc., resulting in a low gas pressure drop. Water-tube WHB are normally built into three sections: a steam generator, a superheater and an economizer. For process streams with catalyst fines, i.e. FCC Gas Cooler, maximum gas velocity across the tube bank should not exceed 100 fps; a deresonating baffle is also furnished to prevent acoustic vibration; cleaning lanes are provided for steam lancing.

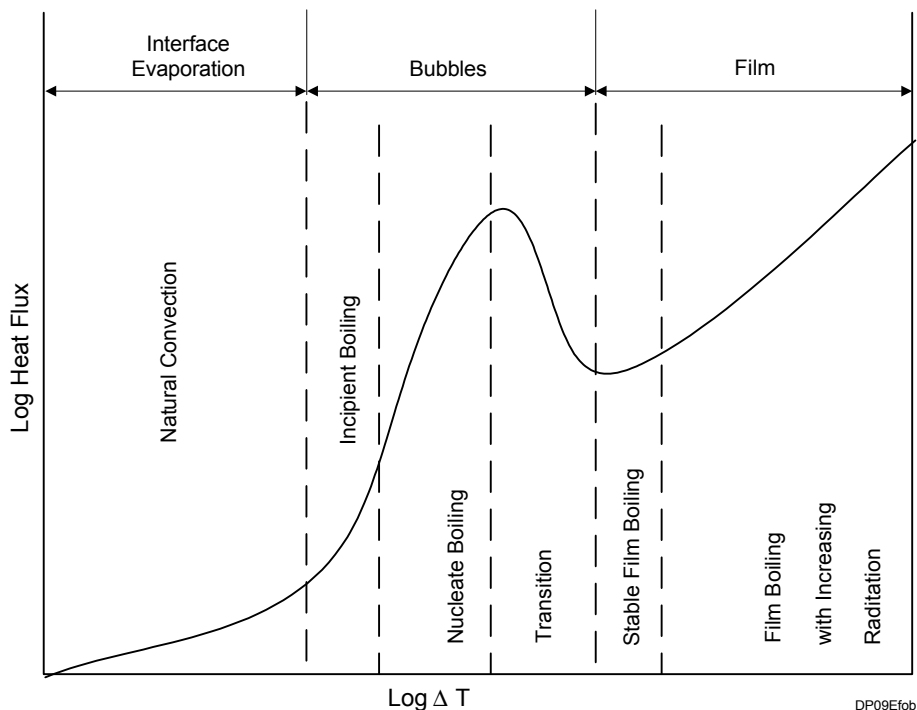
DESIGN CONSIDERATIONS

MAXIMUM AND DESIGN HEAT FLUX (FILM BOILING, VAPOR BLANKETING, CHOKe FLOW)

Pool Boiling

This type of boiling takes place when the heating surface is surrounded by a pool of liquid which is not flowing at any appreciable velocity. The main boiling mechanism is the formation of bubbles and is known as nucleation. Kettles and internal reboilers operate in pool boiling. The typical pool boiling curve, plotted as heat flux vs. the temperature driving force (ΔT), is shown in the figure below. As ΔT is increased heat transfer takes place by natural convection and eventually bubbles are formed at the tube wall and are carried away by convection and buoyancy. This is known as nucleate boiling. Liquid fills in the space vacated by the bubbles so that the tube wall always has liquid in contact with it. As ΔT is further increased (usually in the range of 40 - 90°F for pure hydrocarbons, 22 - 50°C) liquid can no longer exist in contact with the metal surface. Bubbles are formed faster than they can be removed and a vapor film blankets the tube surface. This is known as film boiling. The transition region between nucleate boiling and stable film boiling consists of unstable nucleate and unstable film boiling. This transition begins when the maximum heat flux possible in nucleate boiling, also called the critical heat flux, is reached. Because heat transfer coefficients are highest in the nucleate boiling region for practical values of ΔT , the film boiling region is avoided in design. To ensure operation in nucleate boiling the design heat flux should be limited to 70% of the calculated maximum flux. If the design ΔT is such that film boiling cannot be avoided then the heat flux should be kept above the minimum value required for stable film boiling.

TYPICAL POOL BOILING CURVE



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Flow Boiling

In flow boiling, two-phase fluid is forced over the heating surface. Vertical thermosiphons and pump-through reboilers operate in flow boiling. The same principles of boiling apply to pool and flow boiling. The major difference is that due to the two-phase flow, usually occurring in a restricted channel like a tube, hydrodynamic effects are more pronounced in flow boiling. The limitations on the hydrodynamic effects become even greater at low pressure. The maximum heat flux is limited not only by transition to film boiling but also by mist flow and potential flow instability.

Mist flow occurs when the shear force of the vapor at the vapor-liquid interface is sufficient to strip the liquid film from the wall, producing a dry wall condition. This can be prevented by keeping the exit vapor weight fraction (y_o) less than 0.5. When the pressure drop required for vaporization (acceleration loss) exceeds the available driving head, a backflow condition results that causes flow instability (also known as choke flow). The maximum heat flux under such condition is a function of the operating pressure and a geometry correction factor. The film boiling phenomenon is the same as in pool boiling except that the maximum heat flux may be lower. The calculation procedure for vertical tubeside boiling shows calculations for the mist flow limit (y_{mist}) and the maximum heat flux (q_{max}). For good operation keep $y_o \leq y_{mist}$ and $q \leq (0.6) q_{max}$.

VERY LOW TEMPERATURE DIFFERENCE

When process conditions or economics require the use of a very low ΔT (8°F, 4°C), the ability to produce bubbles may differ drastically for different surfaces. This makes design, performance prediction, and operation difficult. One possible solution is to use tube surfaces which enhance nucleate boiling. Special coatings and specially formed fins are commercially available as described in [Section IX-A](#).

In thermosiphons, a low temperature difference will suppress the pumping action and cause low circulation rates. For vertical tubeside thermosiphons, it is recommended to maintain a minimum heat flux of 2000 Btu/hr ft² (6000 W/m²) to provide good circulation. At lower flux levels it is preferable to use a kettle type exchanger.

VERY HIGH TEMPERATURE DIFFERENCE

A transition flow regime and mist flow condition can occur in reboilers operating at a high temperature difference, leading to local hot spots and thermal cycling; such operating conditions should be avoided. In the transition region, small shifts in operating conditions can cause large changes in performance. Also, this region has reverse control characteristics. That is, because of a mixture of nucleate and film boiling conditions an increase in ΔT may sometimes cause a decrease in the heat transfer. If the reboiler must be designed for film boiling it should be designed to operate in the stable region and not in the transition region.

FOULING

Design fouling factors have a major impact on the operation of reboilers, especially under clean start-up conditions. If a large fouling factor is used, the unit may be so oversurfaced at start-up that the required ΔT will be too small for proper operation. Also, if the design ΔT is applied at start-up it may cause excessively high heat fluxes resulting in film boiling and higher than normal tube wall temperatures, which may promote fouling.

Realistic fouling factors suggested by HTRI are shown in the table below (also refer to [Table 3, Section IX-B](#)). Fouling can be minimized by ensuring good flow distribution on the shell side. Some ways to prevent bad distribution are: use Double Split Flow (TEMA type H) design, avoid large bypass areas, avoid stagnant zones. For fluids with high fouling tendencies, keeping the vapor fraction below 0.2 also helps to minimize fouling.

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DESIGN CONSIDERATIONS (Cont)**SUGGESTED FOULING FACTOR RANGES**

SERVICE	FOULING RESISTANCE, R_F	
	$\text{ft}^2 \text{ F hr/Btu}$	$\text{m}^2 \text{ K/W}$
Boiling Side C ₁ – C ₈ normal hydrocarbons Heavier normal hydrocarbons Diolefins and polymerizing hydrocarbons	0 - 0.0015 0.002 - 0.003 0.003 - 0.005	0 - 0.00026 0.00036 - 0.0005 0.0005 - 0.0009
Heating Side Condensing steam Condensing organics Sensible heating, organic liquids	0 - 0.0005 0.0005 - 0.001 0.001 - 0.002	0 - 0.00009 0.00009 - 0.00018 0.00018 - 0.00035

CALCULATION PROCEDURES**KETTLE AND INTERNAL REBOILERS**

The calculation forms for heat transfer and hydraulic balance for kettle and internal reboilers are given in the Appendix. The heat transfer method is taken from the HTRI Design Manual.

Heat Transfer

The heat transfer procedure assumes a known geometry and calculates the nucleate and film boiling curves for clean and fouled conditions. To rate an existing exchanger, perform the calculation and estimate the required temperature driving force (ΔT) for a given heat duty, or if the ΔT is known, estimate the heat duty. A sample rating calculation is shown following the blank calculation forms. To size a kettle reboiler (design calculation) an iterative procedure is required, as outlined in the steps below. Reboilers should be designed for a heat flux less than 70% of the calculated maximum for nucleate boiling or greater than the calculated minimum for film boiling.

1. Estimate an overall heat transfer coefficient (U , Btu/hr ft^2 °F or W/m² K) from [Table 1, Section IX-B](#).
2. Calculate the required area (A , ft^2 or m^2)

$$A = Q_T / (U \Delta T)$$

where: Q_T is the total heat duty (Btu/hr or W)

ΔT is the difference between the average temperature of the heating medium and the exit vapor temperature (°F or °C)

3. Calculate the number of tubes (NTT) required to provide the area A

$$\text{NTT} = 12(A) / (\pi d_o L) \text{ or } \text{NTT} = 1000(A) / (\pi d_o L)$$

where: d_o is the tube outside diameter (in. or mm)

L is the tube length (ft or m)

4. Perform the calculations and construct a boiling curve.
5. Use the ΔT from step 2 to find an operating heat flux, q (Btu/hr ft^2 or W/m²), using the curve constructed in Step 4.
6. Calculate the required area (A_r)

$$A_r = Q_T / q (\text{ft}^2 \text{ or } \text{m}^2)$$

7. If A_r doesn't match A repeat steps 3-6 using A_r to calculate NTT.

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CALCULATION PROCEDURES (Cont)

Kettle Sizing

The calculation for kettle diameter sizing assumes a known maximum permissible entrainment ratio and calculates the maximum permissible velocity and the actual velocities. For design calculations the kettle size should be guessed and the calculation repeated until the calculated velocities are less than the maximum.

In addition to sizing the kettle diameter to ensure that liquid entrainment is maintained below the design criterion, splash baffles may be provided to prevent splashing of the liquid into the nozzle entrance area. Other methods, such as the "dry-pipe" are widely used in the industry to assure splashed water does not get the outlet nozzle and also achieve a vapor quality of 95%. Higher quality can be achieved by the use of demisters. The size and locations of the splash baffles are shown in **Figure A13**, along with other recommended dimensions for kettle designs. The area A_2 should be about 40% of the area S_h . The 18 in. minimum spacing in the shell should be from the top of the liquid level which is usually at the top of the weir. For high liquid flow rates, it is necessary to calculate a liquid crest height above the weir and size the area S_h above this crest level.

Hydraulic Balance

The hydraulic balance procedure calculates the elevation required for a given mass flow rate. The variable (or unknown) dimension is the height from the tower bottom tangent to the top of the shell. For existing piping, the calculated height should be checked against the actual. For new designs, if the calculated height is greater than desired, line sizes should be increased and a new value calculated.

VERTICAL THERMOSIPHON TUBESIDE REBOILERS

The calculation forms for heat transfer and hydraulic balance for vertical tubeside thermosiphon reboilers are given in the Appendix. The heat transfer method is taken from the HTRI Design Manual.

Heat Transfer

The heat transfer procedure calculates the area required for a given heat flux. To rate an existing exchanger, compare the calculated area to the actual area. A sample rating calculation is shown following the blank calculation forms. To size a new exchanger (design calculation) a geometry must be assumed and an iterative calculation done according to the steps below. Vertical thermosiphons should be designed such that the outlet vapor fraction is less than that required for mist flow and the heat flux is less than 70% of the calculated maximum for choke flow.

1. Estimate an overall heat transfer coefficient (U , Btu/hr ft² F or W/m² K) from **Table 1, Section IX-B**.
2. Calculate the heat flux (q) and the area (A)

$$q = U \Delta T \text{ (Btu/hr ft}^2 \text{ or W/m}^2 \text{)}$$

$$A = Q_T / q \text{ (ft}^2 \text{ or m}^2 \text{)}$$

where: Q_T is the total heat duty (Btu/hr or W)

ΔT is the temperature driving force (defined in the calculation)

3. Calculate the number of tubes (NTT) required to provide the area A

$$NTT = 12(A) / (\pi d_o L) \text{ or } NTT = 1000(A) / (\pi d_o L)$$

where: d_o is the tube outside diameter (in. or mm)

L is the tube length (ft or m)

4. Perform the calculations and find the actual overall heat transfer coefficient (U_o) and required area (A_r).
5. If A_r doesn't match A repeat steps 2-4 using U_o to calculate q .

Hydraulic Balance

The hydraulic balance procedure calculates the elevation required for a given mass flow rate. The variable (or unknown) dimension is the height from the tower bottom tangent to the bottom of the bundle. For existing piping, the calculated height should be checked against the actual. For new designs, if the calculated height is greater than desired, line sizes should be increased and a new value calculated.

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CALCULATION PROCEDURES (Cont)

HORIZONTAL THERMOSIPHON SHELLSIDE REBOILERS

Heat Transfer

The hand calculation method for heat transfer in horizontal shell side thermosiphon reboilers is the same as for kettle reboilers. Additional convective effects due to the flow are usually small but can be significant at high flow rates. These effects are neglected as a safety factor.

For horizontal thermosiphons the heat flux should be limited to 70% of the calculated maximum heat flux to ensure operation in nucleate boiling, and the weight fraction vaporized should be in the range of 0.1 to 0.5 to prevent slug flow at lower weight fractions and mist flow at higher weight fractions.

Hydraulic Balance

The form for hand calculation of the hydraulic balance for shellside reboilers is given in the Appendix. The procedure calculates the elevation required for a given mass flow rate. The variable (or unknown) dimension is the height from the tower bottom tangent to the bottom of the bundle. For existing piping, the calculated height should be checked against the actual. For new designs, if the calculated height is greater than desired, line sizes should be increased and a new value calculated.

PUMP-THROUGH REBOILERS

The calculation form for heat transfer calculations for tubeside pump-through reboilers is given in the Appendix. The method is the same as that for vertical tubeside thermosiphons except that the correction to the boiling coefficient due to the subcooled liquid zone is calculated differently.

WASTE HEAT BOILERS

The following procedure is based on steam generation using conventional tubes (no extended surface).

Exchanger Orientation - The units can be installed in either a vertical or horizontal position, and can have one or more tube passes. In general, one tube pass vertical units have been used most successfully in the past (in cat regenerator flue gas and fixed bed reactor service). However, in recent years, one tube pass horizontal units have been used with increasing frequency (e.g., reformer effluent waste heat boilers).

Shell Side or Tube Side Vaporization - Steam is normally generated in the shell (firetube design) for the following reasons:

1. The hot fluid is frequently so fouling that it must be passed through the tubes.
2. With clean boiler water, it is permissible to use a non-removable tube bundle (fixed tube sheet unit) with triangular tube pitch. This type of construction results in a compact unit with a low initial cost.
3. There is less differential expansion between the tube and the shell.

Tube Selection

1. For installations that recover heat from cat regenerator flue gases or reformer effluent, the number of tubes is based on maintaining an inlet gas velocity of 100 ft/sec (31 m/s). For cat regenerator flue gases use 2-1/2 in. (63.5 mm) O.D., 5 BWG tubes. For reformer effluent, the minimum nominal tube size is 1-1/2 in. (50.8 mm) O.D., 10 BWG. The length of the tubes will vary for each installation, depending on the surface area required. If necessary, the tube length may be 25 ft (7.6 m) or more (see [Section IX-C](#)).
2. Fixed-bed reactors for exothermic processes are often constructed as shell-and-tube exchangers. In these units, the hydrocarbon passes through the tubes, which are filled with catalyst. Steam is generated on the shell side. For this service, use 2-1/2 in. (63.5 mm) O.D., 5 BWG tubes. The number and length of tubes will vary for each installation, depending on process requirements as well as on heat transfer requirements. Steam generator-reactors for polymerization plants have been designed with 4 ft (1.2 m) I.D. shells, and 180 (9.1 m) thirty-foot-long tubes.

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CALCULATION PROCEDURES (Cont)

Water Circulation Rate - To eliminate the possibility of dry tubes, design for a water/steam weight ratio of approximately 15/1 for a heat density of 40,000 Btu/hr ft² (125,000 W/m²). A Baton Rouge design with extended surface tubes requires a 40/1 circulation rate to ensure a wetted wall condition at a heat density of 100,000 Btu/hr ft² (315,000 W/m²). (Note that a wetted wall condition is dependent upon other variables, such as: temperature difference, geometry, orientation, etc., in addition to the circulation rate). Proprietary designs that assure wet-wall conditions at the hot tubesheet for severe services of high temperature and high steam pressure are available from vendors such as Borsig and Schmidte. Transfer line exchangers (TLE), which fall in this category, and are designed for very high heat flux, > 80,000 Btu/hr ft² (250,000 W/m²), are usually specified as double pipes.

Elevation of Steam Disengaging Drum - It is necessary to calculate a detailed pressure balance through the water-steam circuit to determine the elevation of the steam drum above the generator. See the applicable (horizontal or vertical thermosiphon) hydraulic balance calculation form.

Steam Generation on Shell Side

1. **Tube Side Design** - Follow the calculation procedure outlined for no change of phase, [Section IX-D](#), or for condensation, [Section IX-F](#), assuming an overall coefficient, area, etc.
2. **Shell Side Design**
 - a. **Shell Side Pressure Drop** - Calculate the shell side pressure drop using the procedure for reboilers.
 - b. **Shell Side Heat Transfer Coefficient, h_o** - (See Equation 1.)

$$\frac{1}{R_o} = h_o = 80 \left(\frac{Q}{1000 A_s} \right)^{\frac{2}{3}} P^{\frac{1}{3}} \quad \text{(Customary)} \quad \text{Eq. (1)}$$

OR

$$= 353.2 \left(\frac{Q}{5680 A_s} \right)^{\frac{2}{3}} P^{\frac{1}{3}} \quad \text{(Metric)}$$

where: R_o = Outside film resistance to heat transfer corrected to outside area $\left[\frac{\text{hr ft}^2 \text{ } ^\circ\text{F}}{\text{Btu}} \text{ or } \frac{\text{m}^2 \text{ K}}{\text{W}} \right]$

h_o = Outside film coefficient $\left[\frac{\text{Btu}}{\text{hr ft}^2 \text{ } ^\circ\text{F}} \text{ or } \frac{\text{W}}{\text{m}^2 \text{ K}} \right]$

Q = Total rate of heat transfer, [Btu/hr or W]

P = Average pressure in boiler, [psia or kPa abs]

Steam Generation on Tube Side

1. **Tube Side Design**
 - a. **Tube Side Pressure Drop** - Calculate the tube side pressure drop using the procedure for reboilers.
 - b. **Tube Side Heat Transfer Coefficient, h_{io}**

$$h_{io} = \frac{1}{R_{io}} = 1000 \left[\frac{\text{Btu}}{\text{hr ft}^2 \text{ } ^\circ\text{F}} \right]$$

OR

$$= 5680 \left[\frac{\text{W}}{\text{m}^2 \text{ K}} \right]$$

2. **Shell Side Design** - Follow the calculation procedure outlined for no change of phase, [Section IX-D](#), or for condensation, [Section IX-F](#), assuming an overall coefficient, area, etc.

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CALCULATION PROCEDURES (Cont)

Overall Heat Transfer Rate

1. Calculate U_o , the overall heat transfer coefficient:

$$\frac{1}{U_o} = R_t = R_{io} + r_{io} + r_w + r_o + R_o \quad \text{Eq. (2)}$$

where: U_o = Overall coefficient of heat transfer, [Btu/hr ft² °F or W/m² K]

R_t = Total resistance (duty) to heat transfer, $\left[\frac{\text{hr ft}^2 \text{ } ^\circ\text{F}}{\text{Btu}} \text{ or } \frac{\text{m}^2 \text{K}}{\text{W}} \right]$

R_{io} = Inside film resistance to heat transfer corrected to outside area, $\left[\frac{\text{hr ft}^2 \text{ } ^\circ\text{F}}{\text{Btu}} \text{ or } \frac{\text{m}^2 \text{K}}{\text{W}} \right]$

r_{io} = Inside fouling factor referred to the outside surface area, $\left[\frac{\text{hr ft}^2 \text{ } ^\circ\text{F}}{\text{Btu}} \text{ or } \frac{\text{m}^2 \text{K}}{\text{W}} \right]$

r_w = Resistance of tube wall metal at average wall temperature, $\left[\frac{\text{hr ft}^2 \text{ } ^\circ\text{F}}{\text{Btu}} \text{ or } \frac{\text{m}^2 \text{K}}{\text{W}} \right]$

r_o = Outside fouling factor, $\left[\frac{\text{hr ft}^2 \text{ } ^\circ\text{F}}{\text{Btu}} \text{ or } \frac{\text{m}^2 \text{K}}{\text{W}} \right]$

R_o = Outside film resistance to heat transfer based on outside area, $\left[\frac{\text{hr ft}^2 \text{ } ^\circ\text{F}}{\text{Btu}} \text{ or } \frac{\text{m}^2 \text{K}}{\text{W}} \right]$

2. If U_o (calculated) does not agree with U_o (assumed), repeat the calculations with a new U_o (assumed) until agreement is reached.
3. In all cases the designer should satisfy himself that a reasonably good economic balance has been obtained between pressure drop and transfer surface area (see [Section IX-C](#)).
4. Determine the mechanical design features as outlined in [Sections IX-C and IX-D](#).

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NOMENCLATURE

A	Heat transfer surface area, [ft ² , m ²]
AOI	Ratio of outside to inside surface area
A _r	Required heat transfer surface area, [ft ² , m ²]
BR	Boiling range, [°F, °C]
CB	Bundle correction factor
C _{pb}	Heat capacity of bottoms drawoff, [Btu/lb °F, kJ/kg K]
C _{pℓ}	Liquid heat capacity, [Btu/lb °F, kJ/kg K]
DR	Effective temperature range, [°F, °C]
D _b	Bundle diameter, [in., mm]
D _s	Shell diameter, [in., mm]
d _i	Tube inside diameter, [in., mm]
d _o	Tube outside diameter, [in., mm]
dT/dP	Slope of vapor pressure curve, [°F/psi, °C/kPa]
d ₁	Inlet line inside diameter, [in., mm]
d ₂	Outlet line inside diameter, [in., mm]
E	Entrainment ratio, [lb liquid/lb vapor, kg liquid/kg vapor]
F _b	Bundle convection correction factor
F _c	Boiling range (mixture) correction factor
F _d	Sensible heating correction factor
F _f	Fin tube correction factor
F _p	Pressure correction factor for nucleate boiling
F _{ℓz}	Subcooled liquid zone correction for intube boiling
f ₂	Two-phase friction factor
f _ℓ	Liquid friction factor
f _v	Vapor friction factor
H _W	Weir height, [in., mm]
ΔH	Head losses as defined in calculation forms, [ft, m]
ΔH _F	Sum of fixed head losses in hydraulic balance calculations, [ft, m]
h ₁	Single tube nucleate boiling coefficient, [Btu/hr ft ² °F, W/m ² K]
h _{1r}	Single tube boiling reference coefficient, [Btu/hr ft ² °F, W/m ² K]
h _{ab}	Average boiling heat transfer coefficient, [Btu/hr ft ² °F, W/m ² K]
h _n	Heating medium heat transfer coefficient, [Btu/hr ft ² °F, W/m ² K]
h _ℓ	Liquid phase heat transfer coefficient, [Btu/hr ft ² °F, W/m ² K]
h _{nb}	Intube nucleate boiling coefficient, [Btu/hr ft ² °F, W/m ² K]
h _{nc}	Natural convection heat transfer coefficient, [Btu/hr ft ² °F, W/m ² K]
h _{tp}	Convective boiling heat transfer coefficient, [Btu/hr ft ² °F, W/m ² K]

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NOMENCLATURE (Cont)

J	Friction head loss in unknown elevation X, [ft, m]
J'	Head gain in fixed lengths, [ft, m]
K _i	Total inlet piping resistance coefficient
K _o	Total outlet piping resistance coefficient
k _ℓ	Liquid thermal conductivity, [Btu/hr ft °F, W/m K]
k _w	Tube wall thermal conductivity, [Btu/hr ft °F, W/m K]
L	Tube length, [ft, m]
L _c	Chord length above weir, [in., mm]
NN	Number of exit nozzles
NTT	Number of tube holes in tubesheet
P	Pressure, [psi, kPa]
P _c	Critical pressure, [psi, kPa]
P _ℓ	Inlet liquid pressure drop per 100 ft (per 1 m) of line, [psi, kPa]
P _r	Reduced pressure
P _v	Outlet vapor pressure drop per 100 ft (per 1 m) of line, [psi, kPa]
Pr	Prandtl Number, dimensionless
PR	Tube pitch ratio
ΔP _h	Static pressure head [psi, kPa]
ΔP ₁	Inlet line head loss per foot (per m) of line, [ft, m]
ΔP ₂	Outlet line head loss per foot (per m) of line, [ft, m]
Q _s	Sensible heat duty [Btu/hr, W]
Q _T	Total heat duty [Btu/hr, W]
Q _v	Volume flow rate, [gpm, dm ³ /s]
q	Heat flux, [Btu/hr ft ² , W/m ²]
q ₁	First reference heat flux, [Btu/hr ft ² , W/m ²]
q ₂	Second reference heat flux, [Btu/hr ft ² , W/m ²]
q _{max}	Maximum allowable heat flux, [Btu/hr ft ² , W/m ²]
q _{maxr}	Choke flow reference maximum heat flux, [Btu/hr ft ² , W/m ²]
q _{1max}	Single tube maximum allowable heat flux, [Btu/hr ft ² , W/m ²]
q _{bmax}	Bundle maximum allowable heat flux, [Btu/hr ft ² , W/m ²]
q _{bmin}	Minimum film boiling heat flux, [Btu/hr ft ² , W/m ²]
q _n	Effective minimum heat flux for nucleate boiling, [Btu/hr ft ² , W/m ²]
Re _ℓ	Liquid Reynolds number
Re _v	Vapor Reynolds number
r _{fi}	Inside fouling resistance, [hr ft ² °F/Btu, m ² K/W]
r _{fo}	Outside fouling resistance, [hr ft ² °F/Btu, m ² K/W]
r _w	Tube wall resistance, [hr ft ² °F/Btu, m ² K/W]
S _h	Horizontal velocity area, [ft ² , m ²]
S _v	Vertical velocity area, [ft ² , m ²]
S1-S3,	
S4, S6	Distances as defined in calculation forms, [ft, m]
S2	Equivalent length of inlet fittings, [ft, m]
S5	Equivalent length of outlet fittings, [ft, m]
T _{bi}	Boiling side inlet temperature [°F, °C]
T _{bo}	Boiling side outlet temperature [°F, °C]

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NOMENCLATURE (Cont)

T_{bs}	Boiling side saturation temperature [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
T_d	Bottoms drawoff temperature [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
T_f	Feed temperature [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
T_{hi}	Heating medium inlet temperature [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
T_{ho}	Heating medium outlet temperature [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
T_{hs}	Heating medium saturation temperature [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
T_h	Average heating medium temperature [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
T_s	Bubble point temperature [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
T_v	Exit vapor temperature [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
ΔT	Temperature driving force, [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
ΔT_a	Available temperature driving force, [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
ΔT_c	Clean required nucleate boiling temperature driving force, [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
ΔT_f	Fouled required nucleate boiling temperature driving force, [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
ΔT_{fc}	Clean required film boiling temperature driving force, [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
ΔT_{ff}	Fouled required film boiling temperature driving force, [$^{\circ}\text{F}$, $^{\circ}\text{C}$]
U	Overall heat transfer coefficient [Btu/hr ft ² $^{\circ}\text{F}$, W/m ² K]
U_c	Clean overall heat transfer coefficient [Btu/hr ft ² $^{\circ}\text{F}$, W/m ² K]
U_f	Fouled overall heat transfer coefficient [Btu/hr ft ² $^{\circ}\text{F}$, W/m ² K]
U_o	Actual overall heat transfer coefficient [Btu/hr ft ² $^{\circ}\text{F}$, W/m ² K]
V_e	Effective velocity for entrainment, [ft/s, m/s]
V_h	Horizontal velocity for entrainment, [ft/s, m/s]
V_i	Liquid inlet velocity, [ft/s, m/s]
V_{max}	Maximum velocity for entrainment, [ft/s, m/s]
V_v	Vertical velocity for entrainment, or vapor velocity, [ft/s, m/s]
W_B	Bottoms drawoff flowrate [lb/hr, kg/hr or kg/s]
W_T	Total flowrate [lb/hr, kg/hr or kg/s]
W_V	Vapor flowrate [lb/hr, kg/hr or kg/s]
X	Unknown elevation in hydraulic balance calculations, [ft, m]
X_{tt}	Martinelli parameter for two-phase flow
x_r	Vapor to liquid density ratio
y	Weight fraction vapor
y_a	Average weight fraction vapor
y_o	Weight fraction vapor at outlet
y_{mist}	Weight fraction vapor limit for mist flow
Λ	Correction to q_{maxr}
ρ_{ℓ}	Liquid density, [lb/ft ³ , kg/m ³]
ρ_v	Vapor density, [lb/ft ³ , kg/m ³]
ρ_x	Average two-phase mixture density, [lb/ft ³ , kg/m ³]
μ_{ℓ}	Liquid viscosity, [lb/ft hr, kg/m-s]
μ_v	Vapor viscosity, [lb/ft hr, kg/m-s]
σ	Surface tension, [dynes/cm]
θ	Chord angle
λ	Latent heat, [Btu/lb, kJ/kg]

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TABLE 1
OPERATIONAL CHARACTERISTICS FOR VERTICAL THERMOSIPHON REBOILERS

CATEGORY (Examples are Given in Parentheses)	CHARACTERISTICS				
	Heat Transfer Coefficients	Circulation Rate	Maximum Heat Flux Limit	Mean Temperature Difference	Fouling
1. Narrow boiling range saturated hydrocarbons, moderate pressure (C ₅ -C ₇ , 100 psia)	Fairly high	High	High, usually limited by film boiling	No problems	Low
2. Narrow boiling range, aqueous solution of organics (alcohols and water)	High	High	High	Usually non-ideal VLE. Should be carefully calculated	Usually very low
3. Vacuum reboilers with heavy hydrocarbons (mixed xylenes, 1 psia)	Low. Long non-boiling zone. Suppressed nucleation	Low. High friction, momentum and liquid zone static heat losses	Low. Instability or mist flow limitation	Nonlinear. Maximum bulk temp. at top of liquid zone	May be high if mist flow is permitted
4. Very high pressure (near critical). Light hydrocarbon	Very high. Nucleate low due to boiling controlled. Low if in film boiling, but more predictable.	Low due to decreased net driving head. High vapor density.	Low. Film boiling limit	Boiling temp. profile may be difficult to predict because of uncertainty in VLE data	Usually low
5. Wide boiling range mixtures, low to moderate viscosity (pentane-octane, or water glycol)	Fairly low. Nucleate boiling suppressed. Convection controlled.	Can be high with good design	Can be fairly high	Function of circulation rate. Exit temp. increases with vapor fraction	Moderate
6. Very wide boiling range, high viscosity (pentane and 1400 MW butylene polymer oil) BR > 400°F, viscosity > 50 cP	Very low, < 50 Btu/hr ft ² °F (< 280 w/m ² °C. Nucleate boiling suppressed. Poor radial mixing	Very low	Low. Early film boiling due to hindered vapor release	Nonequilibrium effects. Difficult to calculate	Can be high (high wall temperature)
7. Sparged reboilers (steam injected rerun column reboiler)	Can be high, but may have mass transfer limitations at too high sparge rate	Greatly increased by sparging	Mist flow limitation at too high sparge rate	Boiling temp. a function of partial pressure of sparge gas	Lower wall temp. can decrease fouling
8. Total vaporization (LNG revaporizer) (special cases only). Not recommended for general design.	Part of tube in mist flow with low coefficient.	Forced flow	Top of tube above mist flow limit	Nonlinear, superheating above mist flow boundary	Suitable only for non-fouling fluids
9. Polymerizing (unsaturated hydrocarbons (butene-butadiene mixture)	High	High	High	Limit wall temp.	Can be very rapid above certain wall temp. and certain vapor fraction
10. Aqueous solutions of inorganic salts (aluminum sulfate solution)	High, similar to water. Solids do not impede boiling process except at very high solids concentration	High	High	Similar to water except at very high solids concentration	Inverse solubility salts deposits on surface similarly to case of single phase flow. Regular solubility salts may deposit at high vapor fraction
11. Heavy hydrocarbon mixture with heavy fouling (solvent cleanup column bottoms)	Fairly low	Can be low. Should be made as high as possible.	Fairly high. But, due to very high design fouling factor, startup with full steam pressure can cause film boiling.	May be a function of circulation rate.	Fouling factor usually > 0.005 hr ft ² °F/Btu. Sedimentation, polymerization coking. Sensitive to temp., velocity, and trace elements
12. Low temp. boiler-condensers (boiling ethylene-condensing propylene)	Moderately high. Boiling side sensitive to surface characteristics due to low T. Condensing side resistance also significant.	Can be high	High	Very low producing low design heat flux	Very low except for the effects of compressor lubricants
13. Refrigerants (R11, R22, R113)	Fairly high	Generally high but may be lower than hydrocarbons due to higher vapor density	High	No problems if good vapor pressure data are used	Very low except for the effects of compressor lubricants
14. Cryogenics (boiling O ₂ with condensing N ₂)	May be about the same as light hydrocarbons	Expected to be high	T's may be high so watch for film boiling	Adjust for design either in complete nucleate or complete film boiling, not partial film boiling	Very low
15. Waste heat boiler (steam from hot gas)	Very high for boiling side. Heating medium resistance is controlling.	High for good design	Carefully check for mist flow and film boiling as T may be very high	Use concurrent flow to provide maximum T at bottom of tube	Usually low

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TABLE 2
DESIGN SUGGESTIONS FOR VERTICAL THERMOSIPHON REBOILERS

CATEGORY (Examples are Given in Parentheses)	RECOMMENDATIONS (SUGGESTED RANGES AND COMMENTS)			
	Tube Diameter	Tube Length	Static Head	General Comments
1. Narrow boiling range saturated hydrocarbons, moderate pressure (C ₅ -C ₇ , 100 psia)	3/4 - 1 in. (19.1 - 25.4 mm)	Up to 20 ft (6.1 m) if exit weight fraction vapor < 0.35	Equal to tube length.	Few problems and generally good operation
2. Narrow boiling range, aqueous solution of organics (alcohols and water)	3/4 - 1 in. (19.1 - 25.4 mm)	Up to 20 ft (6.1 m) if exit weight fraction vapor < 0.35	Equal to tube length.	High heat transfer coefficients cause design to be sensitive to fouling factor. Feed in boiling temperature profile available.
3. Vacuum reboilers with heavy hydrocarbons (mixed xylenes, 1 psia)	1-1/4 - 1-3/4 in. (31.8 - 44.5 mm)	Usually 12 ft (3.65 m) or less	Less than tube length. Optimum exists	Limit exit vapor weight fraction to less than 0.5. Check for instability. Increase inlet resistance if unstable.
4. Very high pressure (near critical). Light hydrocarbon	1 - 1-1/4 in. (25.4 - 31.8 mm)	Usually 12 ft (3.65 m) or less	As large as possible	Low confidence in physical properties. Operation is sensitive to pressure level. Some safety factor is required. May be better to design for film boiling in critical cases.
5. Wide boiling range mixtures, low to moderate viscosity (pentane-octane, or water glycol)	1 - 1-1/2 in. (25.4 - 38.1 mm). Large enough to allow high circulation rate	Short enough to allow high circulation with tube diameter used	As large as possible	Design for low exit vapor weight fraction (0.1 to 0.25). This increases h, MTD, and confidence in prediction. Always specify boiling range.
6. Very wide boiling range, high viscosity (pentane and 1400 MW butylene polymer oil) BR > 400°F, viscosity > 50 cP	1-3/4 - 2 in. (44.5 - 50.8 mm)	Short	As large as possible. Forced circulation or sparging may be necessary	Thermosiphon reboilers do not operate effectively for these extreme conditions. Consider also shellside horizontal thermosiphons (possibly better mixing) or falling film evaporators.
7. Sparged reboilers (steam injected rerun column reboiler)	1 - 1-1/4 in. (25.4 - 31.8 mm)	Usually 12 ft (3.65 m) or less	Equal to tube length.	Evaluate downstream handling problems. Carefully calculate bulk temperature profile.
8. Total vaporization (LNG revaporizer) (special cases only). Not recommended for general design.	3/4 - 1 in. (19.1 - 25.4 mm)	Length above mist flow boundary is very ineffective	Forced flow only	Use valve in inlet line to damp out instability and regulate location of mist flow boundary. Specify some excess capacity.
9. Polymerizing (unsaturated hydrocarbons (butene-butadiene mixture)	1 - 1-1/4 in. (25.4 - 31.8 mm)	8 ft (2.4 m) or more. Limited by recommended exit vapor fraction	Equal to tube length.	Limit wall temperature and exit vapor fraction below rapid polymerization threshold
10. Aqueous solutions of inorganic salts (aluminum sulfate solution)	1 - 1-1/4 in. (25.4 - 31.8 mm)	Can be long if exit vapor fraction not above 0.1	Equal to tube length.	Design to low vapor fraction. Limit tube wall temperature for inverse solubility.
11. Heavy hydrocarbon mixture with heavy fouling (solvent cleanup column bottoms)	1-1/4 - 2 in. (31.8 - 50.8 mm)	12 ft (3.65 m) or less	May need forced flow for extreme cases	Design for as high velocity as possible with low vapor fraction and low wall temperature. Provide spare reboiler in critical cases with low P valves to isolate unit for cleaning.
12. Low temp. boiler-condensers (boiling ethylene-condensing propylene)	3/4 - 1-1/4 in. (19.1 - 31.8 mm)	Can be long if exit vapor fraction is less than 0.35	Equal to tube length.	Evaluate as a condenser also on CST Program. Feed in nucleate boiling curve if available. Consider enhanced surface for both boiling and condensing sides.
13. Refrigerants (R11, R22, R113)	3/4 - 1-1/4 in. (19.1 - 31.8 mm)	Can be long if exit vapor fraction is less than 0.35	Equal to tube length.	Be specific with properties. Operation near critical pressure is very uncertain due to limited data.
14. Cryogenics (boiling O ₂ with condensing N ₂)	3/4 - 1 in. (19.1 - 25.4 mm)	Can be long if exit vapor fraction is less than 0.35	Equal to tube length.	Little field experience available in applying process reboiler methods to cryogenics. Feed in nucleate boiling curve if available. Check against field experience where possible.
15. Waste heat boiler (steam from hot gas)	1 - 1-1/4 in. (25.4 - 31.8 mm)	Can be long if exit vapor fraction is low	Can be large if subcooled liquid zone is not greater than 1/4 of the tube length	Design for around 10 weight percent vapor. Check instability. Provide additional inlet piping resistance if necessary. Watch for mist flow and for film boiling at high T end. If boiling is shellside, use strip baffles. Design for even flow distribution and eliminate vapor accumulation regions.

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APPENDIX A
HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS

Exchanger No. E - Plant Done by
Service Date

Customary

Metric

A. BOILING SIDE SPECIFICATIONS

1. Pressure, P	<u> </u> psia	<u> </u> kPa
2. Exit Vapor Temperature, T_v	<u> </u> °F	<u> </u> °C
3. Feed Temperature, T_f	<u> </u> °F	<u> </u> °C
4. Bottoms Drawoff Temperature, T_d	<u> </u> °F	<u> </u> °C
5. Bottoms Drawoff Flowrate, WB	<u> </u> lb/hr	<u> </u> kg/hr
6. Average Heating Medium Temperature, T_h	<u> </u> °F	<u> </u> °C
7. Available Temperature Driving Force, $\Delta T_a = T_h - T_v$	<u> </u> °F	<u> </u> °C

B. BOILING FLUID PROPERTIES

1. Boiling Range, BR	<u> </u> °F	<u> </u> °C
2. Bubble Point, T_s	<u> </u> °F	<u> </u> °C
3. Critical Pressure, P_c	<u> </u> psia	<u> </u> kPa
4. Heat Capacity of Bottoms Drawoff, C_{pb}	<u> </u> Btu/lb-°F	<u> </u> kJ/kg-K
5. Reduced Pressure, (P / P_c)	<u> </u>	<u> </u>
6. Average Vapor Density, ρ_v	<u> </u> lb/ft ³	<u> </u> kg/m ³
7. Average Liquid Density, ρ_ℓ	<u> </u> lb/ft ³	<u> </u> kg/m ³
8. Average Latent Heat, λ	<u> </u> Btu/lb	<u> </u> kJ/kg
9. Liquid Surface Tension, σ At exit Temperature [Required for kettle sizing only]	<u> </u> dynes/cm	<u> </u> dynes/cm

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (Cont)

C. EXCHANGER GEOMETRY

- Number of Tube Holes in Tubesheet, NTT

- Bundle Length, L
_____ ft _____ m
- Tube Inside Diameter, d_i
_____ in. _____ mm
- Ratio of Outside to Inside Surface Area, AOI

- Tube Pitch Ratio, PR

- Number of Vapor Exit Nozzles, NN

- Tube Wall Thermal Conductivity, k_w
_____ Btu/hr-ft-°F _____ W/m-K

D. FOULING RESISTANCES

- Tubeside, r_{fi}
_____ hr-ft²-°F/Btu _____ m²-K/W
- Shellside, r_{fo}
_____ hr-ft²-°F/Btu _____ m²-K/W

E. NUCLEATE BOILING CALCULATIONS

- Single Tube Maximum Heat Flux, q_{1max} , **Figure A3**
_____ Btu/hr-ft² _____ W/m²
- Bundle Correction Factor, CB, **Figure A4**

- Bundle Maximum Heat Flux, $q_{bmax} = CB (q_{1max})$
_____ Btu/hr-ft² _____ W/m²
- Reference Heat Fluxes. Use two to construct a plot.

$q_1 = 0.1(q_{bmax})$ _____ Btu/hr-ft² _____ W/m²
 $q_2 = 0.6(q_{bmax})$ _____ Btu/hr-ft² _____ W/m²
- Single Tube Nucleate boiling Reference Coefficient h_{1r} , **Figure A5**

At q_1 , h_{1r1} _____ Btu/hr-ft²-°F _____ W/m²-K
 At q_2 , h_{1r2} _____ Btu/hr-ft²-°F _____ W/m²-K

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (Cont)

6. Pressure Correction, F_p ,

Figure A6

7. Mixture Correction, F_c

- 7.1 Effective Minimum Heat Flux for Nucleate Boiling, q_n

24858 / (BR)^{0.8}
(if > 1000, use 1000)

78365 / (BR)^{0.8}
(if > 3150, use 3150)

_____ Btu/hr-ft² _____ W/m²

- 7.2 (q/q_n) , for **Figure A8**

At q_1

At q_2

- 7.3 Density Ratio, $x_r = (\rho_v / \rho_\ell)$, Limit $x_r \leq 0.2$

- 7.4 Effective Temperature Range, DR, **Figure A7**

_____ °F _____ °C

- 7.5 (BR / DR)

- 7.6 F_c , **Figure A8**

At q_1 , F_{c1}

At q_2 , F_{c2}

8. Single Tube Boiling Coefficients, $h_1 = (h_{1r})(F_p)(F_c)^{0.67}$

At q_1 , h_{11} _____ Btu/hr-ft²-°F _____ W/m²-K

At q_2 , h_{12} _____ Btu/hr-ft²-°F _____ W/m²-K

9. Bundle Convection Correction, F_b ,

Figure A9

10. Sensible Heating Correction, F_d

- 10.1 Sensible Heat Duty, Q_s

(WB)(C_{pb})(T_d - T_f)

(WB)(C_{pb})(T_d - T_f) / 3.6

_____ Btu/hr _____ W

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (Cont)

10.2 Bundle Surface Area, A

$$0.262(d_i)(AOI)(L)(NTT)$$

$$0.00314(d_i)(AOI)(L)(NTT)$$

$$\text{_____ ft}^2$$

$$\text{_____ m}^2$$

10.3 Sensible Heating Correction, $F_d = (1 - Q_s / (q A))^{1/3}$
 [If ≤ 0 , return to step E4, change q_1 to $0.2(q_{bmax})$ and repeat]

At $q = q_1$,

At $q = q_2$,

11. Finned Tube Correction, $F_f = (F_e)(F_s)$, **Figures A10 or A11, and A12**. Use 1.0 for Plain Tubes

12. Average Bundle Boiling Coefficients (Including 0.75 Safety Factor)

$$h_{ab} = (h_1)(F_b)(F_d)(F_f)(0.75)$$

At q_1 , h_{ab1}

_____ Btu/hr-ft²-°F

_____ W/m²-K

At q_2 , h_{ab2}

_____ Btu/hr-ft²-°F

_____ W/m²-K

13. Wall Resistance, r_w , Plain or Finned Tubes

Carbon Steel

_____ 0.0004 hr-ft²-°F/Btu

_____ 0.00007 m²-K/W

Other

_____ 0.0004 (28/ k_w)

_____ 0.00007(48/ k_w)

_____ hr-ft²-°F/Btu

_____ m²-K/W

14. Tubeside Heat Transfer Coefficient, h_h

14.1 For Steam

$$2000 + 0.033[(q)(L)]^{0.8}$$

$$11357 + 0.193[(q)(L)]^{0.8}$$

At q_1

_____ Btu/hr-ft²-°F

_____ W/m²-K

At q_2

_____ Btu/hr-ft²-°F

_____ W/m²-K

14.2 Other than Steam, **Section IX-F**

At q_1

_____ Btu/hr-ft²-°F

_____ W/m²-K

At q_2

_____ Btu/hr-ft²-°F

_____ W/m²-K

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (Cont)

15. Overall Heat Transfer Coefficients

15.1 Clean

$$U_c = \frac{1}{\frac{1}{h_{ab}} + \frac{AOI}{h_h} + R_w}$$

At q_1 $U_{c1} =$ _____ Btu/hr-ft²-°F _____ W/m²-K

At q_2 $U_{c2} =$ _____ Btu/hr-ft²-°F _____ W/m²-K

15.2 Fouled

$$U_f = \frac{1}{\frac{1}{U_c} + (AOI)(r_{fi}) + (r_{fo})}$$

At q_1 $U_{f1} =$ _____ Btu/hr-ft²-°F _____ W/m²-K

At q_2 $U_{f2} =$ _____ Btu/hr-ft²-°F _____ W/m²-K

F. OVERALL ΔT CALCULATION FOR NUCLEATE BOILING

ΔT is the difference between the average heating medium (or condensing) temperature and the exit vapor temperature.

1. Clean

$$\Delta T_c = q / U_c$$

At $q = q_1$ $\Delta T_{c1} =$ _____ °F _____ °C

At $q = q_2$ $\Delta T_{c2} =$ _____ °F _____ °C

2. Fouled

$$\Delta T_f = q / U_f$$

At $q = q_1$ $\Delta T_{f1} =$ _____ °F _____ °C

At $q = q_2$ $\Delta T_{f2} =$ _____ °F _____ °C

G. OVERALL ΔT CALCULATION FOR FILM BOILING

[Calculations may be skipped if ΔT_a is in the nucleate boiling region]

1. Minimum Film Boiling Heat Flux, q_{bmin2}

$q_{bmin} = (q_{1max})(0.12)$ _____ Btu/hr-ft² _____ W/m²

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (Cont)

2. Film Boiling Heat Transfer Coefficients, h_{fb}

At q_{bmin} , h_{fb1}	$50 (P_c)^{0.25} (P/P_c)^{0.43}$	$176 (P_c)^{0.25} (P/P_c)^{0.43}$
	_____ Btu/hr-ft ² -°F	_____ W/m ² -K
At $(2)(q_{bmin})$, $h_{fb2} = 0.71(h_{fb1})$	_____ Btu/hr-ft ² -°F	_____ W/m ² -K

3. Clean ΔT (with the approximation $U = h_{fb}$)

$\Delta T_{fc1} = q_{bmin} / h_{fb1}$	_____ °F	_____ °C
$\Delta T_{fc2} = (2)(q_{bmin}) / h_{fb2}$	_____ °F	_____ °C

4. Fouled ΔT

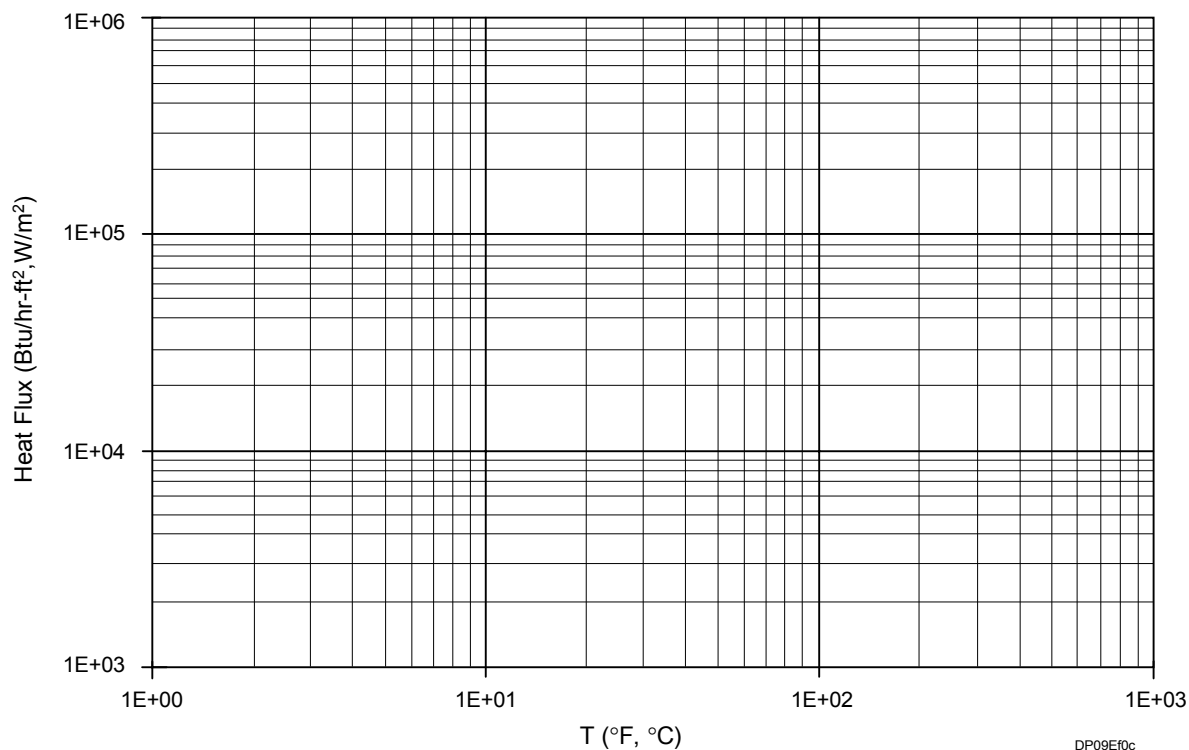
$\Delta T_{ff1} = \Delta T_{fc1} + q_{bmin} [(AOI) (r_{fi}) + r_{fo}]$	_____ °F	_____ °C
$\Delta T_{ff2} = \Delta T_{fc2} + 2 (\Delta T_{ff1} - \Delta T_{fc1})$	_____ °F	_____ °C

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (Cont)

H. CONSTRUCT PLOT

1. Plot heat flux vs. overall ΔT on a log-log graph for each of the four ΔT s in items F and G.



I. CALCULATE HEAT DUTY AND VAPOR FLOW

- | | | |
|---------------------------------------|--------------------------------|--------------------------------------|
| 1. Heat Flux, q_a , at ΔT_a | _____ Btu/hr-ft ² | _____ W/m ² |
| 2. Total Heat Duty, $Q_T = (q_a)(A)$ | _____ Btu/hr | _____ W |
| 3. Vaporizing Duty $Q_V = Q_T - Q_s$ | _____ Btu/hr | _____ W |
| 4. Vapor Flow, W_V | Q_V / λ
_____ lb/hr | $Q_V / \lambda / 1000$
_____ kg/s |

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (Cont)

J. KETTLE SIZE CALCULATION

[See **Figure A13**]

1. Maximum Permissible Entrainment Ratio, E , (lb (kg) liquid/lb (kg) vapor)

2. Entrainment Coefficient, $K = J_2 (E)^{0.33} (\sigma / (\rho_\ell - \rho_v))^{0.25}$

$J_2 = 0.916$

$J_2 = 0.558$

_____ ft/s

_____ m/s

3. Maximum Velocity, $V_{\max} = K(\rho_\ell / \rho_v - 1)^{0.5}$

_____ ft/s

_____ m/s

4. Bundle Diameter, D_b ,

_____ in.

_____ mm

5. Kettle Diameter, D_s ,
[guess for design]

_____ in.

_____ mm

6. Weir Height, H_w

_____ in.

_____ mm

7. Chord Length above Weir,
 $L_c = (4H_w(D_s - H_w))^{0.5}$
[If $D_s > 2D_b$, $L_c = D_s$]

_____ in.

_____ mm

8. Vertical Velocity Area, S_v

$(L_c)(L) / 12$

$(L_c)(L) / 1000$

_____ ft²

_____ m²

9. Chord Angle,

$$\theta' = 2 \sin^{-1} (L_c/D_s)$$

_____ deg

_____ deg

$$\theta = \theta' (\pi/180)$$

_____ rad

_____ rad

10. Horizontal Velocity Area, S_h

$$D_s^2 (\theta - \sin\theta) / 1152$$

$$D_s^2 (\theta - \sin\theta) / (8 \times 10^6)$$

_____ ft²

_____ m²

11. Vertical Velocity, V_v

$$W_v / (3600 \rho_v S_v)$$

$$W_v / (\rho_v S_v)$$

_____ ft/s

_____ m/s

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (Cont)

12. $V_v \leq V_{\max}$? _____

[If NO, then for design cases reguess D_s and repeat from step 7]

13. Horizontal Velocity, V_h $W_v / (3600 \rho_v S_h^{2 NN})$ $W_v / (\rho_v S_h^{2 NN})$
 _____ ft/s _____ m/s

14. Effective Velocity, $V_e = (V_v V_h)^{0.5}$ [limit $V_e \geq V_v$]
 _____ ft/s _____ m/s

15. $V_e \leq V_{\max}$? _____

[If NO, then for design cases reguess D_s and repeat from step J5]

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APPENDIX B

HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (SAMPLE)

Exchanger No. E - (Example 1) Plant _____ Done by _____

Service _____ Date _____

Customary

Metric

A. BOILING SIDE SPECIFICATIONS

1. Pressure, P	<u>172</u> psia	<u>1,187</u> kPa
2. Exit Vapor Temperature, T_v	<u>353</u> °F	<u>179</u> °C
3. Feed Temperature, T_f	<u>337</u> °F	<u>170</u> °C
4. Bottoms Drawoff Temperature, T_d	<u>353</u> °F	<u>179</u> °C
5. Bottoms Drawoff Flowrate, WB	<u>310,000</u> lb/hr	<u>140,572</u> kg/hr
6. Average Heating Medium Temperature, T_h	<u>420</u> °F	<u>216</u> °C
7. Available Temperature Driving Force, $\Delta T_a = T_h - T_v$	<u>67</u> °F	<u>37</u> °C

B. BOILING FLUID PROPERTIES

1. Boiling Range, BR	<u>64</u> °F	<u>35</u> °C
2. Bubble Point, T_s	<u>337</u> °F	<u>170</u> °C
3. Critical Pressure, P_c	<u>456</u> psia	<u>3145</u> kPa
4. Heat Capacity of Bottoms Drawoff, C_{pb}	<u>0.68</u> Btu/lb-°F	<u>2.8</u> kJ/kg-K
5. Reduced Pressure, (P / P_c)	<u>0.38</u>	<u>0.38</u>
6. Average Vapor Density, ρ_v	<u>2.05</u> lb/ft ³	<u>32.8</u> kg/m ³
7. Average Liquid Density, ρ_ℓ	<u>33.9</u> lb/ft ³	<u>542.8</u> kg/m ³
8. Average Latent Heat, λ	<u>125.2</u> Btu/lb	<u>291.0</u> kJ/kg
9. Liquid Surface Tension, σ , At exit Temperature [Required for kettle sizing only]	<u>5.5</u> dynes/cm	<u>5.5</u> dynes/cm

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (SAMPLE) (Cont)

C. EXCHANGER GEOMETRY

1. Number of Tube Holes in Tubesheet, NTT	<u>270</u>		<u>270</u>	
2. Bundle Length, L	<u>12.6</u>	ft	<u>3.84</u>	m
3. Tube Inside Diameter, d_i	<u>0.782</u>	in.	<u>19.86</u>	mm
4. Ratio of Outside to Inside Surface Area, AOI	<u>1.279</u>		<u>1.279</u>	
5. Tube Pitch Ratio, PR	<u>1.25</u>		<u>1.25</u>	
6. Number of Vapor Exit Nozzles, NN	<u>1</u>		<u>1</u>	
7. Tube Wall Thermal Conductivity, k_w	<u>28</u>	Btu/hr-ft-°F	<u>48.5</u>	W/m-K

D. FOULING RESISTANCES

1. Tubeside, r_{fi}	<u>0.002</u>	hr-ft ² -°F/Btu	<u>0.00035</u>	m ² -K/W
2. Shellside, r_{fo}	<u>0.001</u>	hr-ft ² -°F/Btu	<u>0.00018</u>	m ² -K/W

E. NUCLEATE BOILING CALCULATIONS

1. Single Tube Maximum Heat Flux, q_{1max} , Figure A3	<u>1.7 X 10⁵</u>	Btu/hr-ft ²	<u>5.4 X 10⁵</u>	W/m ²
2. Bundle Correction Factor, CB, Figure A4	<u>0.35</u>		<u>0.35</u>	
3. Bundle Maximum Heat Flux, $q_{bmax} = CB(q_{1max})$	<u>59,500</u>	Btu/hr-ft ²	<u>189,000</u>	W/m ²
4. Reference Heat Fluxes. Use two to construct a plot.				
$q_1 = 0.1(q_{bmax})$	<u>5,950</u>	Btu/hr-ft ²	<u>18,900</u>	W/m ²
$q_2 = 0.6(q_{bmax})$	<u>35,700</u>	Btu/hr-ft ²	<u>113,400</u>	W/m ²
5. Single Tube Nucleate boiling Reference Coefficient h_{1r} , Figure A5				
At q_1 , h_{1r1}	<u>290</u>	Btu/hr-ft ² -°F	<u>1,700</u>	W/m ² -K
At q_2 , h_{1r2}	<u>910</u>	Btu/hr-ft ² -°F	<u>5,200</u>	W/m ² -K

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (SAMPLE) (Cont)

6. Pressure Correction, F_p , Figure A6	<u>2.6</u>	<u>2.6</u>
7. Mixture Correction, F_c		
7.1 Effective Minimum Heat Flux for Nucleate Boiling, q_n		
	$24858 / (BR)^{0.8}$ (if > 1000, use 1000)	$78365 / (BR)^{0.8}$ (if > 3150, use 3150)
	<u>892</u> Btu/hr-ft ²	<u>3150</u> W/m ²
7.2 (q / q_n) , for Figure A8		
At q_1	<u>6.7</u>	<u>6</u>
At q_2	<u>40.0</u>	<u>36</u>
7.3 Density Ratio, $x_r = (\rho_v / \rho_\ell)$, Limit $x_r \leq 0.2$	<u>0.06</u>	<u>0.06</u>
7.4 Effective Temperature Range, DR, Figure A7	<u>90</u> °F	<u>50</u> °C
7.5 (BR / DR)	<u>0.711</u>	<u>0.711</u>
7.6 F_c , Figure A8		
At q_1 , F_{c1}	<u>0.26</u>	<u>0.26</u>
At q_2 , F_{c2}	<u>0.19</u>	<u>0.19</u>
8. Single Tube Boiling Coefficients, $h_1 = (h_{1r})(F_p)(F_c)^{0.67}$		
At q_1 , h_{11}	<u>306</u> Btu/hr-ft ² -°F	<u>1,838</u> W/m ² -K
At q_2 , h_{12}	<u>778</u> Btu/hr-ft ² -°F	<u>4,599</u> W/m ² -K
9. Bundle Convection Correction, F_b , Figure A9	<u>1.8</u>	<u>1.8</u>
10. Sensible Heating Correction, F_d		
10.1 Sensible Heat Duty, Q_s	$(WB)(C_{pb})(T_d - T_f)$ <u>3,372,800</u> Btu/hr	$(WB)(C_{pb})(T_d - T_f)/3.6$ <u>973,000</u> W

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (SAMPLE) (Cont)

10.2 Bundle Surface Area, A

$$0.262(d_i)(AOI)(L)(NTT)$$

$$0.00314(d_i)(AOI)(L)(NTT)$$

$$\underline{891} \text{ ft}^2$$

$$\underline{82.7} \text{ m}^2$$

10.3 Sensible Heating Correction, $F_d = (1 - Q_s / (q A))^{1/3}$
[If ≤ 0 , return to step E4, change q_1 to $0.2(q_{bmax})$ and repeat]

At $q = q_1$,

$$\underline{0.71}$$

$$\underline{0.71}$$

At $q = q_2$,

$$\underline{0.96}$$

$$\underline{0.96}$$

11. Finned Tube Correction, $F_f = (F_e)(F_s)$, **Figures A10 or A11**, and **A12**. Use 1.0 for Plain Tubes.

$$\underline{1.0}$$

$$\underline{1.0}$$

12. Average Bundle Boiling Coefficients (Including 0.75 Safety Factor)

$$h_{ab} = (h_1)(F_b)(F_d)(F_f)(0.75)$$

At q_1 , h_{ab1}

$$\underline{293} \text{ Btu/hr-ft}^2\text{-}^\circ\text{F}$$

$$\underline{1,837} \text{ W/m}^2\text{-K}$$

At q_2 , h_{ab2}

$$\underline{1,008} \text{ Btu/hr-ft}^2\text{-}^\circ\text{F}$$

$$\underline{6,126} \text{ W/m}^2\text{-K}$$

13. Wall Resistance, r_w , Plain or Finned Tubes

Carbon Steel

$$\underline{0.0004} \text{ hr-ft}^2\text{-}^\circ\text{F/Btu}$$

$$\underline{0.00007} \text{ m}^2\text{-K/W}$$

Other

$$0.0004 (28/k_w)$$

$$0.00007 (48/k_w)$$

$$\underline{\hspace{1cm}} \text{ hr-ft}^2\text{-}^\circ\text{F/Btu}$$

$$\underline{\hspace{1cm}} \text{ m}^2\text{-K/W}$$

14. Tubeside Heat Transfer Coefficient, h_h

14.1 For Steam

$$2000 + 0.033[(q)(L)]^{0.8}$$

$$11357 + 0.193[(q)(L)]^{0.8}$$

At q_1

$$\underline{2,262} \text{ Btu/hr-ft}^2\text{-}^\circ\text{F}$$

$$\underline{12,850} \text{ W/m}^2\text{-K}$$

At q_2

$$\underline{3,099} \text{ Btu/hr-ft}^2\text{-}^\circ\text{F}$$

$$\underline{17,616} \text{ W/m}^2\text{-K}$$

14.2 Other than Steam, **Section IX-F**

At q_1

$$\underline{\hspace{1cm}} \text{ Btu/hr-ft}^2\text{-}^\circ\text{F}$$

$$\underline{\hspace{1cm}} \text{ W/m}^2\text{-K}$$

At q_2

$$\underline{\hspace{1cm}} \text{ Btu/hr-ft}^2\text{-}^\circ\text{F}$$

$$\underline{\hspace{1cm}} \text{ W/m}^2\text{-K}$$

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (SAMPLE) (Cont)

15. Overall Heat Transfer Coefficients

15.1 Clean

$$U_c = \frac{1}{\frac{1}{h_{ab}} + \frac{AOI}{h_h} + R_w}$$

At q_1 $U_{c1} = \underline{228}$ Btu/hr-ft²-°F $\underline{1,401}$ W/m²-K

At q_2 $U_{c2} = \underline{554}$ Btu/hr-ft²-°F $\underline{3,270}$ W/m²-K

15.2 Fouled

$$U_f = \frac{1}{\frac{1}{U_c} + (AOI)(r_{fi}) + (r_{fo})}$$

At q_1 $U_{f1} = \underline{126}$ Btu/hr-ft²-°F $\underline{746}$ W/m²-K

At q_2 $U_{f2} = \underline{186}$ Btu/hr-ft²-°F $\underline{1,073}$ W/m²-K

F. OVERALL ΔT CALCULATION FOR NUCLEATE BOILING

ΔT is the difference between the average heating medium (or condensing) temperature and the exit vapor temperature.

1. Clean

$$\Delta T_c = q / U_c$$

At $q = q_1$ $\Delta T_{c1} = \underline{26.1}$ °F $\underline{13.5}$ °C

At $q = q_2$ $\Delta T_{c2} = \underline{64.4}$ °F $\underline{34.7}$ °C

2. Fouled

$$\Delta T_f = q / U_f$$

At $q = q_1$ $\Delta T_{f1} = \underline{47.2}$ °F $\underline{25.3}$ °C

At $q = q_2$ $\Delta T_{f2} = \underline{192.0}$ °F $\underline{105.7}$ °C

G. OVERALL ΔT CALCULATION FOR FILM BOILING

[Calculations may be skipped if ΔT_a is in the nucleate boiling region]

1. Minimum Film Boiling Heat Flux, q_{bmin2}

$q_{bmin} = (q_{1max})(0.12)$ $\underline{20,400}$ Btu/hr-ft² $\underline{64,800}$ W/m²

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (SAMPLE) (Cont)

2. Film Boiling Heat Transfer Coefficients, h_{fb}

$At\ q_{bmin}, h_{fb1}$	$50 (P_c)^{0.25} (P/P_c)^{0.43}$	$176 (P_c)^{0.25} (P/P_c)^{0.43}$
	<u>152</u> Btu/hr-ft ² -°F	<u>869</u> W/m ² -K

$At\ (2)(q_{bmin}), h_{fb2} = 0.71(h_{fb1})$	<u>108</u> Btu/hr-ft ² -°F	<u>617</u> W/m ² -K
--	---------------------------------------	--------------------------------

3. Clean ΔT (with the approximation $U = h_{fb}$)

$\Delta T_{fc1} = q_{bmin} / h_{fb1}$	<u>134.2</u> °F	<u>74.5</u> °C
---------------------------------------	-----------------	----------------

$\Delta T_{fc2} = (2)(q_{bmin}) / h_{fb2}$	<u>377.8</u> °F	<u>209.9</u> °C
--	-----------------	-----------------

4. Fouled ΔT

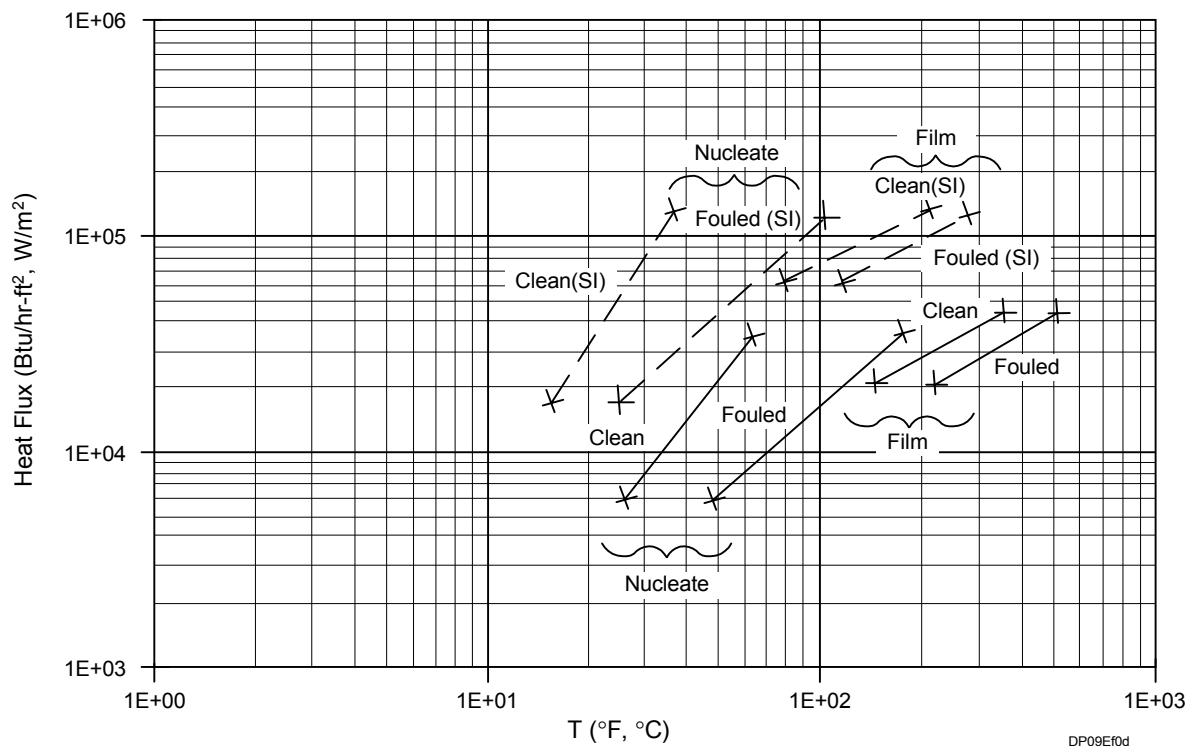
$\Delta T_{ff1} = \Delta T_{fc1} + q_{bmin} [(AOI) (r_{fi}) + r_{fo}]$	<u>206.8</u> °F	<u>115.0</u> °C
--	-----------------	-----------------

$\Delta T_{ff2} = \Delta T_{fc2} + 2 (\Delta T_{ff1} - \Delta T_{fc1})$	<u>523.0</u> °F	<u>290.0</u> °C
---	-----------------	-----------------

HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (SAMPLE) (Cont)

H. CONSTRUCT PLOT

- Plot heat flux vs. overall ΔT on a log-log graph for each of the four ΔT 's in items F and G.



I. CALCULATE HEAT DUTY AND VAPOR FLOW

1. Heat Flux, q_a , at ΔT_a	<u>9,200</u> Btu/hr-ft ²	<u>28,000</u> W/m ²
2. Total Heat Duty, $Q_T = (q_a)(A)$	<u>8,197,200</u> Btu/hr	<u>2,314,480</u> W
3. Vaporizing Duty, $Q_V = Q_T - Q_s$	<u>4,824,400</u> Btu/hr	<u>1,341,480</u> W
4. Vapor Flow, W_V	Q_V / λ	$Q_V / \lambda / 1000$
	<u>38,534</u> lb/hr	<u>4.61</u> kg/s

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (SAMPLE) (Cont)

J. KETTLE SIZE CALCULATION

[See **Figure A13**]

1. Maximum Permissible Entrainment Ratio, E, (lb (kg) liquid/lb (kg) vapor)

0.01
0.01

2. Entrainment Coefficient, $K = J_2 (E)^{0.33} (\sigma / (\rho_\ell - \rho_v))^{0.25}$

$J_2 = 0.916$
 $J_2 = 0.558$

0.129 ft/s
0.039 m/s

3. Maximum Velocity, $V_{\max} = K (\rho_\ell / \rho_v - 1)^{0.5}$

0.51 ft/s
0.155 m/s

4. Bundle Diameter, D_b ,

26 in.
660 mm

5. Kettle Diameter, D_s ,
[guess for design]

52 in.
1,321 mm

6. Weir Height, H_w

31 in.
787 mm

7. Chord Length above Weir,
 $L_c = (4H_w(D_s - H_w))^{0.5}$
[If $D_s > 2D_b$, $L_c = D_s$]

51.0 in.
1,297 mm

8. Vertical Velocity Area S_v

$(L_c)(L) / 12$
 $(L_c)(L) / 1000$

53.6 ft²
4.98 m²

9. Chord Angle,
 $\theta' = 2 \sin^{-1} (L_c/D_s)$

158 deg
158 deg

$$\theta = \theta' (\pi/180)$$

2.75 rad
2.75 rad

10. Horizontal Velocity Area, S_h

$D_s^2 (\theta - \sin\theta) / 1152$
 $D_s^2 (\theta - \sin\theta) / (8 \times 10^6)$

5.56 ft²
0.517 m²

11. Vertical Velocity, V_v

$W_v / (3600 \rho_v S_v)$
 $W_v / (\rho_v S_v)$

0.097 ft/s
0.030 m/s

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HEAT TRANSFER CALCULATION FORM FOR KETTLE AND INTERNAL REBOILERS (SAMPLE) (Cont)

12. $V_v \leq V_{\max}$? yes yes
 [If NO, then for design cases reguess D_s and repeat from step 7]
13. Horizontal Velocity, V_h $W_v / (3600 \rho_v S_h^2 NN)$ $W_v / (\rho_v S_h^2 NN)$
0.47 ft/s 0.14 m/s
14. Effective Velocity, $V_e = (V_v V_h)^{0.5}$ [limit $V_e \geq V_v$]
0.21 ft/s 0.065 m/s
15. $V_e \leq V_{\max}$? yes yes
 [If NO, then for design cases reguess D_s and repeat from step J5]

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APPENDIX C
HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS

Exchanger No. E - Plant Done by
Service Date

Customary

Metric

A. BOILING SIDE SPECIFICATIONS

- | | | |
|-------------------------------------|------------------------------------|-----------------------------------|
| 1. Pressure, P | <u> </u> psia | <u> </u> kPa |
| 2. Inlet Temperature, T_{bi} | <u> </u> °F | <u> </u> °C |
| 3. Exit Temperature, T_{bo} | <u> </u> °F | <u> </u> °C |
| 4. Saturation Temperature, T_{bs} | <u> </u> °F | <u> </u> °C |
| 5. Total Heat Duty, Q_T | <u> </u> Btu/hr | <u> </u> kW |
| 6. Flow Rate, W | <u> </u> lb/hr | <u> </u> kg/hr |
| 7. Outlet Vapor Fraction, y_o | <u> </u> | <u> </u> |
- [Enter either item 6 or item 7. The other will be calculated.]

B. HEATING MEDIUM SPECIFICATIONS

- | | | |
|--|--------------------------------|--------------------------------|
| 1. Inlet Temperature, T_{hi} | <u> </u> °F | <u> </u> °C |
| 2. Exit Temperature, T_{ho} | <u> </u> °F | <u> </u> °C |
| 3. Saturation Temperature, T_{hs}
(for condensing streams only) | <u> </u> °F | <u> </u> °C |

C. AVAILABLE ΔT

[For pure component boiling vs. condensing heating medium use (average heating medium temp minus exit vapor temp); for mixture boiling vs. condensing heating medium use log-mean ΔT with hot inlet at saturation; for single phase heating medium use log-mean ΔT].

 °F °C

D. BOILING FLUID PROPERTIES

- | | | |
|--|----------------------------------|---------------------------------|
| 1. Boiling Range, BR,
(Dew point - Bubble point)
at inlet pressure | <u> </u> °F | <u> </u> °C |
| 2. Critical Pressure, P_c | <u> </u> psia | <u> </u> kPa |

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HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS (Cont)

- | | | |
|---|--------------------------|-------------------------|
| 3. Latent Heat, λ | _____ Btu/lb | _____ kJ/kg |
| 4. Liquid Density, ρ_ℓ | _____ lb/ft ³ | _____ kg/m ³ |
| 5. Vapor Density, ρ_v | _____ lb/ft ³ | _____ kg/m ³ |
| 6. Liquid Viscosity, μ_ℓ | _____ lb/ft-hr | _____ kg/m-s |
| 7. Vapor Viscosity, μ_v | _____ lb/ft-hr | _____ kg/m-s |
| 8. Liquid Thermal Conductivity, k_ℓ | _____ Btu/hr-ft-°F | _____ W/m-K |
| 9. Liquid Heat Capacity, $C_{p\ell}$ | _____ Btu/lb-°F | _____ kJ/kg-K |
| 10. Liquid Prandtl Number,
$Pr_\ell = C_{p\ell} \mu_\ell / k_\ell$ | _____ | _____ |
| 11. Reduced Pressure, $P_r = P / P_c$ | _____ | _____ |

E. EXCHANGER GEOMETRY

For design cases start with steps 1 - 3 in item B under the heading **CALCULATION PROCEDURES, VERTICAL THERMOSIPHON TUBESIDE REBOILERS**

- | | | |
|---|---|--|
| 1. Number of Tubes, NTT | _____ | _____ |
| 2. Bundle Length, L | _____ ft | _____ m |
| 3. Tube Inside Diameter, d_i | _____ in. | _____ mm |
| 4. Tube Outside Diameter, d_o
[Equivalent diameter for finned tubes] | _____ in. | _____ mm |
| 5. Outside Surface Area, A | $\pi (d_o / 12)(L)(NTT)$
_____ ft ² | $\pi (d_o / 1000)(L)(NTT)$
_____ m ² |
| 6. Tube Wall Thermal Conductivity, k_w | _____ Btu/hr-ft-°F | _____ W/m-K |

F. FOULING RESISTANCES

- | | | |
|------------------------|----------------------------------|---------------------------|
| 1. Tubeside, r_{fi} | _____ hr-ft ² -°F/Btu | _____ m ² -K/W |
| 2. Shellside, r_{fo} | _____ hr-ft ² -°F/Btu | _____ m ² -K/W |

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HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS (Cont)

G. CALCULATE OVERALL HEAT TRANSFER COEFFICIENT

1. Heat Flux, $q = Q_T / A$ _____ Btu/hr-ft² _____ W/m²
2. Choke Flow Limit Heat Flux, $q_{\max}$
 - 2.1 Reference Maximum $q_{\max}$, **Figure A14** _____ Btu/hr-ft² _____ W/m²
 - 2.2 Correction, Δ , **Figure A15** _____
 - 2.3 $q_{\max} = 0.7(q_{\max})\Delta$ _____ Btu/hr-ft² _____ W/m²
 - 2.4 $q \leq q_{\max}$? _____
3. Mixture Correction, F_c
 - 3.1 Effective Minimum Heat Flux for Nucleate Boiling, q_n

$24858 / (BR)^{0.8}$
 (if > 1000, use 1000)

$78365 / (BR)^{0.8}$
 (if > 3150, use 3150)

_____ Btu/hr-ft²

_____ W/m²
 - 3.2 (q / q_n) , for **Figure A8** _____
 - 3.3 Density Ratio, $x_r = (\rho_v / \rho_\ell)$, Limit $x_r \leq 0.2$

 - 3.4 Effective Temperature Range, DR, **Figure A7**
_____ °F _____ °C
 - 3.5 (BR / DR) _____
 - 3.6 F_c , **Figure A8** _____
4. Single Tube Nucleate Boiling Coefficient, h_{nb1} , **Figure A5**
_____ Btu/hr-ft²-°F _____ W/m²-K
5. Natural Convection Coefficient, h_{nc} , **Figure A16**
_____ Btu/hr-ft²-°F _____ W/m²-K

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HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS (Cont)

6. Nucleate Boiling Heat Transfer Coefficient, $h_{nb} = h_{nb1} (F_c)^{0.67} + h_{nc}$

_____ Btu/hr-ft²-°F

_____ W/m²-K

7. Sensible Heat Duty, Q_S

$(W)(C_{p\ell})(T_{bs} - T_{bi})$

$(W)(C_{p\ell})(T_{bs} - T_{bi}) / 3.6$

_____ Btu/hr

_____ W

8A. For known flow rate calculate outlet vapor fraction

$y_o = (Q_T - Q_S) / (W \lambda)$

$y_o = 3.6 (Q_T - Q_S) / (W \lambda)$

8B. For known vapor fraction calculate flow rate, $W = (Q_T - Q_S) / (y_o \lambda)$

9. Inlet Liquid Velocity, v_i

$(0.051 W) / (\rho_\ell d_i^2 NTT)$

$(353.7 W) / (\rho_\ell d_i^2 NTT)$

_____ ft/s

_____ m/s

10. For **Figure A17**

$(q) / (\lambda \rho_\ell)$

$(q) / (\lambda \rho_\ell) / 1000$

_____ ft/hr

_____ m/s

11. Check y_o for good operation, **Figure A17**

OK?

12. Average Vapor Fraction, $y_a = y_o / 2$

13. Estimated Mist Flow Vapor Fraction, $y_{mist} = 1 - J1 (q / \lambda)(L / d_i)(\rho_\ell / \rho_v)^{0.5}$

$J1 = 2.1E-05$

$J1 = 1.29E-03$

14. $y_o \leq y_{mist}$?

15. Liquid Phase Reynolds Number,
 Re_ℓ

$15.28 W (1 - y_a) / (d_i \mu_\ell NTT)$

$0.353 W (1 - y_a) / (d_i \mu_\ell NTT)$

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HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS (Cont)

16. Liquid Phase Convective Heat Transfer Coefficient, h_ℓ

$$0.264 (Re_\ell)^{0.8} (Pr_\ell)^{0.4} (k_\ell / d_i) \quad 22 (Re_\ell)^{0.8} (Pr_\ell)^{0.4} (k_\ell / d_i)$$

_____ Btu/hr-ft²-°F _____ W/m²-K

17. Martinelli Parameter, $X_{tt} = 1.3 [(1 - y_a) / y_a](\rho_v / \rho_\ell)^{0.5}$

18. Two Phase Convection Factor, $F_{tp} = (1 + 3.4 / X_{tt})^{0.667}$

19. Convective Boiling Heat Transfer Coefficient, h_{tp}

$F_{mv} = 1.0$ (BR > 5°F, 3°C); 0.8 (BR 5-50°F, 3-28°C); 0.6 (BR > 50°F, 28°C)

$$h_{tp} = F_{mv} F_{tp} h_\ell$$

_____ Btu/hr-ft²-°F _____ W/m²-K

20. Static Pressure Head, ΔP_h

$$L(\rho_\ell) / 144 \quad 9.807 L(\rho_\ell) / 1000$$

_____ psi _____ kPa

21. Subcooled Liquid Zone Correction, $F_{LZ} = 0.15 + 0.85 (P / (\Delta P_h + P))^{0.5}$

22. Average Boiling Heat Transfer Coefficient, $h_{ab} = (h_{tp} + 0.6h_{nb}) F_{LZ}$

_____ Btu/hr-ft²-°F _____ W/m²-K

23. Condensing Heating Medium Heat Transfer Coefficient based on Outside Area, h_h .
For steam use 1500 Btu/hr-ft²-°F, 8500 W/m²-K

For other fluids

$$h_h = 0.18 k_\ell (\rho_\ell (\rho_\ell - \rho_v) g Pr_\ell / \mu_\ell^2)^{1/3}$$

where: g = acceleration of gravity, 4.17 x 10⁸ ft/hr², 9.81 m/s²

_____ Btu/hr-ft²-°F _____ W/m²-K

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HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS (Cont)

24. Wall Resistance, r_w , Plain or Finned Tubes

Carbon Steel 0.0004 hr-ft²-°F/Btu 0.00007 m²-K/W

Other 0.0004 (28 / k_w) 0.00007 (48 / k_w)

 hr-ft²-°F/Btu m²-K/W

25. Overall Heat Transfer Coefficient based on Outside Area, U_o

$$U_o = ((d_o / d_i)(1 / h_{ab}) + (1 / h_h) + (d_o / d_i) r_{fi} + r_{fo} + r_w)^{-1}$$

 Btu/hr-ft²-°F W/m²-K

26. Required Area, $A_r = Q_T / (U_o \Delta T)$

 ft² m²

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APPENDIX D
HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS
(SAMPLE)

Exchanger No. E - (Example 2) Plant _____ Done by _____

Service _____ Date _____

Customary

Metric

A. BOILING SIDE SPECIFICATIONS

1. Pressure, P	<u>45</u> psia	<u>310</u> kPa
2. Inlet Temperature, T_{bi}	<u>199.0</u> °F	<u>92.8</u> °C
3. Exit Temperature, T_{bo}	<u>202.3</u> °F	<u>94.6</u> °C
4. Saturation Temperature, T_{bs}	<u>200.8</u> °F	<u>93.8</u> °C
5. Total Heat Duty, Q_T	<u>33.306×10^6</u> Btu/hr	<u>9,762</u> kW
6. Flow Rate, W_T	<u>940,262</u> lb/hr	<u>426,500</u> kg/hr
7. Outlet Vapor Fraction, y_o	<u>-----</u>	<u>-----</u>

[Enter either item 6 or item 7. The other will be calculated.]

B. HEATING MEDIUM SPECIFICATIONS

1. Inlet Temperature, T_{hi}	<u>259.3</u> °F	<u>126.3</u> °C
2. Exit Temperature, T_{ho}	<u>257.6</u> °F	<u>125.3</u> °C
3. Saturation Temperature, T_{hs} (for condensing streams only)	<u>259.3</u> °F	<u>126.3</u> °C

C. AVAILABLE ΔT

[For pure component boiling vs. condensing heating medium use (average heating medium temp minus exit vapor temp); for mixture boiling vs. condensing heating medium use log-mean ΔT with hot inlet at saturation; for single phase heating medium use log-mean ΔT].

<u>57.8</u> °F	<u>32.1</u> °C
----------------	----------------

D. BOILING FLUID PROPERTIES

1. Boiling Range, BR, (Dew point - Bubble point) at inlet pressure	<u>0.73</u> °F	<u>0.41</u> °C
2. Critical Pressure, P_c	<u>499</u> psia	<u>3,441</u> kPa

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HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS (SAMPLE) (Cont)

3. Latent Heat, λ	<u>123.2</u>	Btu/lb	<u>286.5</u>	kJ/kg
4. Liquid Density, ρ_ℓ	<u>40.8</u>	lb/ft ³	<u>653.9</u>	kg/m ³
5. Vapor Density, ρ_v	<u>0.62</u>	lb/ft ³	<u>9.85</u>	kg/m ³
6. Liquid Viscosity, μ_ℓ	<u>0.424</u>	lb/ft-hr	<u>0.175 x 10⁻³</u>	kg/m-s
7. Vapor Viscosity, μ_v	<u>0.0217</u>	lb/ft-hr	<u>0.009 x 10⁻³</u>	kg/m-s
8. Liquid Thermal Conductivity, k_ℓ	<u>0.0472</u>	Btu/hr-ft-°F	<u>0.0817</u>	W/m-K
9. Liquid Heat Capacity, $C_{p\ell}$	<u>0.568</u>	Btu/lb-°F	<u>2.376</u>	kJ/kg-K
10. Liquid Prandtl Number, $Pr_\ell = C_{p\ell} \mu_\ell / k_\ell$	<u>5.1</u>		<u>5.1</u>	
11. Reduced Pressure, $P_r = P / P_c$	<u>0.09</u>		<u>0.09</u>	

E. EXCHANGER GEOMETRY

For design cases start with steps 1 - 3 in item B under the heading Calculation Procedures

1. Number of Tubes, NTT	<u>1,479</u>		<u>1,479</u>	
2. Bundle Length, L	<u>10.0</u>	ft	<u>3.048</u>	m
3. Tube Inside Diameter, d_i	<u>0.834</u>	in.	<u>21.18</u>	mm
4. Tube Outside Diameter, d_o [Equivalent diameter for finned tubes]	<u>1.00</u>	in.	<u>25.4</u>	mm
5. Outside Surface Area, A	$\pi (d_o / 12)(L)(NTT)$		$\pi (d_o / 1000)(L)(NTT)$	
	<u>3,872</u>	ft ²	<u>360</u>	m ²
6. Tube Wall Thermal Conductivity, k_w	<u>28</u>	Btu/hr-ft-°F	<u>48</u>	W/m-K

F. FOULING RESISTANCES

1. Tubeside, r_{fi}	<u>0.002</u>	hr-ft ² -°F/Btu	<u>0.0004</u>	m ² -K/W
2. Shellside, r_{fo}	<u>0.0005</u>	hr-ft ² -°F/Btu	<u>0.00009</u>	m ² -K/W

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HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS (SAMPLE) (Cont)

G. CALCULATE OVERALL HEAT TRANSFER COEFFICIENT

1. Heat Flux, $q = Q_T / A$ 8,602 Btu/hr-ft² 27,097 W/m²

2. Choke Flow Limit Heat Flux, $q_{\max}$
 - 2.1 Reference Maximum $q_{\max}$, **Figure A14** 2.8×10^4 Btu/hr-ft² 90×10^3 W/m²
 - 2.2 Correction, Δ , **Figure A15** 0.85 0.85
 - 2.3 $q_{\max} = 0.7(q_{\max}) \Delta$ 16,660 Btu/hr-ft² 53,500 W/m²
 - 2.4 $q \leq q_{\max}$? yes yes

3. Mixture Correction, F_c
 - 3.1 Effective Minimum Heat Flux for Nucleate Boiling, q_n

$24858 / (BR)^{0.8}$
 (if > 1000, use 1000)

$78365 / (BR)^{0.8}$
 (if > 3150, use 3150)

1,000 Btu/hr-ft²

3,150 W/m²
 - 3.2 (q / q_n) , for **Figure A8** 8.6 8.6
 - 3.3 Density Ratio, $x_r = (\rho_v / \rho_\ell)$, Limit $x_r \leq 0.2$ 0.015 0.015
 - 3.4 Effective Temperature Range, DR, **Figure A7**

215 °F

119 °C
 - 3.5 (BR / DR) 0.003 0.003
 - 3.6 F_c , **Figure A8** 1.0 1.0

4. Single Tube Nucleate Boiling Coefficient, h_{nb1} , **Figure A5**

360 Btu/hr-ft²-°F

2,200 W/m²-K

5. Natural Convection Coefficient, h_{nc} , **Figure A16**

65 Btu/hr-ft²-°F

369 W/m²-K

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HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS (SAMPLE) (Cont)

6. Nucleate Boiling Heat Transfer Coefficient, $h_{nb} = h_{nb1}(F_c)^{0.67} + h_{nc}$

425 Btu/hr-ft²-F 2,569 W/m²-K

7. Sensible Heat Duty, Q_S $(W)(C_{p\ell})(T_{bs} - T_{bi})$ $(W)(C_{p\ell})(T_{bs} - T_{bi}) / 3.6$

961,324 Btu/hr 281,490 W

- 8A. For known flow rate calculate outlet vapor fraction

$$y_o = (Q_T - Q_S) / (W \lambda) \quad y_o = 3.6(Q_T - Q_S) / (W \lambda)$$

0.279 0.279

- 8B. For known vapor fraction calculate flow rate, $W = (Q_T - Q_S) / (y_o \lambda)$

----- -----

9. Inlet Liquid Velocity, v_i $(0.051 W) / (\rho_\ell d_i^2 NTT)$ $(353.7 W) / (\rho_\ell d_i^2 NTT)$

1.14 ft/s 0.35 m/s

10. For **Figure A17** $(q) / (\lambda \rho_\ell)$ $(q) / (\lambda \rho_\ell) / 1000$

1.71 ft/hr 1.45×10^{-4} m/s

11. Check y_o for good operation, **Figure A17**

OK? ok ok

12. Average Vapor Fraction, $y_a = y_o / 2$ 0.14 0.14

13. Estimated Mist Flow Vapor Fraction, $y_{mist} = 1 - J1 (q / \lambda)(L / d_i)(\rho_\ell / \rho_v)^{0.5}$

$$J1 = 2.1E-05 \quad J1 = 1.29E-03$$

0.86 0.86

14. $y_o \leq y_{mist}$? yes yes

15. Liquid Phase Reynolds Number, Re_ℓ

$$15.28 W (1 - y_a) / (d_i \mu_\ell NTT) \quad 0.353 W (1 - y_a) / (d_i \mu_\ell NTT)$$

23,625 23,625

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HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS (SAMPLE) (Cont)

16. Liquid Phase Convective Heat Transfer Coefficient, h_ℓ

$$0.264 (Re_\ell)^{0.8} (Pr_\ell)^{0.4} (k_\ell / d_i) \qquad 22 (Re_\ell)^{0.8} (Pr_\ell)^{0.4} (k_\ell / d_i)$$

90 Btu/hr-ft²-°F
513 W/m²-K

17. Martinelli Parameter, $X_{tt} = 1.3 [(1 - y_a) / y_a] (\rho_v / \rho_\ell)^{0.5}$

0.98
0.98

18. Two Phase Convection Factor

$$F_{tp} = (1 + 3.4 / X_{tt})^{0.667}$$

271
271

19. Convective Boiling Heat Transfer Coefficient, h_{tp}

$$F_{mv} = 1.0 \text{ (BR < 5°F, 3°C); } 0.8 \text{ (BR 5-50°F, 3-28°C); } 0.6 \text{ (BR > 50°F, 28°C)}$$

$$h_{tp} = F_{mv} F_{tp} h_\ell \qquad \qquad \qquad \text{244 Btu/hr-ft}^2\text{-°F} \qquad \qquad \qquad \text{1,394 W/m}^2\text{-K}$$

20. Static Pressure Head, ΔP_h

$$L(\rho_\ell) / 144$$

$$9.807L(\rho_\ell) / 1000$$

2.833 psi
19.5 kPa

21. Subcooled Liquid Zone Correction, $F_{LZ} = 0.15 + 0.85 (P / (\Delta P_h + P))^{0.5}$

0.97
0.97

22. Average Boiling Heat Transfer Coefficient, $h_{ab} = (h_{tp} + 0.6h_{nb}) F_{LZ}$

484 Btu/hr-ft²-°F
2,847 W/m²-K

23. Condensing Heating Medium Heat Transfer Coefficient based on Outside Area, h_h .

For steam use 1500 Btu/hr-ft²-°F, 8500 W/m²-K

For other fluids

$$h_h = 0.18 k_\ell (\rho_\ell (\rho_\ell - \rho_v) g Pr_\ell / \mu_\ell^2)^{1/3}$$

where: g = acceleration of gravity, 4.17 x 10⁸ ft/hr², 9.81 m/s²

1,500 Btu/hr-ft²-°F
8,500 W/m²-K

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HEAT TRANSFER CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS (SAMPLE) (Cont)

24. Wall Resistance, r_w , Plain or Finned Tubes

Carbon Steel 0.0004 hr-ft²-°F/Btu 0.00007 m²-K/W

Other 0.0004 (28 / k_w) 0.00007 (48 / k_w)

----- hr-ft²-°F/Btu ----- m²-K/W

25. Overall Heat Transfer Coefficient based on Outside Area, U_o

$$U_o = ((d_o / d_i)(1/h_{ab}) + (1/h_h) + (d_o / d_i) r_{fi} + r_{to} + r_w)^{-1}$$

155 Btu/hr-ft²-°F 848 W/m²-K

26. Required Area, $A_r = Q_T / (U_o \Delta T)$

3,718 ft² 359 m²

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APPENDIX E
HEAT TRANSFER CALCULATION FORM FOR TUBESIDE PUMP-THROUGH REBOILERS

Exchanger No. E - Plant Done by
Service Date

A. FOLLOW STEPS A1 THROUGH F20 IN CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS (APPENDIX C).

B. CALCULATE AVERAGE BOILING HEAT TRANSFER COEFFICIENT

	Customary	Metric
1. Pressure Head Supplied by Pump, ΔP_h	<u> </u> psi	<u> </u> kPa
2. Fraction of Tube Length in Liquid Zone, $L_{\ell Z} = 1 - (P / (\Delta P_h + P))^{0.5}$	<u> </u>	<u> </u>
3. Liquid Reynolds Number, $Re_{\ell Z}$	$15.28 W / d_i / \mu_\ell / NTT$ <u> </u>	$0.353 W / d_i / \mu_\ell / NTT$ <u> </u>
4. Liquid Convective Heat Transfer Coefficient, $h_{\ell Z}$	$0.264 (Re_{\ell Z})^{0.8} (Pr_\ell)^{0.4} (k_\ell / d_i)$ <u> </u> Btu/hr-ft ² -°F	$22 (Re_{\ell Z})^{0.8} (Pr_\ell)^{0.4} (k_\ell / d_i)$ <u> </u> W/m ² -K
5. Average Boiling Heat Transfer Coefficient, $h_{ab} = (h_{tp} + 0.6h_{nb}) (1 - L_{\ell Z}) + h_{\ell Z} L_{\ell Z}$	<u> </u> Btu/hr-ft ² -°F	<u> </u> W/m ² -K

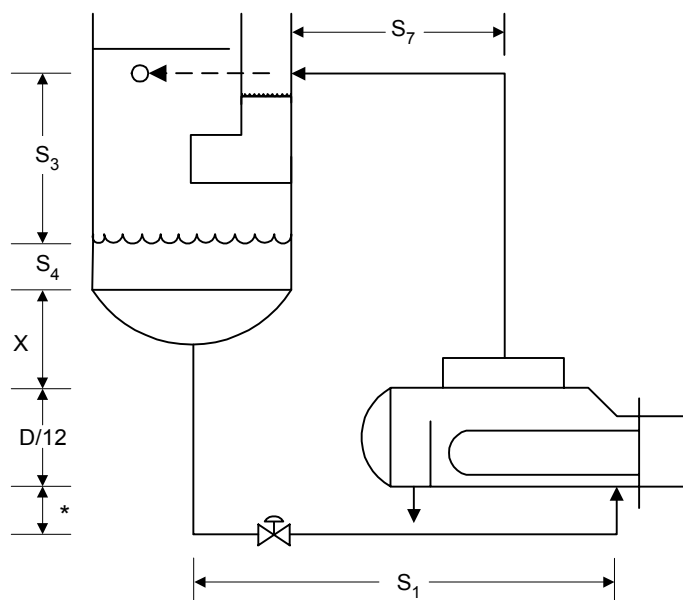
C. GO TO STEP G24 IN CALCULATION FORM FOR VERTICAL TUBESIDE THERMOSIPHON REBOILERS.

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APPENDIX F
HYDRAULIC BALANCE CALCULATION FORM FOR KETTLE REBOILERS

Exchanger No. E - _____ Plant _____ Done by _____
 Service _____ Date _____

A. PIPING LAYOUT AND DIMENSIONS



	Customary		Metric
S1 =	_____ ft	_____	m
S3 =	_____ ft	_____	m
S4 =	_____ ft	_____	m
S7 =	_____ ft	_____	m
X will be calculated			
Shell and Bundle Diameter			
D _s =	_____ in.	_____	m
D _b =	_____ in.	_____	m

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* This dimension is usually about 3ft (1m)

B. INLET LINE (LIQUID) PIPE FITTING RESISTANCE COEFFICIENTS (SECTION XIV-B)

- | | | |
|------------------------------------|-------|-----|
| 1. Tower liquid outlet | _____ | 0.5 |
| 2. Exchanger inlet nozzle | _____ | 2.0 |
| 3. Total for bends, ells, and tees | _____ | |
| 4. Total for valves | _____ | |
| 5. Sum 1 - 4, K _i | _____ | |

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HYDRAULIC BALANCE CALCULATION FORM FOR KETTLE REBOILERS (Cont)

C. OUTLET LINE (VAPOR) PIPE FITTING RESISTANCE COEFFICIENTS (SECTION XIV-B)

1. Tower vapor inlet 1.0
2. Exchanger outlet nozzle 1.0
3. Total for bends, ells, and tees _____
4. Total for valves _____
5. Sum 1 - 4, K_o _____

D. BOILING FLUID PROPERTIES

1. Liquid Viscosity, μ_ℓ _____ lb/ft-hr _____ kg/m-s
2. Vapor Viscosity, μ_v _____ lb/ft-hr _____ kg/m-s
3. Total Flow Rate, $W_T = W_B + W_V$, _____ lb/hr _____ kg/s

E. INLET LINE SIZE AND FRICTION PRESSURE DROP

1. Line Diameter, d_1 _____ in. _____ mm
[For design, estimate from **Figure A18**, based on 5 ft/s (1.52 m/s) liquid velocity]
2. Liquid Velocity, V_ℓ $(0.0509 W_T) / (\rho_\ell d_1^2)$ $(1.273 E6 W_T) / (\rho_\ell d_1^2)$
_____ ft/s _____ m/s
3. Reynolds Number, Re_ℓ $(15.28 W_T) / (\mu_\ell d_1)$ $(1.273E3 W_T) / (\mu_\ell d_1)$

4. Friction Factor, f_ℓ _____
[**Figure 1 or 2, Section XIV-B**]
5. Equivalent Length of Fittings, S_2 $(K_i d_1) / (48 f_\ell)$ $(K_i d_1) / (4000 f_\ell)$
_____ ft _____ m

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HYDRAULIC BALANCE CALCULATION FORM FOR KETTLE REBOILERS (Cont)

6. Volume Flow Rate, Q_V $(0.125 W_T) / (\rho_\ell)$ $(1000 W_T) / (\rho_\ell)$
 _____ gpm _____ dm³/s
7. Pressure Drop per 100 ft (per 1 m), P_ℓ , [Figure 3, Section XIV-B]
 _____ psi _____ kPa
8. Head Loss, ΔP_1 , (ft (m) of liquid per foot (meter) of line)
 $(1.44 P_\ell) / (\rho_\ell)$ $(102 P_\ell) / (\rho_\ell)$
 _____ ft liquid _____ m liquid

F. OUTLET LINE SIZE AND FRICTION PRESSURE DROP

1. Line Diameter, d_2 _____ in. _____ mm
 [For design, estimate one or two line sizes larger than d_1]
2. Vapor Velocity, V_v $(0.0509 W_V) / (\rho_v d_2^2)$ $(1.273E6 W_V) / (\rho_v d_2^2)$
 _____ ft/s _____ m/s
3. Reynolds Number, Re_v $(15.28 W_V) / (\mu_v d_2)$ $(1.273E3 W_V) / (\mu_v d_2)$

4. Friction Factor, f_v _____
 [Figure 1 or 2, Section XIV-B]
5. Equivalent Length of Fittings, S_5 $(K_o d_2) / (48 f_v)$ $(K_o d_2) / (4000 f_v)$
 _____ ft _____ m
6. Pressure Drop per 100 ft (per 1 m), P_v $(0.52 f_v \rho_v V_v^2) / d_2$ $(2 f_v \rho_v V_v^2) / d_2$
 _____ psi _____ kPa
7. Head Loss, ΔP_2 , (ft (m) of liquid per foot (meter) of line)
 $(1.44 P_v) / (\rho_\ell)$ $(102 P_v) / (\rho_\ell)$
 _____ ft liquid _____ m liquid

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HYDRAULIC BALANCE CALCULATION FORM FOR KETTLE REBOILERS (Cont)

G. FIXED HEAD LOSSES

1. Piping Low Point to top of Bundle, ΔH_2 ,

$$(D_b / 12) + 2$$

$$(D_b / 1000) + 0.61$$

_____ ft liquid

_____ m liquid

2. Tower Inlet to Tower Bottom Tangent, $\Delta H_3 = (S_3 + S_4)(\rho_v / \rho_\ell)$

_____ ft liquid

_____ m liquid

3. Friction in Liquid Line, ΔH_4 ,

$$\Delta P_1 (S_1 + S_2 + (D_s / 12) + 4)$$

$$\Delta P_1 (S_1 + S_2 + (D_s / 1000) + 1.22)$$

_____ ft liquid

_____ m liquid

4. Friction in Vapor Line, $\Delta H_5 = \Delta P_2 (S_3 + S_4 + S_5 + S_7)$

_____ ft liquid

_____ m liquid

Total Fixed Head Loss, ΔH_F , Sum 1-4

_____ ft liquid

_____ m liquid

H. FIXED HEAD GAIN

1. Liquid Head, J'

$$(S_4 + (D_s / 12) + 2)$$

$$(S_4 + (D_s / 1000) + 0.61)$$

_____ ft liquid

_____ m liquid

I. VARIABLE (UNKNOWN) HEAD LOSSES

1. Friction in Unknown Length X, per foot of line

$$J = \Delta P_1 + \Delta P_2 + (\rho_v / \rho_\ell)$$

_____ ft liquid

_____ m liquid

J. UNKNOWN ELEVATION, X

1. $X = (\Delta H_F - J') / (1 - J)$

_____ ft

_____ m

If required elevation is not acceptable repeat calculation with different line sizes.

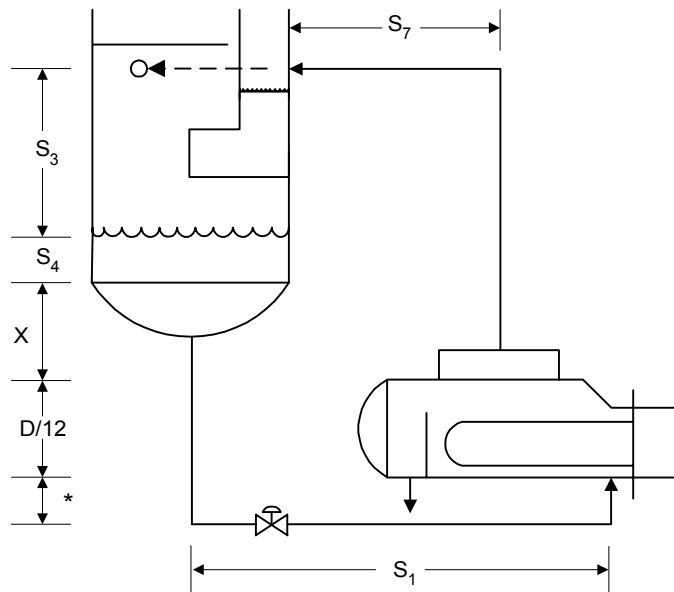
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APPENDIX G
HYDRAULIC BALANCE CALCULATION FORM FOR KETTLE REBOILERS (SAMPLE)

Exchanger No. E - (Example 3) Plant Done by

Service Date

A. PIPING LAYOUT AND DIMENSIONS



	Customary		Metric
S1 =	<u>20</u>	ft	<u>6.1</u> m
S3 =	<u>10</u>	ft	<u>3.05</u> m
S4 =	<u>1</u>	ft	<u>0.305</u> m
S7 =	<u>15</u>	ft	<u>4.57</u> m
X will be calculated			
Shell and Bundle Diameter			
D _s =	<u>52</u>	in.	<u>1321</u> m
D _b =	<u>26</u>	in.	<u>660</u> m

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* This dimension is usually about 3ft (1m)

B. INLET LINE (LIQUID) PIPE FITTING RESISTANCE COEFFICIENTS (SECTION XIV-B)

1. Tower liquid outlet	<u>0.5</u>
2. Exchanger inlet nozzle	<u>2.0</u>
3. Total for bends, ells, and tees	<u>3</u>
4. Total for valves	<u>1</u>
5. Sum 1 - 4, K _i	<u>6.5</u>

C. OUTLET LINE (VAPOR) PIPE FITTING RESISTANCE COEFFICIENTS (SECTION XIV-B)

1. Tower vapor inlet	<u>1.0</u>
2. Exchanger outlet nozzle	<u>1.0</u>
3. Total for bends, ells, and tees	<u>3</u>

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HYDRAULIC BALANCE CALCULATION FORM FOR KETTLE REBOILERS (SAMPLE) (Cont)

4. Total for valves 1

5. Sum 1 - 4, K_o 6.0

D. BOILING FLUID PROPERTIES

1. Liquid Viscosity, μ_ℓ 0.264 lb/ft-hr 1.093×10^{-4} kg/m-s

2. Vapor Viscosity, μ_v 0.0263 lb/ft-hr 1.087×10^{-5} kg/m-s

3. Total Flow Rate, $W_T = W_B + W_V$ 348.534 lb/hr 43.7 kg/s

E. INLET LINE SIZE AND FRICTION PRESSURE DROP

1. Line Diameter, d_1 10 in. 254 mm
[For design, estimate from **Figure A18**, based on 5 ft/s (1.52 m/s) liquid velocity]

2. Liquid Velocity, V_ℓ $(0.0509 W_T) / (\rho_\ell d_1^2)$ $(1.273E6 W_T) / (\rho_\ell d_1^2)$
5.23 ft/s 1.59 m/s

3. Reynolds Number, Re_ℓ $(15.28 W_T) / (\mu_\ell d_1)$ $(1.273E3 W_T) / (\mu_\ell d_1)$
2,017,000 2,017,000

4. Friction Factor, f_ℓ 0.0034 0.0034
[**Figure 1 or 2, Section XIV-B**]

5. Equivalent Length of Fittings, S_2 $(K_i d_1) / (48 f_\ell)$ $(K_i d_1) / (4000 f_\ell)$
398 ft 121 m

6. Volume Flow Rate, Q_v $(0.125 W_T) / (\rho_\ell)$ $(1000 W_T) / (\rho_\ell)$
1285 gpm 80.5 dm³/s

7. Pressure Drop per 100 ft (per 1 m), P_ℓ , [**Figure 3, Section XIV-B**]
0.3 psi 0.065 kPa

8. Head Loss, ΔP_1 , (ft (m) of liquid per foot (meter) of line)
 $(1.44 P_\ell) / (\rho_\ell)$ $(102 P_\ell) / (\rho_\ell)$
0.0128 ft liquid 0.0122 m liquid

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HYDRAULIC BALANCE CALCULATION FORM FOR KETTLE REBOILERS (SAMPLE) (Cont)

F. OUTLET LINE SIZE AND FRICTION PRESSURE DROP

- Line Diameter, d_2 14 in. 355.6 mm
[For design, estimate one or two line sizes larger than d_1]
- Vapor Velocity, V_v $(0.0509 W_v) / (\rho_v d_2^2)$ $(1.273E6 W_v) / (\rho_v d_2^2)$
4.9 ft/s 1.415 m/s
- Reynolds Number, $(15.28 W_v) / (\mu_v d_2)$ $(1.273E3 W_v) / (\mu_v d_2)$
1,599,000 1,518,000
- Friction Factor, f_v 0.0034 0.0034
[Figure 1 or 2, Section XIV-B]
- Equivalent Length of Fittings, S_5 $(K_o d_2) / (48 f_v)$ $(K_o d_2) / (4000 f_v)$
515 ft 157 m
- Pressure Drop per 100 ft (per 1 m), P_v $(0.52 f_v \rho_v V_v^2) / d_2$ $(2 f_v \rho_v V_v^2) / d_2$
0.0062 psi 0.00162 kPa
- Head Loss, ΔP_2 , (ft (m) of liquid per foot (meter) of line)
 $(1.44 P_v) / (\rho_\ell)$ $(102 P_v) / (\rho_\ell)$
0.00026 ft liquid 0.00024 m liquid

G. FIXED HEAD LOSSES

- Piping Low Point to top of Bundle, ΔH_2 , $(D_b / 12) + 2$ $(D_b / 1000) + 0.61$
4.2 ft liquid 1.27 m liquid
- Tower Inlet to Tower Bottom Tangent, $\Delta H_3 = (S_3 + S_4)(\rho_v / \rho_\ell)$
0.67 ft liquid 0.203 m liquid

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HYDRAULIC BALANCE CALCULATION FORM FOR KETTLE REBOILERS (SAMPLE) (Cont)

3. Friction in Liquid Line, ΔH_4 ,

$$\Delta P_1 (S1 + S2 + (D_s / 12) + 4)$$

$$\Delta P_1 (S1 + S2 + (D_s / 1000) + 1.22)$$

5.46 ft liquid

1.58 m liquid

4. Friction in Vapor Line, $\Delta H_5 = \Delta P_2 (S3 + S4 + S5 + S7)$

0.14 ft liquid

0.0396 m liquid

Total Fixed Head Loss, ΔH_F , Sum 1 - 4

10.47 ft liquid

3.093 m liquid

H. FIXED HEAD GAIN

1. Liquid Head, J'

$$(S4 + (D_s / 12) + 2)$$

$$(S4 + (D_s / 1000) + 0.61)$$

7.33 ft liquid

2.24 m liquid

I. VARIABLE (UNKNOWN) HEAD LOSSES

1. Friction in Unknown Length X, per foot of line

$$J = \Delta P_1 + \Delta P_2 + (\rho_v / \rho_\ell)$$

0.0735 ft liquid

0.0729 m liquid

J. UNKNOWN ELEVATION, X

1. $X = (\Delta H_F - J') / (1 - J)$

3.39 ft

0.92 m

If required elevation is not acceptable repeat calculation with different line sizes.

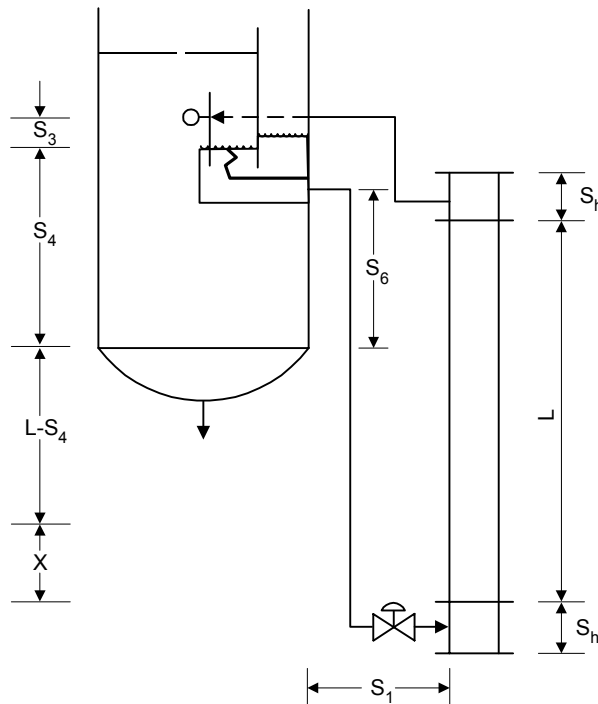
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APPENDIX H
HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS

Exchanger No. E - _____ Plant _____ Done by _____

Service _____ Date _____

A. PIPING LAYOUT AND DIMENSIONS



	Customary		Metric	
S1 =	_____	ft	_____	m
S3 =	_____	ft	_____	m
S4 =	_____	ft	_____	m
S6 =	_____	ft	_____	m
L =	_____	ft	_____	m
Sh =	_____	ft	_____	m
X will be calculated				

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B. INLET LINE (LIQUID) PIPE FITTING RESISTANCE COEFFICIENTS (SECTION XIV-B)

- | | | |
|------------------------------------|-----|-------|
| 1. Tower liquid outlet | 0.5 | _____ |
| 2. Exchanger inlet nozzle | 1.1 | _____ |
| 3. Total for bends, ells, and tees | | _____ |
| 4. Total for valves | | _____ |
| 5. Sum 1 - 4, K_i | | _____ |

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HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS (Cont)

C. OUTLET LINE (VAPOR) PIPE FITTING RESISTANCE COEFFICIENTS (SECTION XIV-B)

1. Tower inlet
(1.0 for flush nozzle) _____
For a perforated pipe calculate pressure drop using [Section XIV-D](#) and enter in step G4.
2. Exchanger outlet nozzle _____ 0.7
3. Total for bends, ells, and tees in vertical portion _____
4. Total for valves in vertical portion _____
5. Sum 1 - 4, K_{ov} _____
6. Total for bends, ells, and tees in horizontal portion _____
7. Total for valves in horizontal portion _____
8. Sum 6 -7, K_{oh} _____

D. BOILING FLUID PROPERTIES

Note: For viscosity lb/ft-hr = 2.42 x cP, Kg/m-s = 0.001 x cP

1. Liquid Viscosity, μ_ℓ _____ lb/ft-hr _____ kg/m-s
2. Vapor Viscosity, μ_v _____ lb/ft-hr _____ kg/m-s
3. Total Flow Rate, W _____ lb/hr _____ kg/s
4. Slope of Vapor Pressure Curve, dT / dP _____ °F/psi _____ °C/kPa

E. INLET LINE SIZE AND FIXED PRESSURE DROP

1. Line Diameter, d_1 _____ in. _____ mm
For design, estimate $d_1 = 0.5 (d_i) (NTT)^{0.5}$, round off to nearest standard size]
2. Liquid Velocity, V_ℓ _____ $(0.0509 W) / (\rho_\ell d_1^2)$ _____ $(1.273E6 W) / (\rho_\ell d_1^2)$
_____ ft/s _____ m/s

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HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS (Cont)

3. Reynolds Number, Re_ℓ $(15.28 W) / (\mu_\ell d_1)$ $(1.273E3 W) / (\mu_\ell d_1)$

4. Friction Factor, f_ℓ _____
[Figure 1 or 2, Section XIV-B] _____
5. Equivalent Length of Fittings, S_2 $(K_i d_1) / (48 f_\ell)$ $(K_i d_1) / (4000 f_\ell)$
 _____ ft _____ m
6. Volume Flow Rate, Q_v $(0.125 W_T) / (\rho_\ell)$ $(1000 W) / (\rho_\ell)$
 _____ gpm _____ dm^3/s
7. Pressure Drop per ft (per m), P_ℓ ,

$$\frac{f_\ell V_\ell^2 \rho_\ell}{193 d}$$

$$\frac{2f_\ell V_\ell^2 \rho_\ell}{d}$$

 _____ psi/ft _____ kPa/m
8. Head Loss, ΔP_1 , (ft of liquid per ft of line, or m of liquid per m of line)
 $(144 P_\ell) / (\rho_\ell)$ $(102 P_\ell) / (\rho_\ell)$
 _____ ft liquid/ft _____ m liquid/m
9. Total
 $\Delta H_i = \Delta P_1 [S_1 + S_2 + S_6 + (L - S_4) + S_h/2]$
 _____ ft liquid _____ m liquid
 [ΔH_i recommended to be in the range of 0.3(L) to 0.5(L)]

F. OUTLET LINE SIZE AND FIXED PRESSURE DROP

1. Line Diameter, d_2 _____ in. _____ mm
 [For design, estimate $d_2 = (d_i) (NTT)^{0.5}$, round off to nearest standard size]

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HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS (Cont)

2. Pressure drop in outlet piping. Calculate using [Section XIV-D](#), Method B. Separate calculations are required for the horizontal (friction only) and vertical (friction and elevation) portions. Calculate pressure drops for a length of 1 ft or 1 m.

Horizontal P_h _____ psi/ft _____ kPa/m

Vertical friction P_{vf} _____ psi/ft _____ kPa/m

Vertical elevation P_{ve} _____ psi/ft _____ kPa/m

3. Head losses, ft of liquid per ft of line or m of liquid per m of line.

$\Delta P_h = 144 P_h / \rho_\ell$ _____ ft liquid/ft = $102 P_h / \rho_\ell$ _____ m liquid/m

$\Delta P_{vf} = 144 P_{vf} / \rho_\ell$ _____ ft liquid/ft = $102 P_{vf} / \rho_\ell$ _____ m liquid/m

$\Delta P_{ve} = 144 P_{ve} / \rho_\ell$ _____ ft liquid/ft = $102 P_{ve} / \rho_\ell$ _____ m liquid/m

4. Equivalent Lengths of Fittings

S_5 $K_{oh} d_2 / 48 f_{2h}$ _____ ft $K_{oh} d_2 / 4000 f_{2h}$ _____ m

S_8 $K_{ov} d_2 / 48 f_{2v}$ _____ ft $K_{ov} d_2 / 4000 f_{2v}$ _____ m

f_{2h} and f_{2v} are the two phase friction factors, for horizontal and vertical flow respectively, calculated in step F2.

5. Total Head Loss

$\Delta H_o = \Delta P_h (S1 + S5) + \Delta P_{vf} (S3 - S_h/2 + S8) + \Delta P_{ve} (S3 - S_h/2)$
 _____ ft liquid _____ m liquid

If the horizontal length of the return piping is different from S1, use the actual length.

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HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS (Cont)

G. FIXED HEAD LOSSES

1. Exchanger pressure drop. Calculate using [Section XIV-D](#), Method B, or use value from a computer program output.

Friction loss P_{ef} _____ psi _____ kPa

Elevation loss P_{ee} _____ psi _____ kPa

2. Friction Head Loss ΔH_1 $144 P_{ef} / \rho_\ell$ $102 P_{ef} / \rho_\ell$
 _____ ft liquid _____ m liquid

3. Elevation Head Loss ΔH_2 $144 P_{ee} / \rho_\ell$ $102 P_{ee} / \rho_\ell$
 _____ ft liquid _____ m liquid

4. Perforated pipe distributor pressure drop

P_p _____ psi _____ kPa

Head loss ΔH_p $144 P_p / \rho_\ell$ $102 P_p / \rho_\ell$
 _____ ft liquid _____ m liquid

H. TOTAL FIXED HEAD LOSS

$$\Delta H_F = \Delta H_i + \Delta H_o + \Delta H_1 + \Delta H_2 + \Delta H_p$$

_____ ft liquid _____ m liquid

I. VARIABLE HEAD LOSSES

1. Friction in Variable Length X, per foot (per m) of line

$J = \Delta P_1 + \Delta P_{vf} + \Delta P_{ve}$ _____ ft liquid/ft _____ m liquid/m

J. FIXED HEAD GAIN

(skip to item L for vacuum services or if the slope of the vapor pressure curve is $> 0.16^\circ\text{F/psi}$ [0.013°C/kPa])

1. Liquid Head, $J' = L$ _____ ft liquid _____ m liquid

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HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS (Cont)

K. VARIABLE ELEVATION, X

1. $X = (\Delta H_F - J') / (1 - J)$ _____ ft _____ m

If required elevation is not acceptable repeat calculation with different line sizes.

L. FIXED HEAD GAIN

(for vacuum services or if the slope of the vapor pressure curve is $> 0.16^\circ\text{F/psi}$ [0.013°C/kPa])

1. Inlet Liquid Reynolds Number, Re_ℓ $15.28 W / (d_i \mu_\ell \text{ NTT})$ (1.273E3) $W / (d_i \mu_\ell \text{ NTT})$

2. Liquid Convective Heat Transfer Coefficient, h_ℓ

$0.264 (Re_\ell)^{0.8} (Pr_\ell)^{0.4} (k_\ell / d_i)$ $22 (Re_\ell)^{0.8} (Pr_\ell)^{0.4} (k_\ell / d_i)$

_____ Btu/hr-ft²-°F _____ W/m²-K

3. Overall Heat Transfer Coefficient in liquid zone, U_o

$U_o = ((d_o/d_i)(1/h_\ell) + 1/h_h + (d_o/d_i) r_{fi} + r_{fo} + r_w)^{-1}$

_____ Btu/hr-ft²-°F _____ W/m²-K

All variables are as defined in the heat transfer calculation form.

4. Temperature increase per unit tube length

$dT/dL = (U_o \pi d_o \text{ NTT} (T_{ho} - T_{bi})) / (W C_{p\ell})$

_____ °F/in. _____ °C/mm

5. Pressure change per unit length, dP/dL

_____ psi/ft _____ kPa/m

6. Length of liquid zone, $Y = [(L) (dT/dP)] / [dT/dP + (dT/dL) / (dP/dL)]$

_____ ft _____ m

M. VARIABLE ELEVATION, X

1. $X = (\Delta H_F - L + Y) / (1 - J)$ _____ ft _____ m

If required elevation is not acceptable repeat calculation with different line sizes.

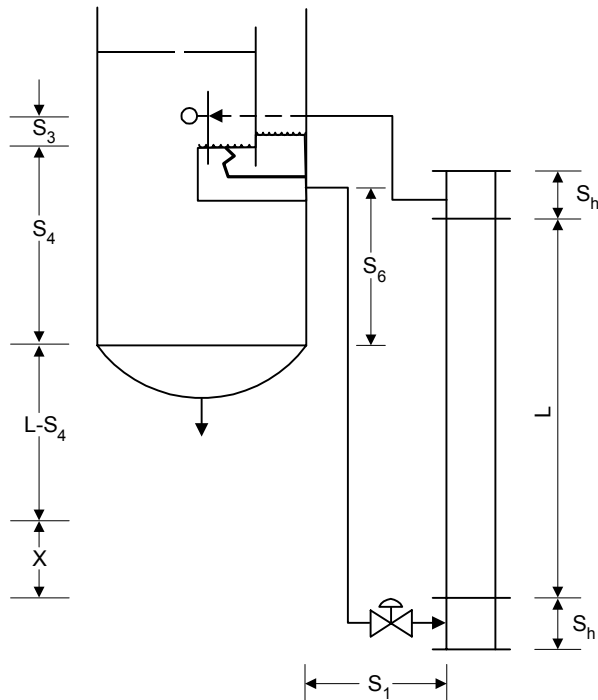
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APPENDIX I
HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS
(SAMPLE)

Exchanger No. E - (Example 4) Plant Done by

Service Date

A. PIPING LAYOUT AND DIMENSIONS



	Customary		Metric
S1 =	<u>3</u>	ft	<u>0.9144</u> m
S3 =	<u>1</u>	ft	<u>0.3048</u> m
S4 =	<u>6</u>	ft	<u>1.8288</u> m
S6 =	<u>4</u>	ft	<u>1.2192</u> m
L =	<u>10</u>	ft	<u>3.048</u> m
S _h =	<u>2</u>	ft	<u>0.6096</u> m
X will be calculated			

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B. INLET LINE (LIQUID) PIPE FITTING RESISTANCE COEFFICIENTS (SECTION XIV-B)

1. Tower liquid outlet	<u>0.5</u>
2. Exchanger inlet nozzle	<u>1.1</u>
3. Total for bends, ells, and tees	<u>0.3</u>
4. Total for valves	<u>1.0</u>
5. Sum 1 - 4, K _i	<u>2.9</u>

C. OUTLET LINE (VAPOR) PIPE FITTING RESISTANCE COEFFICIENTS (SECTION XIV-B)

1. Tower inlet	<u>1.0</u>
(1.0 for flush nozzle)	
For a perforated pipe calculate pressure drop using Section XIV-D and enter in step G4.	

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HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS (SAMPLE) (Cont)

2. Exchanger outlet nozzle	0.7
3. Total for bends, ells, and tees in vertical portion	0.15
4. Total for valves in vertical portion	0.0
5. Sum 1 - 4, K_{OV}	1.85
6. Total for bends, ells, and tees in horizontal portion	0.15
7. Total for valves in horizontal portion	0.0
8. Sum 6 - 7, K_{OH}	0.15

D. BOILING FLUID PROPERTIES

Note: For viscosity lb/ft-hr = 2.42 x cP, Kg/m-s = 0.001 x cP

1. Liquid Viscosity, μ_ℓ	0.424	lb/ft-hr	0.175×10^{-3}	kg/m-s
2. Vapor Viscosity, μ_v	0.0217	lb/ft-hr	9.0×10^{-5}	kg/m-s
3. Total Flow Rate, W	940,262	lb/hr	118.5	kg/s
4. Slope of Vapor Pressure Curve, dT/dP		°F/psi		°C/kPa

E. INLET LINE SIZE AND FIXED PRESSURE DROP

1. Line Diameter, d_1	12	in.	304.8	mm
-------------------------	----	-----	-------	----

[For design, estimate $d_1 = 0.5 (d_i) (NTT)^{0.5}$, round off to nearest standard size]

2. Liquid Velocity, V_ℓ	$(0.0509 W) / (\rho_\ell d_1^2)$		$(1.273E6 W) / (\rho_\ell d_1^2)$	
	8.15	ft/s	2.48	m/s
3. Reynolds Number, Re_ℓ	$(15.28 W) / (\mu_\ell d_1)$		$(1.273E3 W) / (\mu_\ell d_1)$	
	2,824,000		2,824,000	

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HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS (SAMPLE) (Cont)

4. Friction Factor, f_ℓ 0.0033 0.0033
[Figure 1 or 2, Section XIV-B]
5. Equivalent Length of Fittings, S_2 $(K_i d_1) / (48 f_\ell)$ $(K_i d_1) / (4000 f_\ell)$
220.0 ft 67.0 m
6. Volume Flow Rate, Q_V , $(0.125 W) / (\rho_\ell)$ $(1000 W) / (\rho_\ell)$
2,881 gpm 181.2 dm³/s
7. Pressure Drop per ft (per m), P_ℓ , $\frac{f_\ell V_\ell^2 \rho_\ell}{193 d}$ $\frac{2f_\ell V_\ell^2 \rho_\ell}{d}$
0.0039 psi/ft 0.087 kPa/m
8. Head Loss, ΔP_1 , (ft of liquid per ft of line, or m of liquid per m of line)
 $(144 P_\ell) / (\rho_\ell)$ $(102 P_\ell) / (\rho_\ell)$
0.0136 ft liquid/ft 0.0136 m liquid/m
9. Total
 $\Delta H_i = \Delta P_1 [S_1 + S_2 + S_6 + (L - S_4) + S_h/2]$
3.15 ft liquid 0.96 m liquid
[ΔH_i recommended to be in the range of 0.3(L) to 0.5(L)]

F. OUTLET LINE SIZE AND FIXED PRESSURE DROP

1. Line Diameter, d_2 24 in. 609.6 mm
[For design, estimate $d_2 = (d_i) (NTT)^{0.5}$, round off to nearest standard size]
2. Pressure drop in outlet piping. Calculate using [Section XIV-D](#), method B. Separate calculations are required for the horizontal (friction only) and vertical (friction and elevation) portions. Calculate the pressure drops for a length of 1 ft or 1 m.
- Horizontal P_h 0.0035 psi/ft 0.0804 kPa/m
- Vertical P_{vf} 0.0032 psi/ft 0.0716 kPa/m
- Vertical elevation P_{ve} 0.0169 psi/ft 0.381 kPa/m

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HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS (SAMPLE) (Cont)

3. Head losses, ft of liquid per ft of line or m of liquid per m of line.

$$\Delta P_h = 144 P_h / \rho_\ell \quad \underline{0.0125} \text{ ft liquid/ft} = 102 P_h / \rho_\ell \quad \underline{0.0125} \text{ m liquid/m}$$

$$\Delta P_{vf} = 144 P_{vf} / \rho_\ell \quad \underline{0.0112} \text{ ft liquid} = 102 P_{vf} / \rho_\ell \quad \underline{0.0112} \text{ m liquid/m}$$

$$\Delta P_{ve} = 144 P_{ve} / \rho_\ell \quad \underline{0.0595} \text{ ft liquid} = 102 P_{ve} / \rho_\ell \quad \underline{0.0595} \text{ m liquid/m}$$

4. Equivalent Lengths of Fittings, S_5

$$K_{oh} d_2 / 48 f_{2h}$$

$$K_{oh} d_2 / 4000 f_{2h}$$

$$\underline{14.9} \text{ ft}$$

$$\underline{4.5} \text{ m}$$

S_8

$$K_{ov} d_2 / 48 f_{2v}$$

$$K_{ov} d_2 / 4000 f_{2v}$$

$$\underline{181.4} \text{ ft}$$

$$\underline{55.2} \text{ m}$$

f_{2h} and f_{2v} are the two phase friction factors, for horizontal and vertical flow respectively, calculated in step F2.

5. Total Head Loss

$$\Delta H_o = \Delta P_h (S1 + S5) + \Delta P_{vf} (S3 - S_h/2 + S8) + \Delta P_{ve} (S3 - S_h/2)$$

$$\underline{2.26} \text{ ft liquid}$$

$$\underline{0.685} \text{ m liquid}$$

If the horizontal length of the return piping is different from $S1$, use the actual length.

G. FIXED HEAD LOSSES

1. Exchanger pressure drop. Calculate using [Section XIV-D](#), Method B, or use value from a computer program output.

$$\text{Friction loss } P_{ef} \quad \underline{1.86} \text{ psi} \quad \underline{12.65} \text{ kPa}$$

$$\text{Elevation loss } P_{ee} \quad \underline{0.24} \text{ psi} \quad \underline{1.63} \text{ kPa}$$

2. Friction Head Loss ΔH_1

$$144 P_{ef} / \rho_\ell$$

$$102 P_{ef} / \rho_\ell$$

$$\underline{6.56} \text{ ft liquid}$$

$$\underline{2.0} \text{ m liquid}$$

3. Elevation Head Loss ΔH_2

$$144 P_{ee} / \rho_\ell$$

$$102 P_{ee} / \rho_\ell$$

$$\underline{0.84} \text{ ft liquid}$$

$$\underline{0.26} \text{ m liquid}$$

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HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS (SAMPLE) (Cont)

4. Perforated pipe distributor pressure drop

P_p	_____ psi	_____ kPa
Head loss ΔH_p	$144 P_p / \rho_\ell$	$102 P_p / \rho_\ell$
	_____ ft liquid	_____ m liquid

H. TOTAL FIXED HEAD LOSS

$$\Delta H_F = \Delta H_i + \Delta H_0 + \Delta H_1 + \Delta H_2 + \Delta H_p$$

_____ 12.81 _____ ft liquid	_____ 3.94 _____ m liquid
-----------------------------	---------------------------

I. VARIABLE HEAD LOSSES

1. Friction in Variable Length X, per foot (m) of line

$J = \Delta P_1 + \Delta P_{vf} + \Delta P_{ve}$	_____ 0.0843 _____ ft liquid	_____ 0.0843 _____ m liquid
--	------------------------------	-----------------------------

J. FIXED HEAD GAIN

(skip to item L for vacuum services or if the slope of the vapor pressure curve is $> 0.16^\circ\text{F/psi}$ [0.013°C/kPa])

1. Liquid Head, $J' = L$	_____ 10.0 _____ ft liquid	_____ 3.05 _____ m liquid
--------------------------	----------------------------	---------------------------

K. VARIABLE ELEVATION, X

1. $X = (\Delta H_F - J') / (1 - J)$	_____ 3.07 _____ ft	_____ 0.97 _____ m
--------------------------------------	---------------------	--------------------

If required elevation is not acceptable repeat calculation with different line sizes.

L. FIXED HEAD GAIN

(for vacuum services or if the slope of the vapor pressure curve is $> 0.16^\circ\text{F/psi}$ [0.013°C/kPa])

1. Inlet Liquid Reynolds Number, Re_ℓ

15.28 W / ($d_i \mu_\ell$ NTT)	(1.273E3) W / ($d_i \mu_\ell$ NTT)
---------------------------------	-------------------------------------

_____	_____
-------	-------

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HYDRAULIC BALANCE CALCULATION FORM FOR VERTICAL TUBESIDE REBOILERS (SAMPLE) (Cont)

2. Liquid Convective Heat Transfer Coefficient, h_ℓ

$$0.264 (Re_\ell)^{0.8} (Pr_\ell)^{0.4} (k_\ell / d_i)$$

$$22 (Re_\ell)^{0.8} (Pr_\ell)^{0.4} (k_\ell / d_i)$$

_____ Btu/hr-ft²-°F

_____ W/m²-K

3. Overall Heat Transfer Coefficient in liquid zone, U_o

$$U_o = ((d_o/d_i)(1/h_\ell) + 1/h_h + (d_o/d_i) r_{fi} + r_{fo} + r_w)^{-1}$$

_____ Btu/hr-ft²-°F

_____ W/m²-K

All variables are as defined in the heat transfer calculation form.

4. Temperature increase per unit tube length

$$dT / dL = (U_o \pi d_o NTT (T_{ho} - T_{bi})) / (W C_{p\ell})$$

_____ °F/in.

_____ °C/mm

5. Pressure change per unit length, dP / dL

_____ psi/ft

_____ kPa/m

6. Length of liquid zone, $Y = [(L) (dT/dP)] / [dT/dP + (dT/dL) / (dP/dL)]$

_____ ft

_____ m

M. VARIABLE ELEVATION, X

1. $X = (\Delta H_F - L + Y) / (1 - J)$

_____ ft

_____ m

If required elevation is not acceptable repeat calculation with different line sizes.

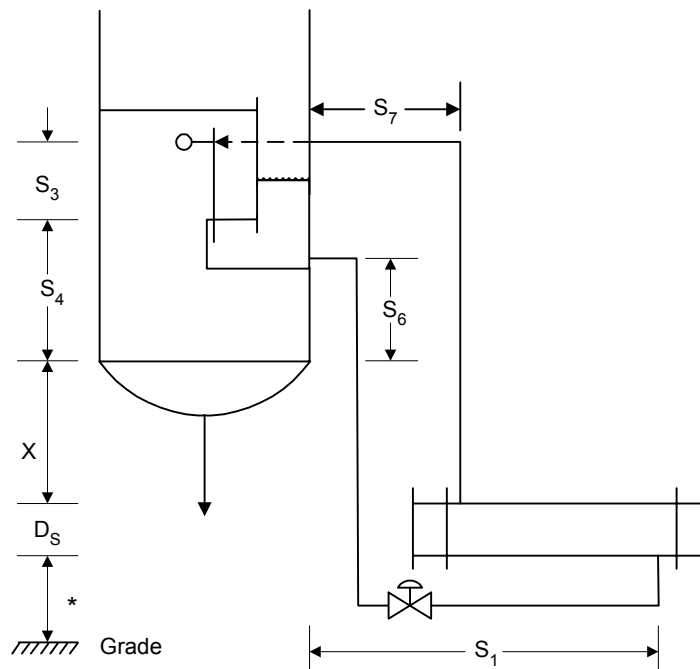
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APPENDIX J
HYDRAULIC BALANCE CALCULATION FORM FOR HORIZONTAL SHELLSIDE REBOILERS

Exchanger No. E - _____ Plant _____ Done by _____

Service _____ Date _____

A. PIPING LAYOUT AND DIMENSIONS



	Customary		Metric
S1 =	_____ ft		_____ m
S2 =	_____ ft		_____ m
S3 =	_____ ft		_____ m
S4 =	_____ ft		_____ m
S6 =	_____ ft		_____ m
S7 =	_____ ft		_____ m

X will be calculated

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* This dimension is usually about 3 ft (1 m).

B. INLET LINE (LIQUID) PIPE FITTING RESISTANCE COEFFICIENTS (SECTION XIV-B)

- | | |
|------------------------------------|-----------------|
| 1. Tower liquid outlet | _____ 0.5 _____ |
| 2. Exchanger inlet nozzle | _____ 2.0 _____ |
| 3. Total for bends, ells, and tees | _____ |
| 4. Total for valves | _____ |
| 5. Sum 1 - 4, K_i | _____ |

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HYDRAULIC BALANCE CALCULATION FORM FOR HORIZONTAL SHELLSIDE REBOILERS (Cont)

C. OUTLET LINE (VAPOR) PIPE FITTING RESISTANCE COEFFICIENTS (SECTION XIV-B)

1. Tower inlet _____
(1.0 for flush nozzle)
For a perforated pipe calculate pressure drop using [Section XIV-D](#) and enter in step G4.
2. Exchanger outlet nozzle 2.0
3. Total for bends, ells, and tees in vertical portion _____
4. Total for valves in vertical portion _____
5. Sum 1 - 4, K_{OV} _____
6. Total for bends, ells, and tees in horizontal portion _____
7. Total for valves in horizontal portion _____
8. Sum 6 - 7, K_{OH} _____

D. BOILING FLUID PROPERTIES

Note: For viscosity lb/ft-hr = 2.42 x cP, Kg/m-s = 0.001 x cP

1. Liquid Viscosity, μ_ℓ _____ lb/ft-hr _____ kg/m-s
2. Vapor Viscosity, μ_v _____ lb/ft-hr _____ kg/m-s
3. Total Flow Rate, W _____ lb/hr _____ kg/s

E. INLET LINE SIZE AND FIXED PRESSURE DROP

1. Line Diameter, d_1 _____ in. _____ mm

[For design, estimate from [Figure A18](#), based on 3 to 5 ft/s (0.91 to 1.52 m/s) liquid velocity]

2. Liquid Velocity, V_ℓ $(0.0509 W) / (\rho_\ell d_1^2)$ $(1.273E6 W) / (\rho_\ell d_1^2)$
_____ ft/s _____ m/s

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HYDRAULIC BALANCE CALCULATION FORM FOR HORIZONTAL SHELLSIDE REBOILERS (Cont)

3. Reynolds Number, Re_ℓ $(15.28 W) / (\mu_\ell d_1)$ $(1.273E3 W) / (\mu_\ell d_1)$
4. Friction Factor, f_ℓ $\underline{\hspace{2cm}}$ $\underline{\hspace{2cm}}$
[Figure 1 or 2, Section XIV-B] $\underline{\hspace{2cm}}$ $\underline{\hspace{2cm}}$
5. Equivalent Length of Fittings, S_2 $(K_i d_1) / (48 f_\ell)$ $(K_i d_1) / (4000 f_\ell)$
 $\underline{\hspace{2cm}}$ ft $\underline{\hspace{2cm}}$ m
6. Volume Flow Rate, Q_V $(0.125 W) / (\rho_\ell)$ $(1000 W) / (\rho_\ell)$
 $\underline{\hspace{2cm}}$ gpm $\underline{\hspace{2cm}}$ dm³/s
7. Pressure Drop per ft (per m), P_ℓ , $\frac{f_\ell V_\ell^2 \rho_\ell}{193 d}$ $\frac{2f_\ell V_\ell^2 \rho_\ell}{d}$
 $\underline{\hspace{2cm}}$ psi/ft $\underline{\hspace{2cm}}$ kPa/m
8. Head Loss, ΔP_1 , (ft of liquid per foot of line, or m of liquid per m of line) $(144 P_\ell) / (\rho_\ell)$ $(102 P_\ell) / (\rho_\ell)$
 $\underline{\hspace{2cm}}$ ft liquid/ft $\underline{\hspace{2cm}}$ m liquid/m
9. Total Inlet Line Friction Head Loss, ΔH_i $\Delta P_1 (S_1 + S_2 + D_s/12 + 4 + S_6)$ $\Delta P_1 (S_1 + S_2 + D_s/1000 + 1.22 + S_6)$
 $\underline{\hspace{2cm}}$ ft liquid $\underline{\hspace{2cm}}$ m liquid

F. OUTLET LINE SIZE AND FIXED PRESSURE DROP (INCLUDING FITTINGS)

Note: For TEMA J type shells with two outlets it is necessary to calculate the outlet pressure drop for only half the flow rate for part (or all) of the return piping, depending on how the return to the tower is designed.

1. Line Diameter, d_2 $\underline{\hspace{2cm}}$ in. $\underline{\hspace{2cm}}$ mm
 [For design, estimate d_2 to be two line sizes larger than d_1]

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HYDRAULIC BALANCE CALCULATION FORM FOR HORIZONTAL SHELLSIDE REBOILERS (Cont)

2. Pressure drop in outlet piping. Calculate using [Section XIV-D](#), Method B. Separate calculations are required for the horizontal (friction only) and vertical (friction and elevation) portions. Calculate the pressure drops for a length of 1 ft or 1 m.

Horizontal P_h _____ psi/ft _____ kPa/m

Vertical friction P_{vf} _____ psi/ft _____ kPa/m

Vertical elevation P_{ve} _____ psi/ft _____ kPa/m

3. Head losses, ft of liquid per ft of line or m of liquid per m of line.

$\Delta P_h = 144 P_h / \rho_\ell$ _____ ft liquid/ft = $102 P_h / \rho_\ell$ _____ m liquid/m

$\Delta P_{vf} = 144 P_{vf} / \rho_\ell$ _____ ft liquid/ft = $102 P_{vf} / \rho_\ell$ _____ m liquid/m

$\Delta P_{ve} = 144 P_{ve} / \rho_\ell$ _____ ft liquid/ft = $102 P_{ve} / \rho_\ell$ _____ m liquid/m

4. Equivalent Lengths of Fittings, S5

$K_{oh} d_2 / 48 f_{2h}$

$K_{oh} d_2 / 4000 f_{2h}$

_____ ft

_____ m

S8

$K_{ov} d_2 / 48 f_{2v}$

$K_{ov} d_2 / 4000 f_{2v}$

_____ ft

_____ m

f_{2h} and f_{2v} are the two phase friction factors, for horizontal and vertical flow respectively, calculated in step F2.

5. Total Head Loss

$$\Delta H_o = \Delta P_h (S7 + S5) + \Delta P_{vf} (S3 + S4 + S8) + \Delta P_{ve} (S3 + S4)$$

_____ ft liquid

_____ m liquid

G. OTHER FIXED HEAD LOSSES

1. Exchanger pressure drop, ΔP_f (estimate or from CST2 computer program).

_____ psi

_____ kPa

2. Exchanger friction head loss, ΔH_1 .

$(144 \Delta P_f) / (\rho_\ell)$

$(102 \Delta P_f) / (\rho_\ell)$

_____ ft liquid

_____ m liquid

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HYDRAULIC BALANCE CALCULATION FORM FOR HORIZONTAL SHELLSIDE REBOILERS (Cont)

3. Piping Low Point to top of Exchanger, ΔH_2 .

$$\text{Average mixture density, } \rho_x = (W / [W / \rho_\ell + 0.16(W_v)(1 - \rho_v / \rho_\ell) / \rho_v])$$

	_____ lb/ft ³	_____ kg/m ³
ΔH_2	$(D_s / 12)(\rho_x / \rho_\ell) + 2$	$(D_s / 1000)(\rho_x / \rho_\ell) + 0.61$
	_____ ft liquid	_____ m liquid

4. Perforated pipe distributor pressure drop,

$\Delta P_p =$	_____ psi	_____ kPa
Head loss ΔH_p	$144 \Delta P_p / \rho_\ell$	$102 \Delta P_p / \rho_\ell$
	_____ ft liquid	_____ m liquid

H. TOTAL FIXED HEAD LOSS

$$\text{Total Fixed Head Loss, } \Delta H_F = \Delta H_i + \Delta H_o + \Delta H_1 + \Delta H_2 + \Delta H_p$$

_____ ft liquid	_____ m liquid
-----------------	----------------

I. FIXED HEAD GAIN

1. Liquid Head, J',	$S_4 + D_s / 12 + 2$	$S_4 + D_s / 1000 + 0.61$
	_____ ft liquid	_____ m liquid

J. VARIABLE HEAD LOSSES

1. Friction in Unknown Length X, per foot (per m) of line

$J = \Delta P_1 = \Delta P_{vf} + \Delta P_{ve}$	_____ ft liquid/ft	_____ m liquid/m
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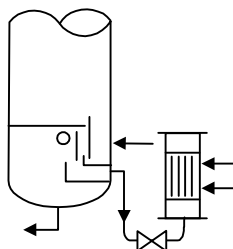
K. VARIABLE ELEVATION, X

1. $X = (\Delta H_F - J') / (1 - J)$	_____ ft	_____ m
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If required elevation is not acceptable repeat calculation with different line sizes.

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FIGURE A1 - REBOILER TYPES



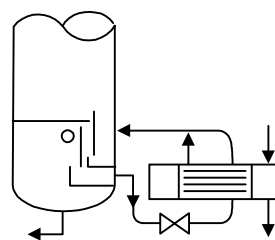
VERTICLE RECIRCULATING THERMOSIPHON

Advantages

1. Cheapest (Normally).
2. Process side easily cleaned.
3. Compact plot area.
4. Easily supported.
5. Low process side residence time (tends to be less fouling).
6. Good controllability (via valve in feed line).

Disadvantages

1. Circulation requires elevated tower.
2. Requires fixed head of feed liquid.
3. Tower requires internal baffling to approach theoretical tray.
4. Should be limited to 50% max. vaporization. However, lower % gives smoother operation.



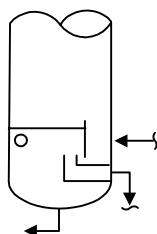
HORIZONTAL RECIRCULATING THERMOSIPHON

Advantages

1. Heating side easily cleaned.
2. Requires lower tower elevation than vehicle thermosiphon.
3. Low process side residence time.
4. Good controllability (via valve in feed line).

Disadvantages

1. Larger plot area required.
2. Vapor piping may be complex (16 ft & 20 ft bundles should have two vapor outlets).
3. Process side hard to clean.
4. Should be limited to 50% vaporization.
5. Tower requires internal baffling to approach theoretical tray.



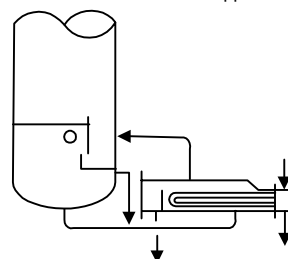
ONCE THROUGH

Advantages

1. Same as recirculating.
2. Equivalent to theoretical.

Disadvantages

1. Same as recirculating.
2. Danger of excessive vaporization per pass (higher fouling).



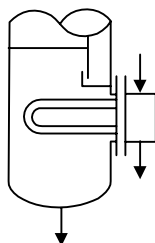
KETTLE REBOILER

Advantages

1. Can handle high vaporization loads (80%-normal feeds and 100%-clean feeds).
2. Simple column internals.
3. Provides theoretical tray.
4. Requires lowest tower elevation.

Disadvantages

1. Expensive exchanger construction.
2. Bottoms product holdup low.
3. High process fluid holdup (fouling).
4. Process side is difficult to clean.
5. Tends to be dirt trap.
6. Requires larger plot area.



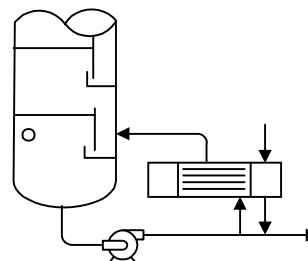
INTERNAL REBOILER

Advantages

1. No shell required.
2. Circulation piping eliminated.
3. Simple column internals.
4. No plot area required.

Disadvantages

1. Bundle length limited.
2. Larger column flange required.
3. Process side difficult to clean.



PUMP THROUGH REBOILER

Advantages

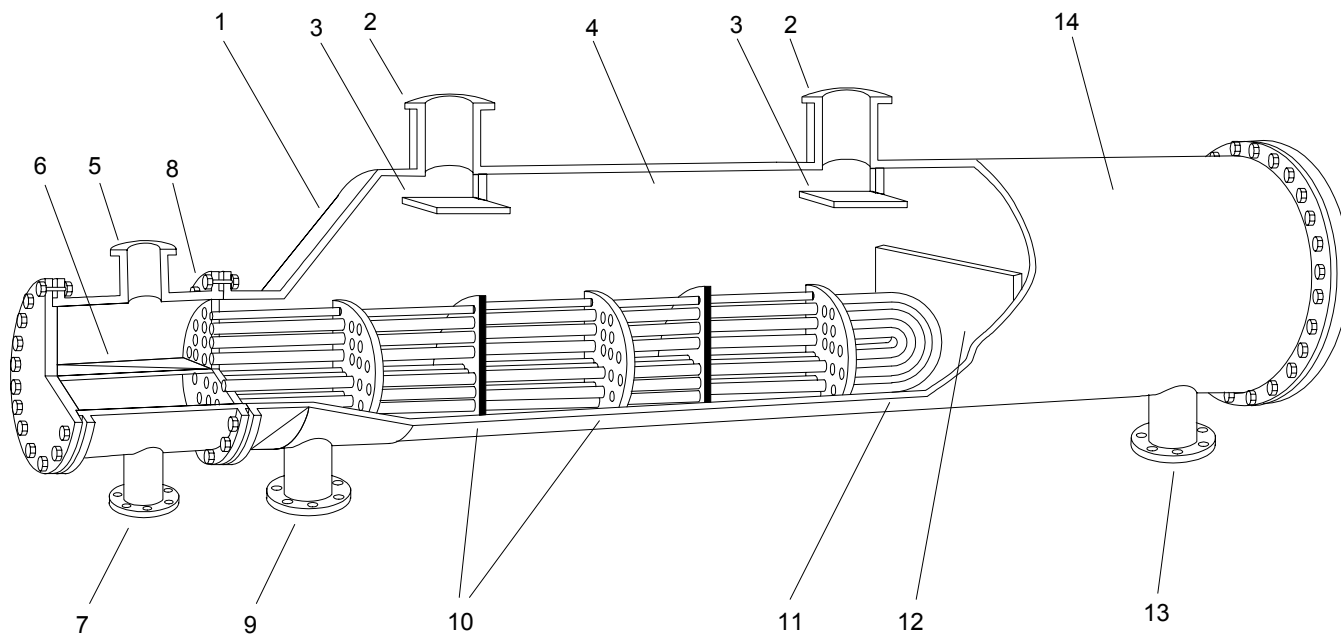
1. Often best for fouling or viscous service.
2. Full control of circulation rate.
3. Capable of high circulation rate.
4. Best for large surface requirements.
5. Operable at very low circulation rates.
6. Can use standard exchangers and smaller piping.

Disadvantages

1. Pumping costs.
2. Requires tower elevation for pump suction head.

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**FIGURE A2
COMPONENTS OF A KETTLE REBOILER**



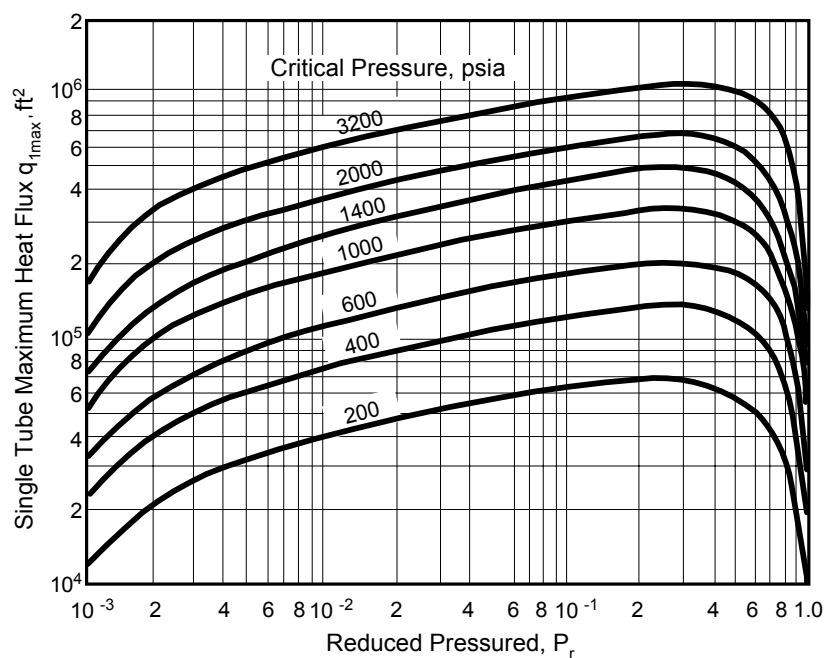
- | | | |
|-------------------------------|-------------------------|----------------------------------|
| 1 SHELL | 6 CHANNEL PARTITION | 11 U-TUBE RETURNS |
| 2 SHELL OUTLET NOZZLE (VAPOR) | 7 CHANNEL OUTLET NOZZLE | 12 WEIR |
| 3 ENTRAINMENT BAFFLES | 8 TUBE SHEET | 13 SHELL OUTLET NOZZLE (LIQUID) |
| 4 VAPOR DISENGAGEMENT SPACE | 9 SHELL INLET NOZZLE | 14 LIQUID HOLDUP (SURGE) SECTION |
| 5 CHANNEL INLET NOZZLE | 10 TUBE SUPPORT PLATES | |

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HEAT EXCHANGE EQUIPMENT

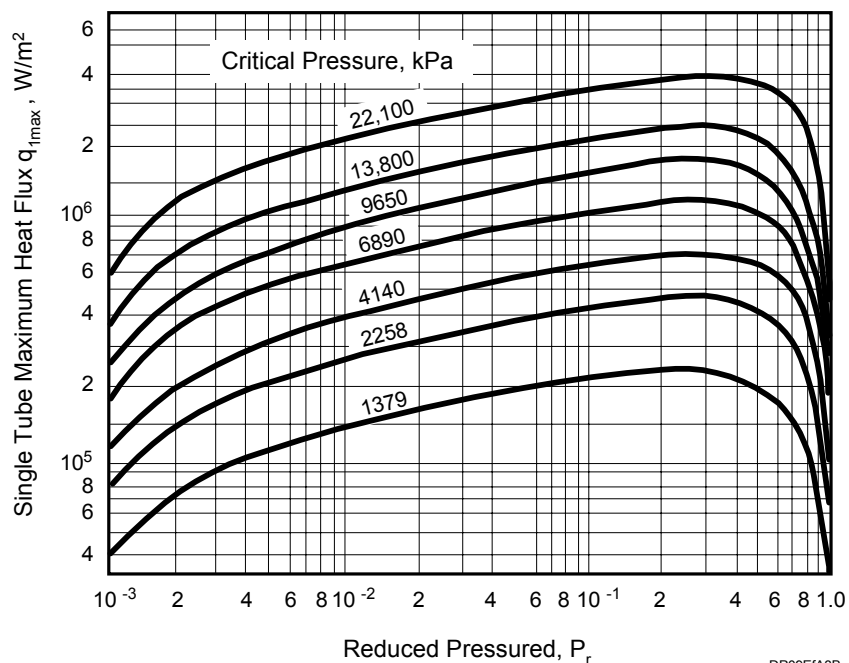
CALCULATION PROCEDURE, VAPORIZATION

FIGURE A3A
SINGLE TUBE MAXIMUM HEAT FLUX (CUSTOMARY)



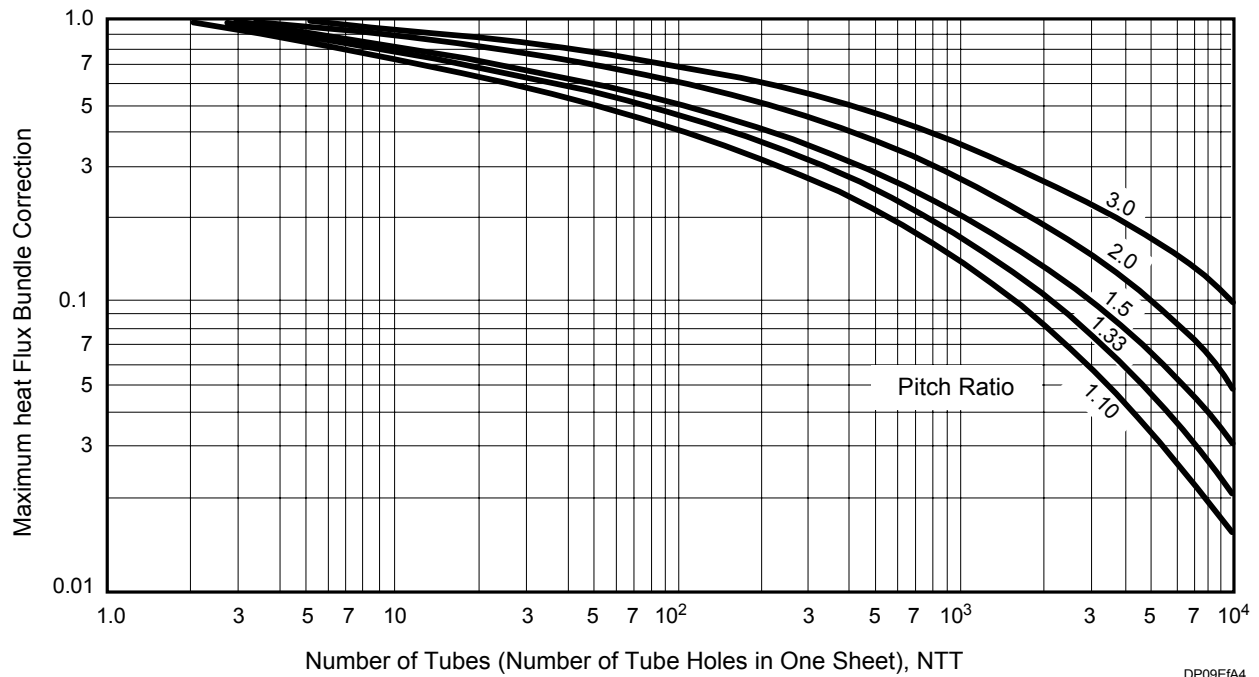
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FIGURE A3B
SINGLE TUBE MAXIMUM HEAT FLUX (METRIC)



DP09EFA3B

FIGURE A4
BUNDLE CORRECTION FACTOR (CB)



DP09EfA4

HEAT EXCHANGE EQUIPMENT

CALCULATION PROCEDURE, VAPORIZATION

FIGURE A5A
SINGLE TUBE NUCLEATE BOILING COEFFICIENTS (CUSTOMARY)

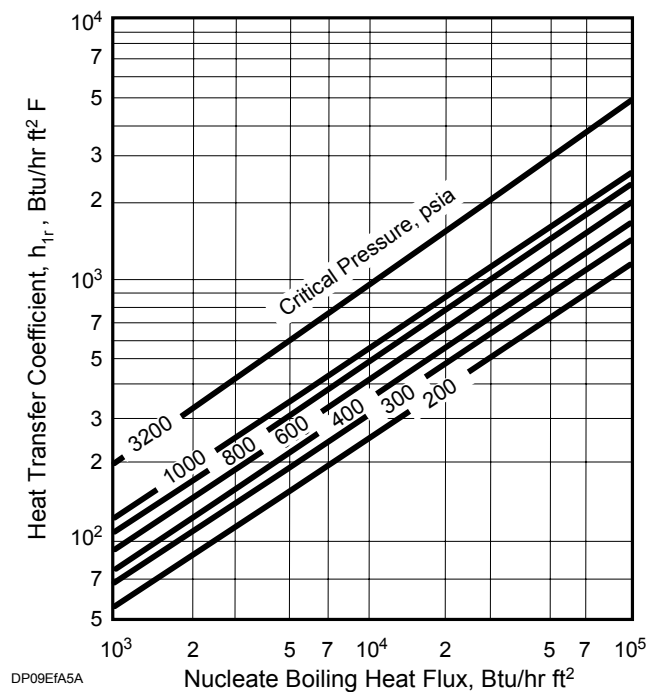


FIGURE A5B
SINGLE TUBE NUCLEATE BOILING COEFFICIENTS (METRIC)

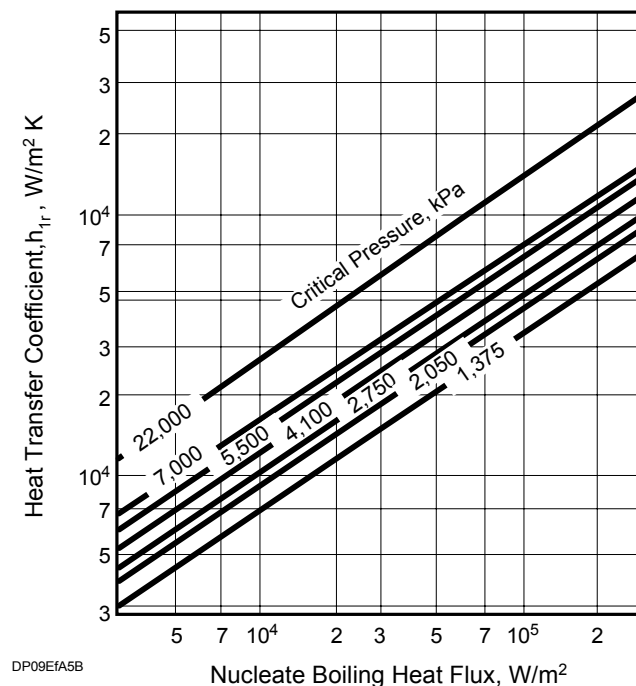
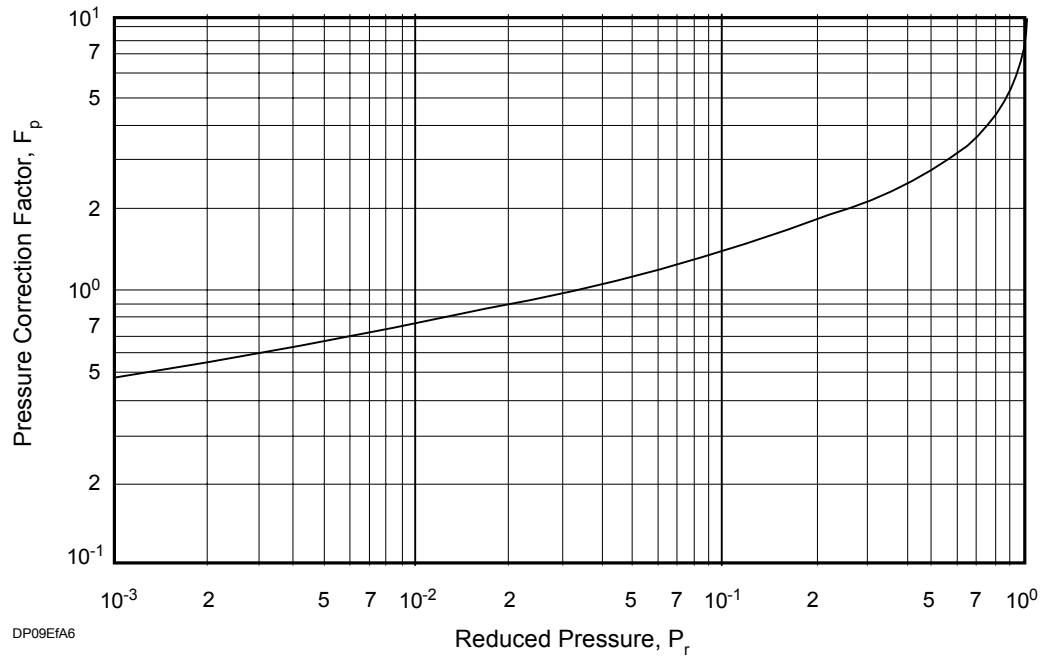


FIGURE A6
NUCLEATE BOILING PRESSURE CORRECTION, F_p



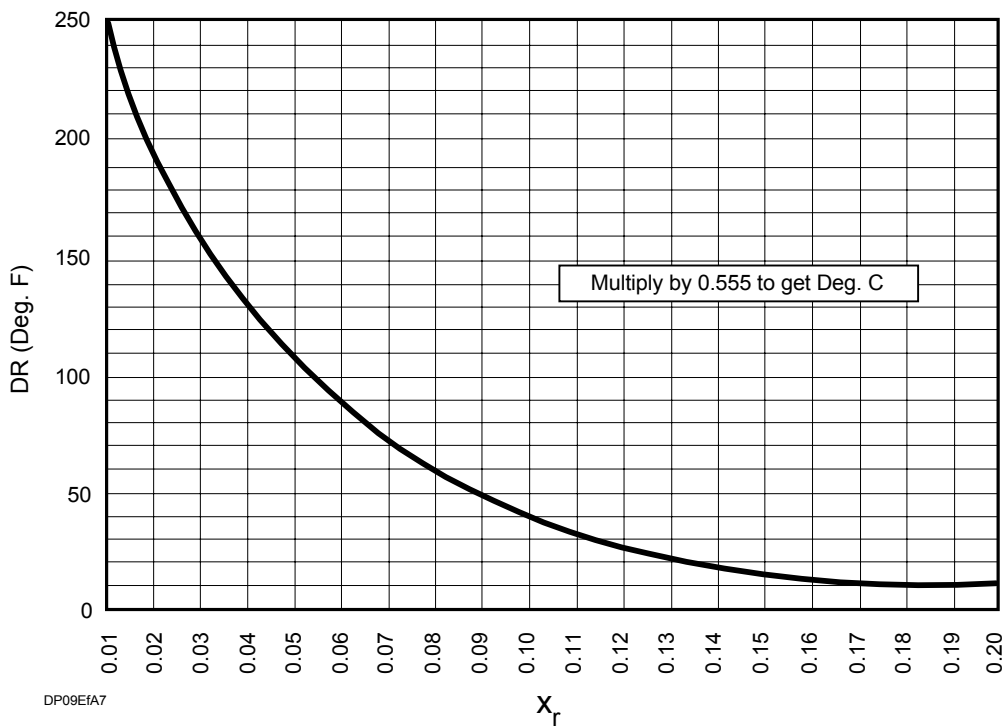
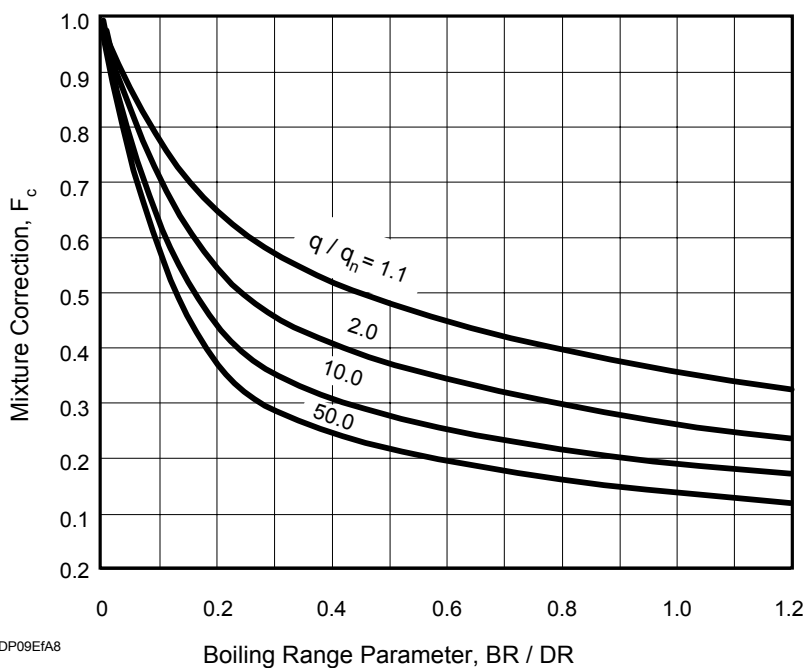
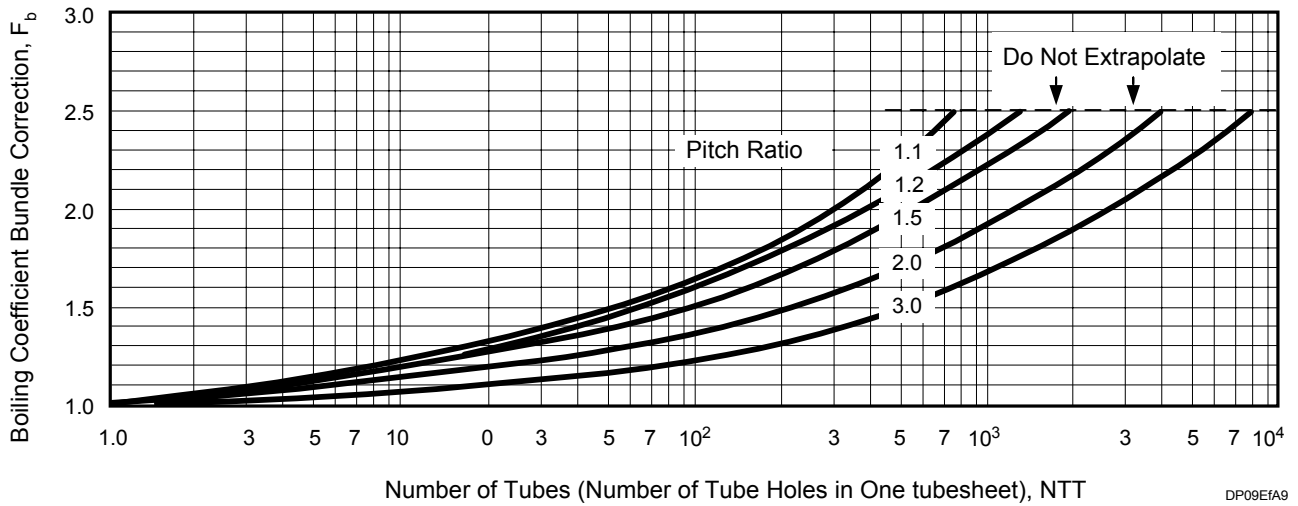
HEAT EXCHANGE EQUIPMENT
CALCULATION PROCEDURE, VAPORIZATION**FIGURE A7**
EFFECTIVE TEMPERATURE RANGE, DR**FIGURE A8**
MIXTURE CORRECTION, F_c 

FIGURE A9
BUNDLE CORRECTION FACTOR, F_b



HEAT EXCHANGE EQUIPMENT

CALCULATION PROCEDURE, VAPORIZATION

FIGURE A10A
FINNED TUBE SURFACE FACTOR, F_s , FOR PURE HYDROCARBONS (CUSTOMARY)

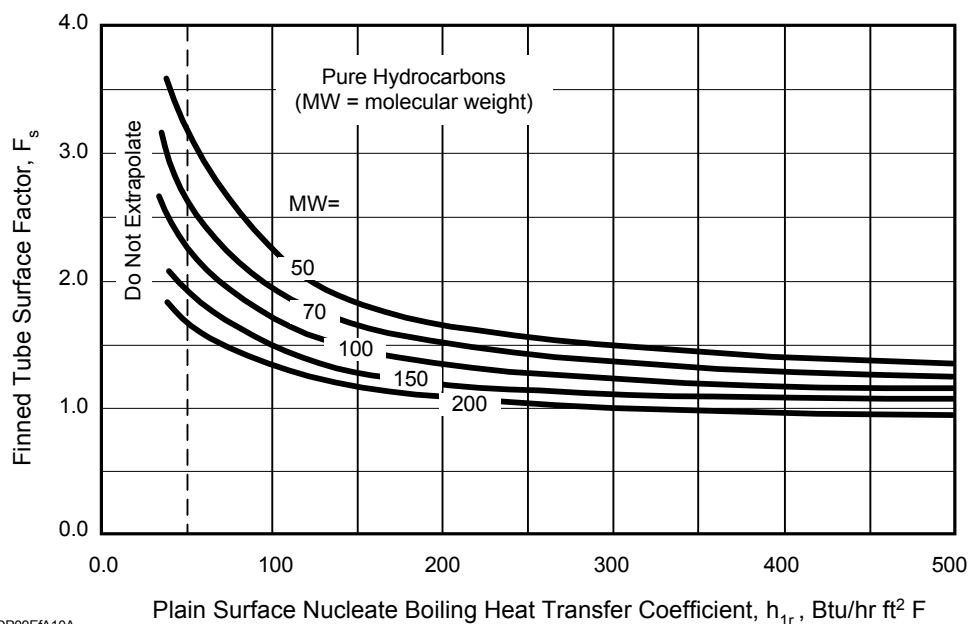


FIGURE A10B
FINNED TUBE SURFACE FACTOR, F_s , FOR PURE HYDROCARBONS (METRIC)

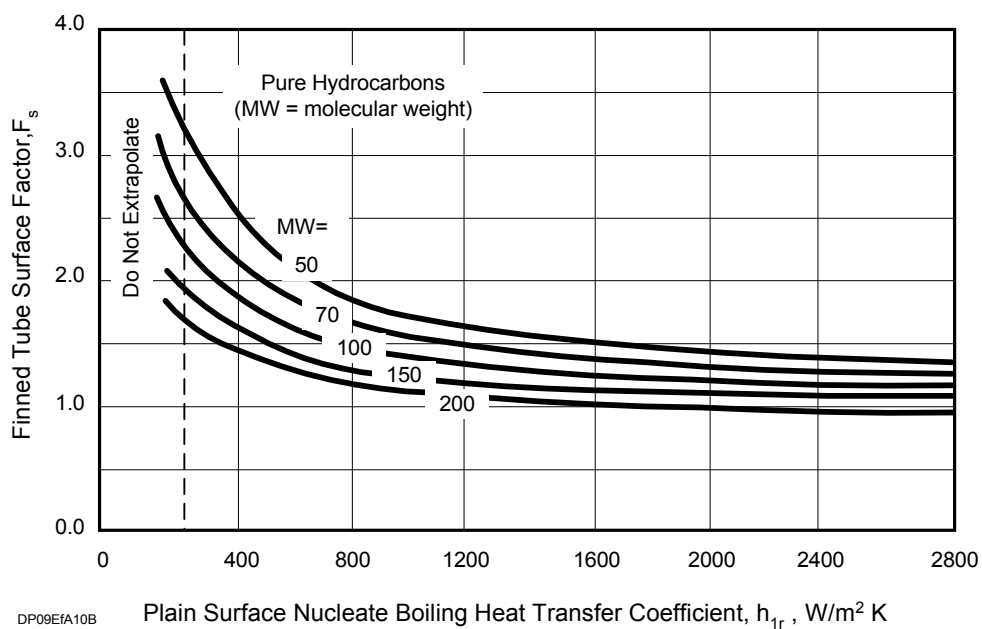


FIGURE A11A
FINNED TUBE SURFACE FACTOR, F_s , FOR MIXED HYDROCARBONS (CUSTOMARY)

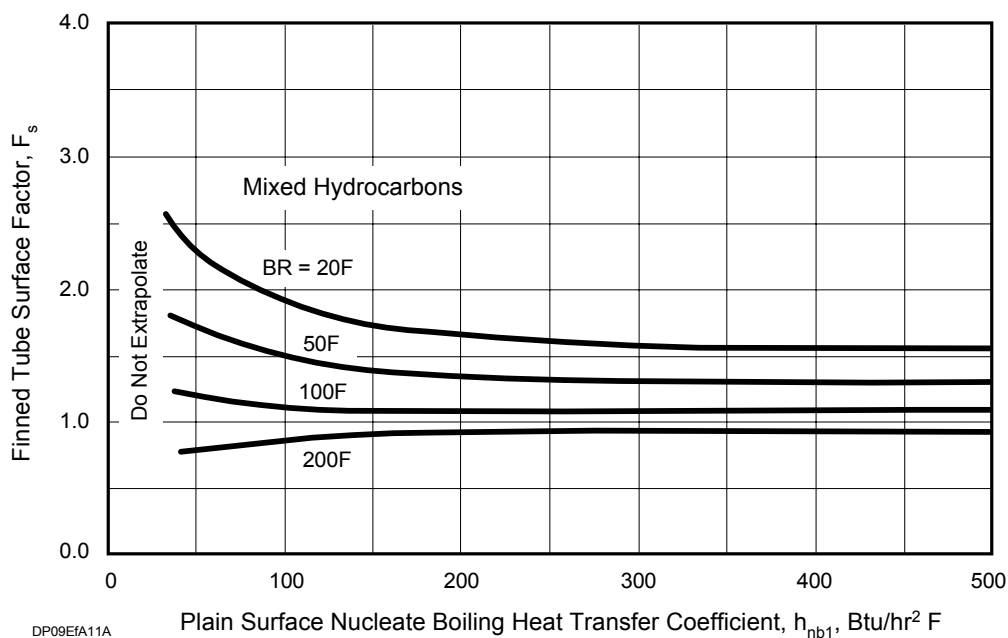
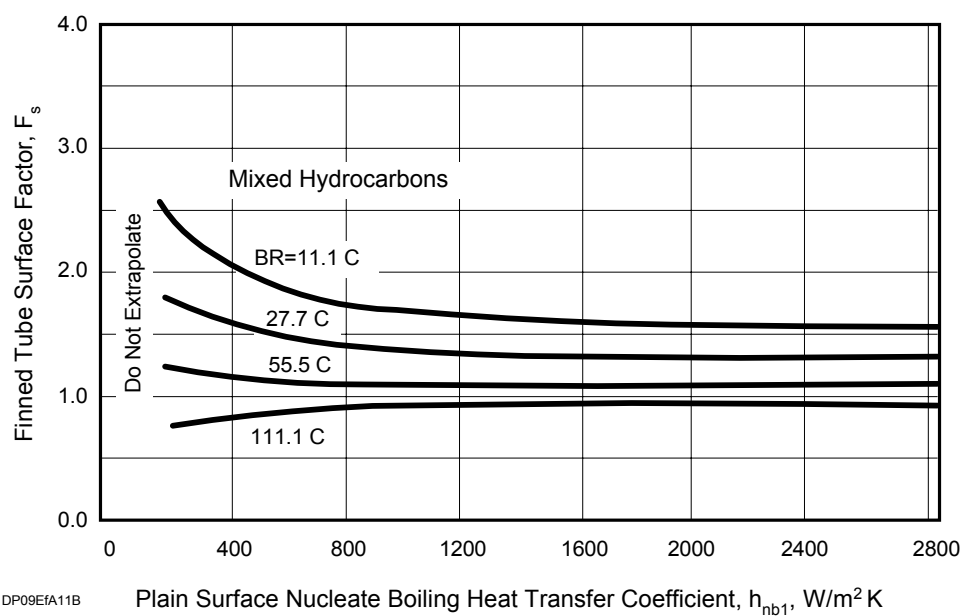
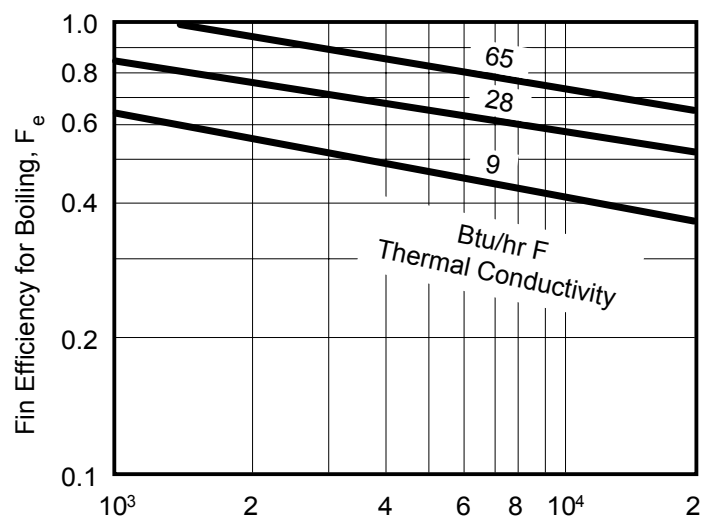
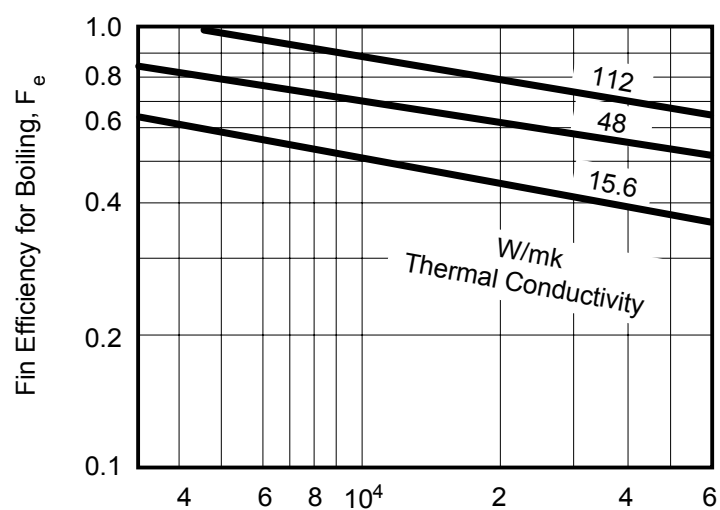


FIGURE A11B
FINNED TUBE SURFACE FACTOR, F_s , FOR MIXED HYDROCARBONS (METRIC)



HEAT EXCHANGE EQUIPMENT
CALCULATION PROCEDURE, VAPORIZATION**FIGURE A12A**
FIN EFFICIENCY FACTOR, F_e (CUSTOMARY)

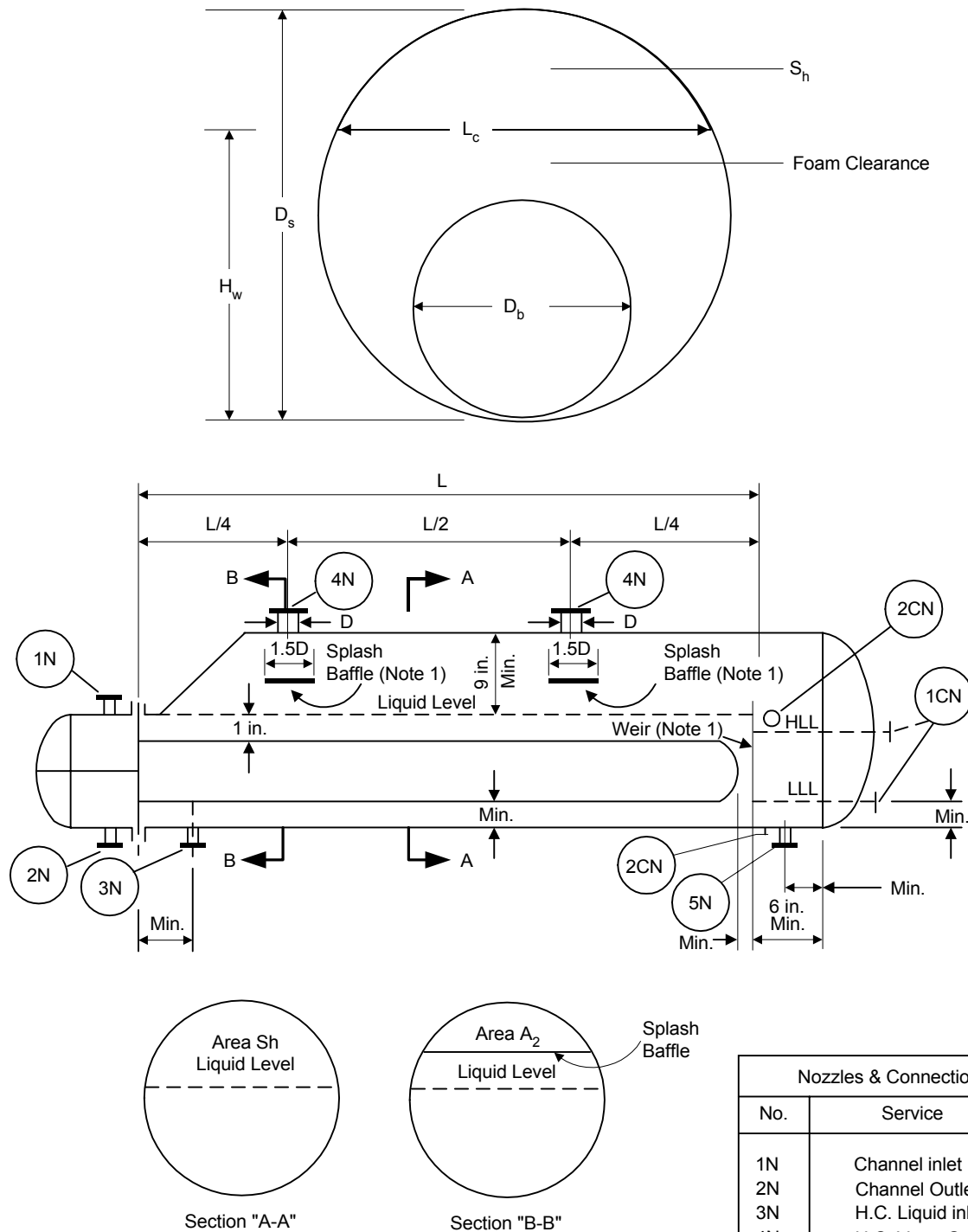
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FIGURE A12B
FIN EFFICIENCY FACTOR, F_e (METRIC)

DP09EfA12B

HEAT EXCHANGE EQUIPMENT
CALCULATION PROCEDURE, VAPORIZATION

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FIGURE A13
KETTLE FLOW AREAS


DP9EFA13

HEAT EXCHANGE EQUIPMENT
CALCULATION PROCEDURE, VAPORIZATION

FIGURE A14A
CHOKE FLOW REFERENCE MAXIMUM HEAT FLUX FOR VERTICAL THERMOSIPHON REBOILERS

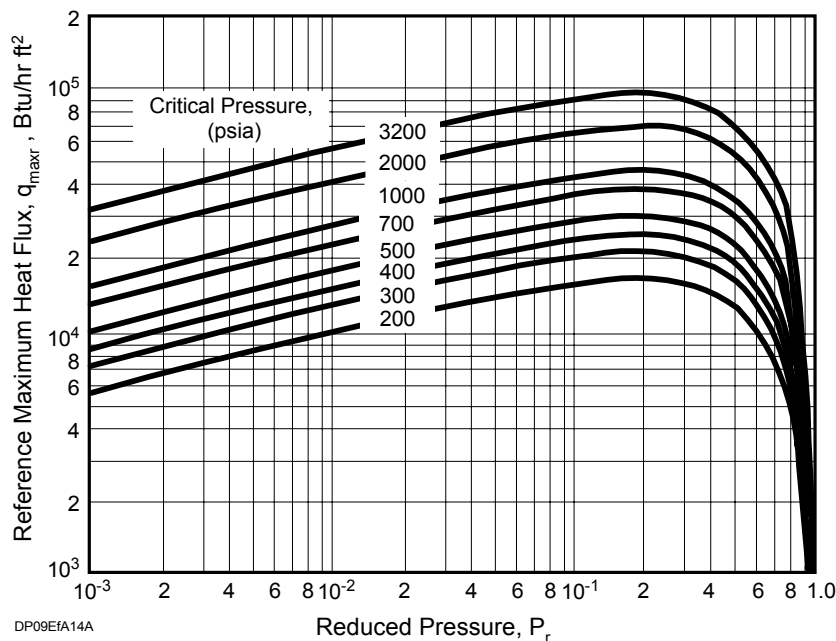


FIGURE A14B
CHOKE FLOW REFERENCE MAXIMUM HEAT FLUX FOR VERTICAL THERMOSIPHON REBOILERS

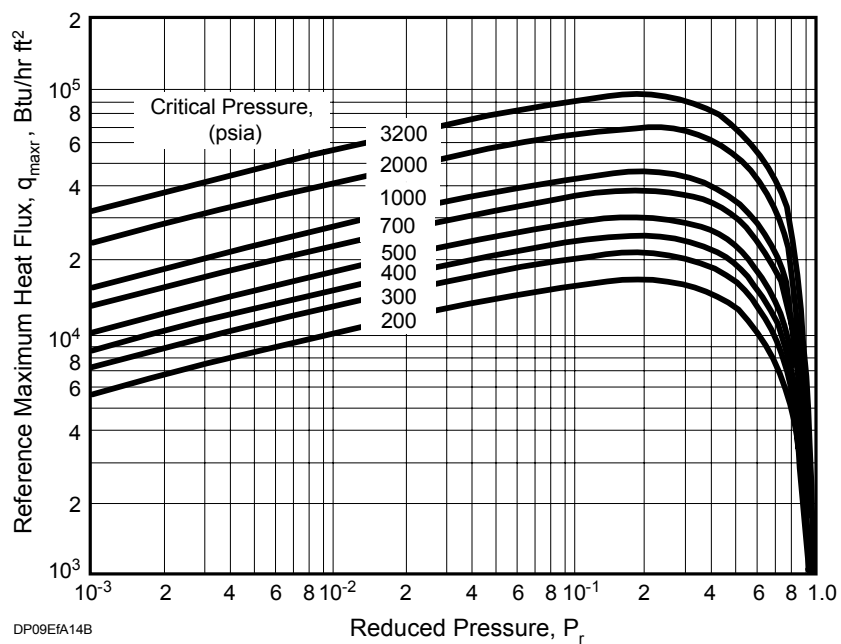


FIGURE A15A
TUBE GEOMETRY CORRECTION TO MAXIMUM HEAT FLUX (CUSTOMARY)

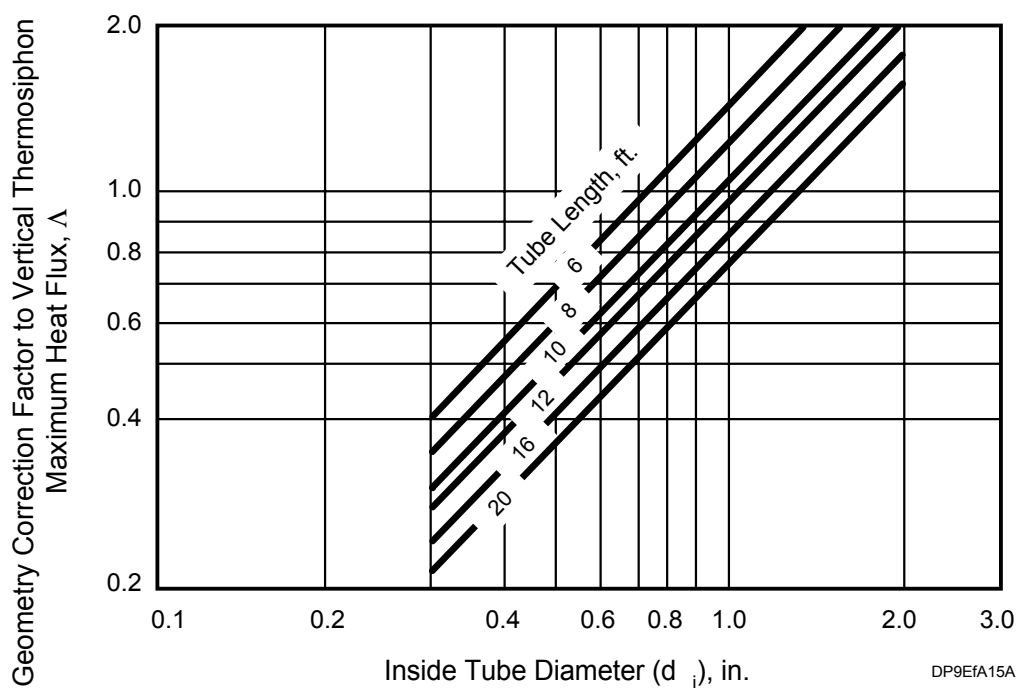
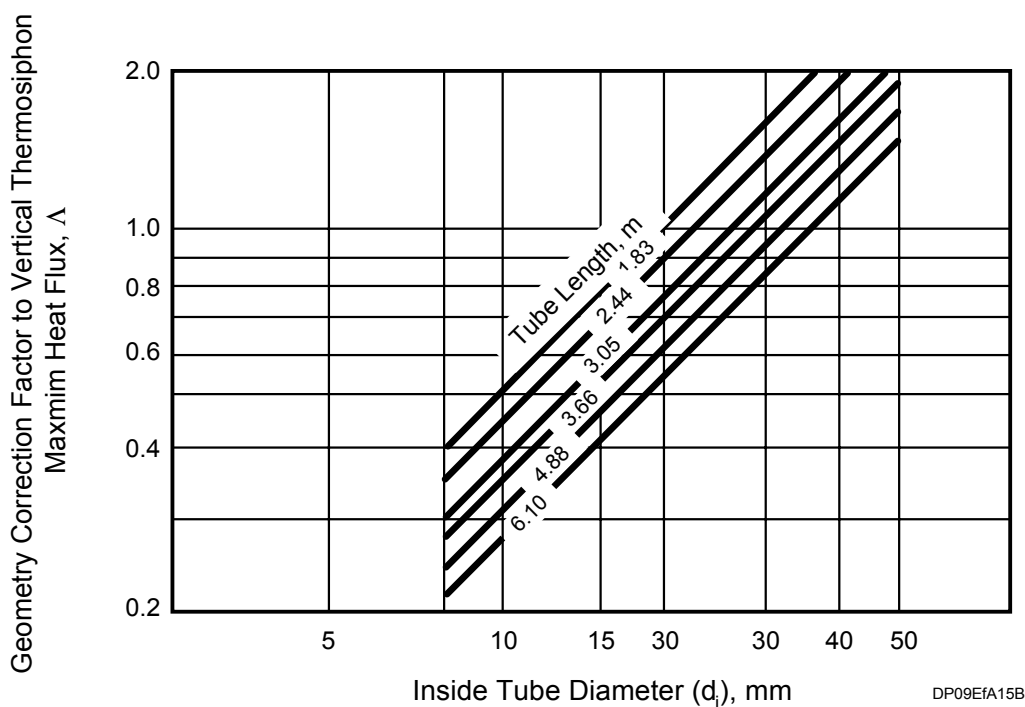


FIGURE A15B
TUBE GEOMETRY CORRECTION TO MAXIMUM HEAT FLUX (METRIC)



HEAT EXCHANGE EQUIPMENT
CALCULATION PROCEDURE, VAPORIZATION

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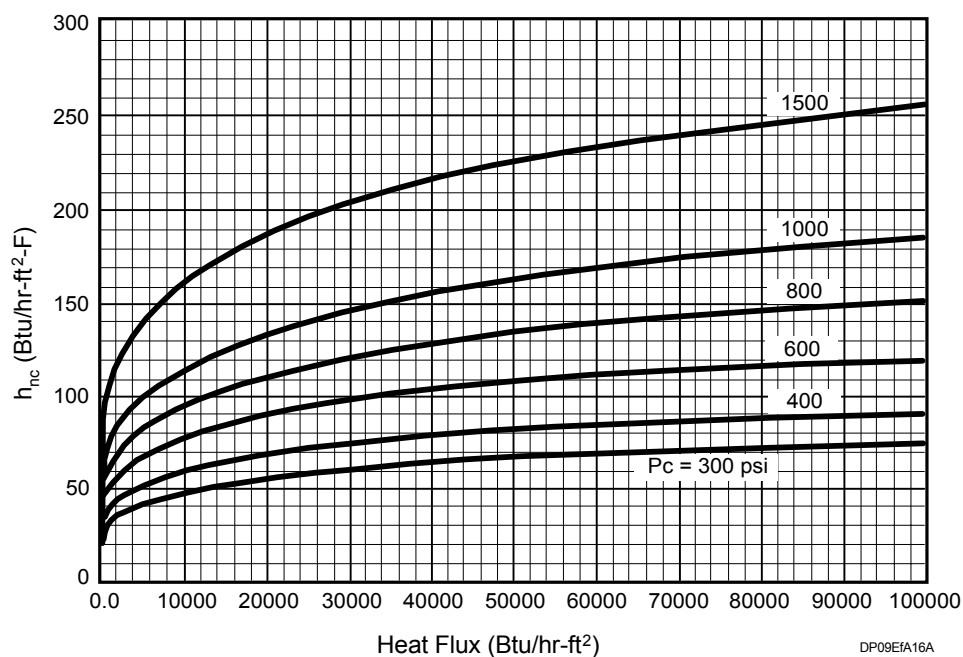
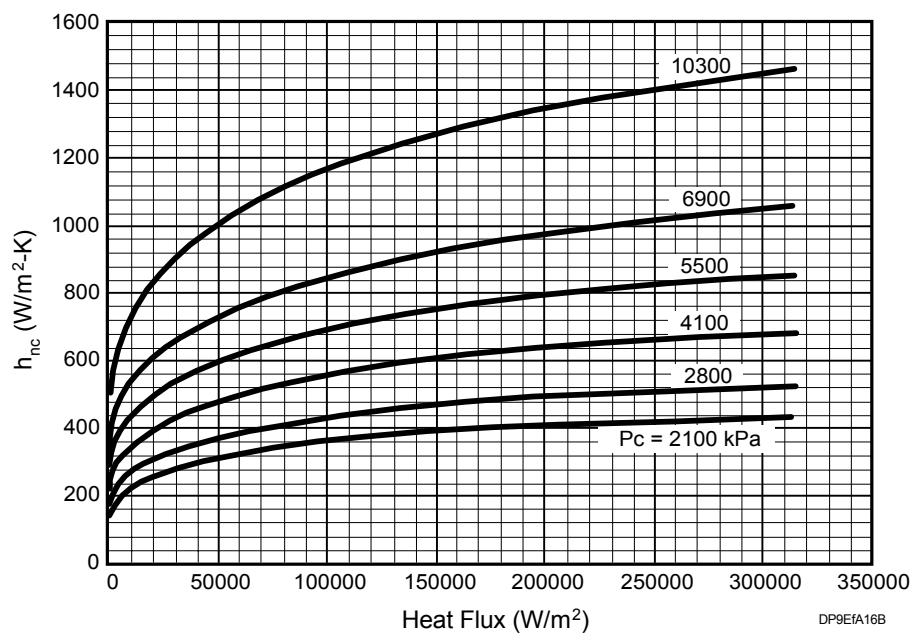
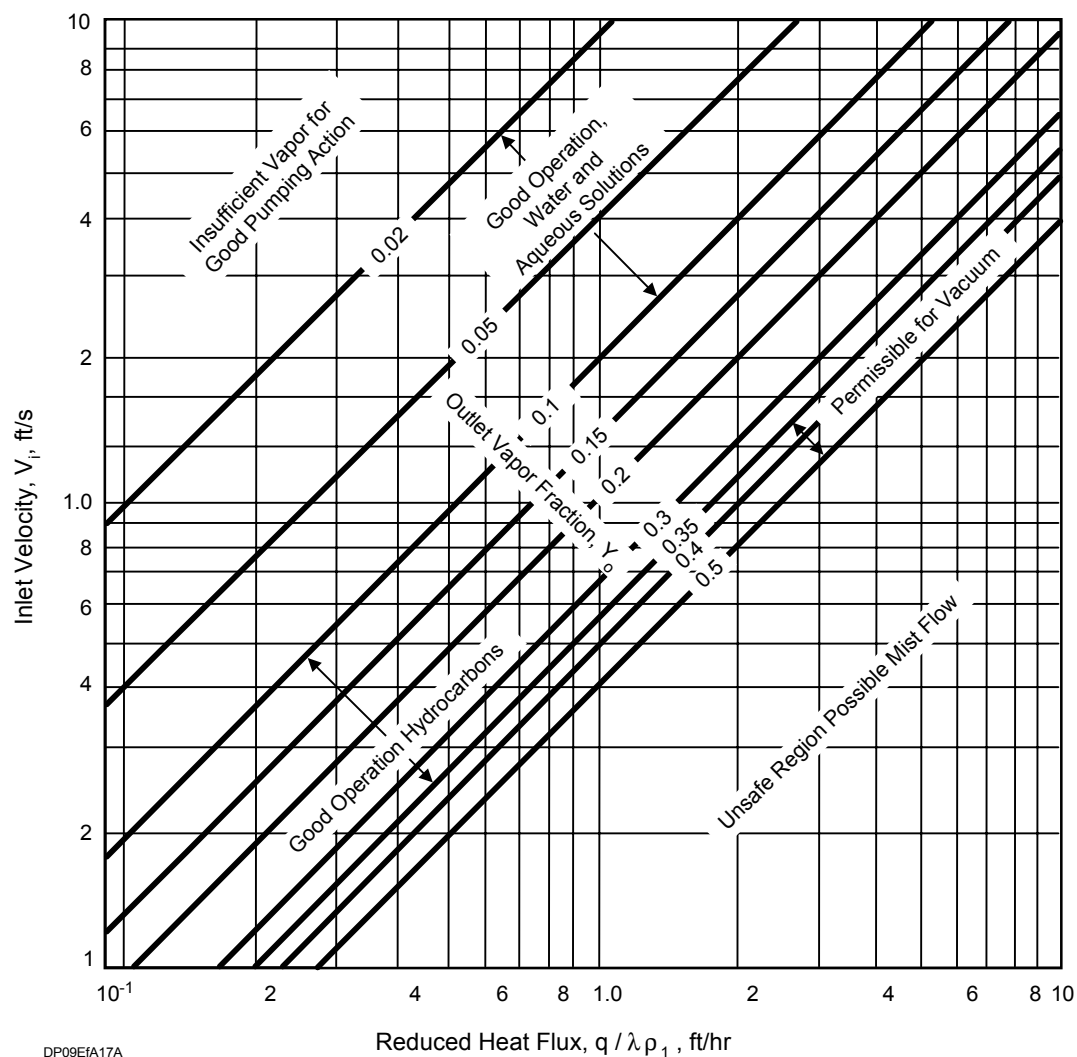
FIGURE A16A
NATURAL CONVECTION COEFFICIENT (CUSTOMARY)**FIGURE A16B**
NATURAL CONVECTION COEFFICIENT (METRIC)

FIGURE A17A
OUTLET VAPOR FRACTION, y_o , AS A FUNCTION OF
REDUCED HEAT FLUX AND INLET VELOCITY (CUSTOMARY)



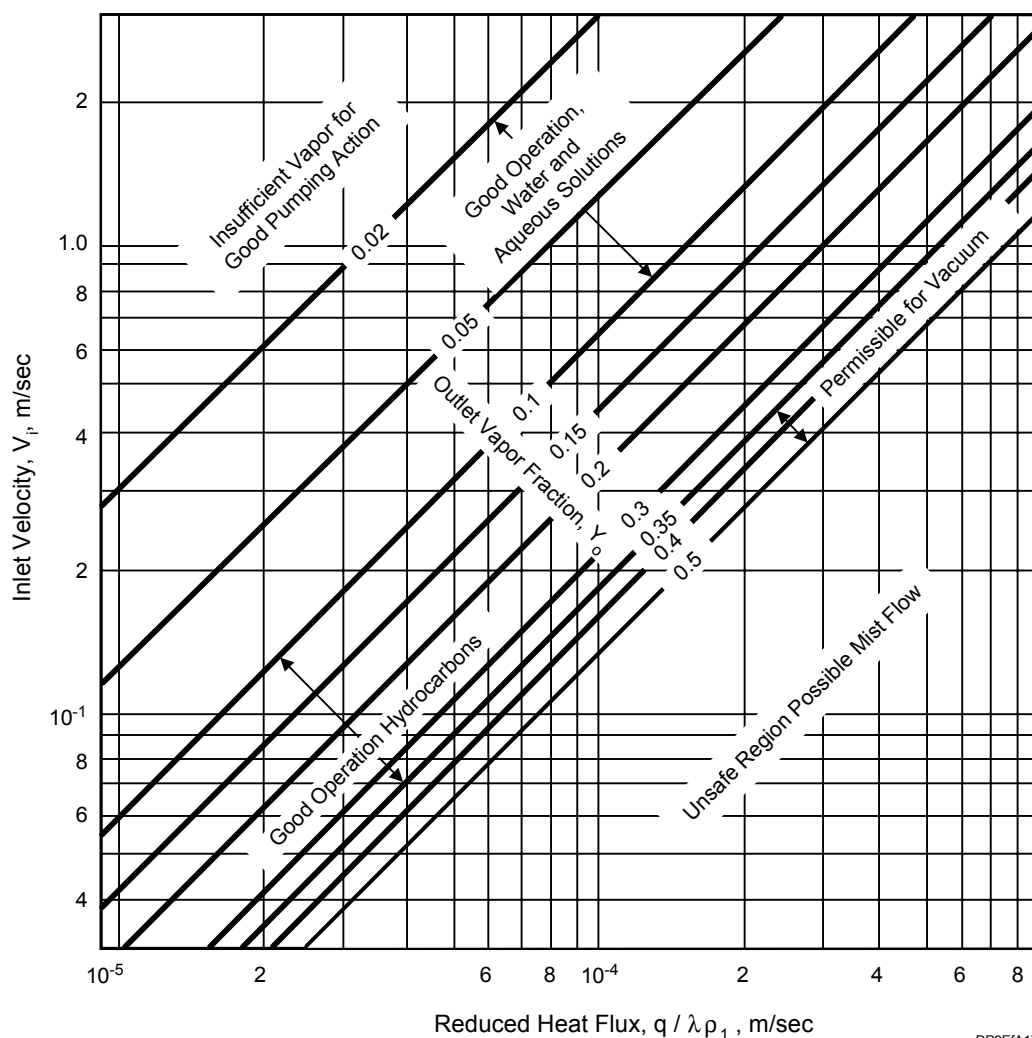
HEAT EXCHANGE EQUIPMENT

CALCULATION PROCEDURE, VAPORIZATION

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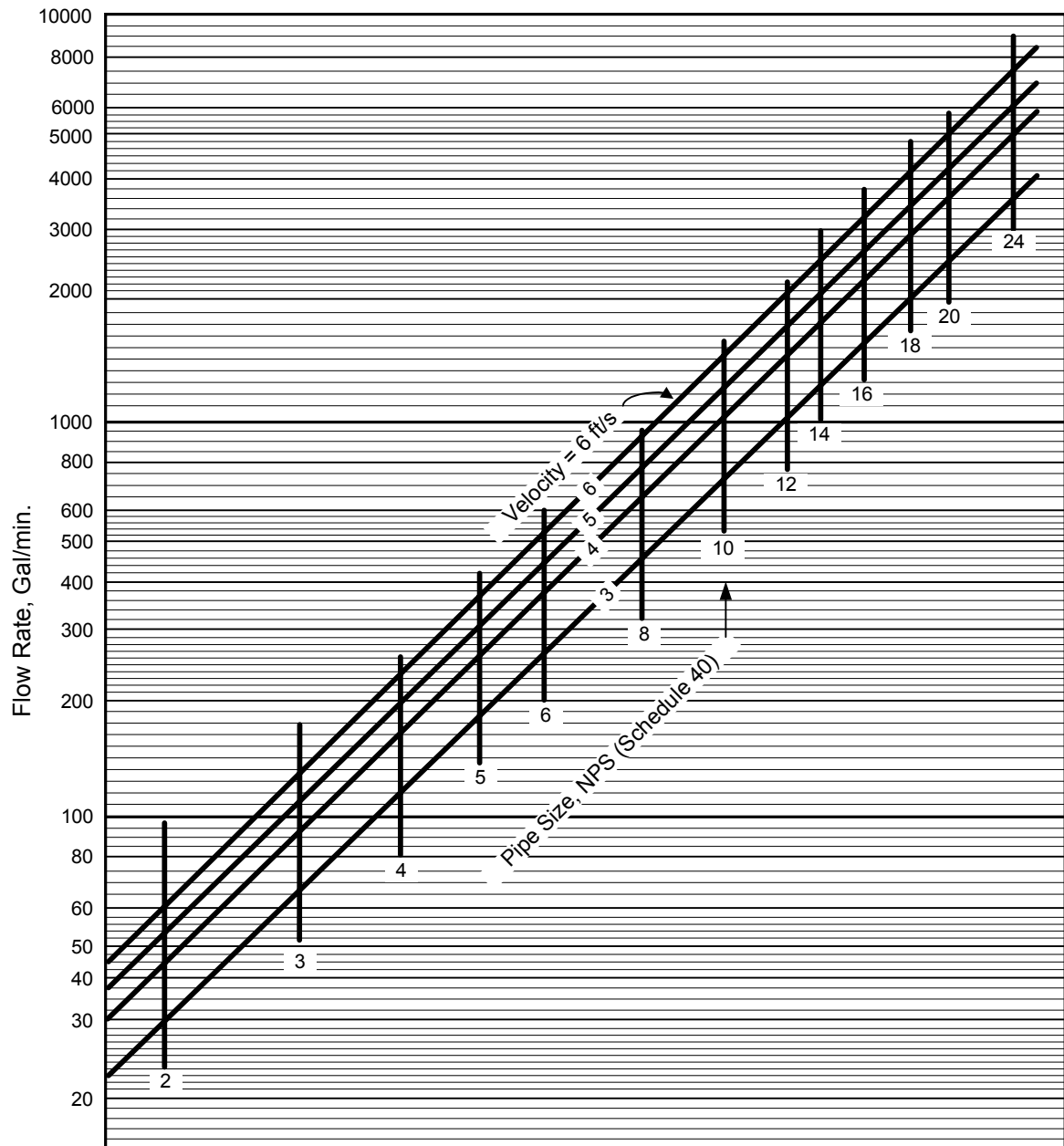
DESIGN PRACTICES

FIGURE A17B
OUTLET VAPOR FRACTION, y_o , AS A FUNCTION OF
REDUCED HEAT FLUX AND INLET VELOCITY (METRIC)

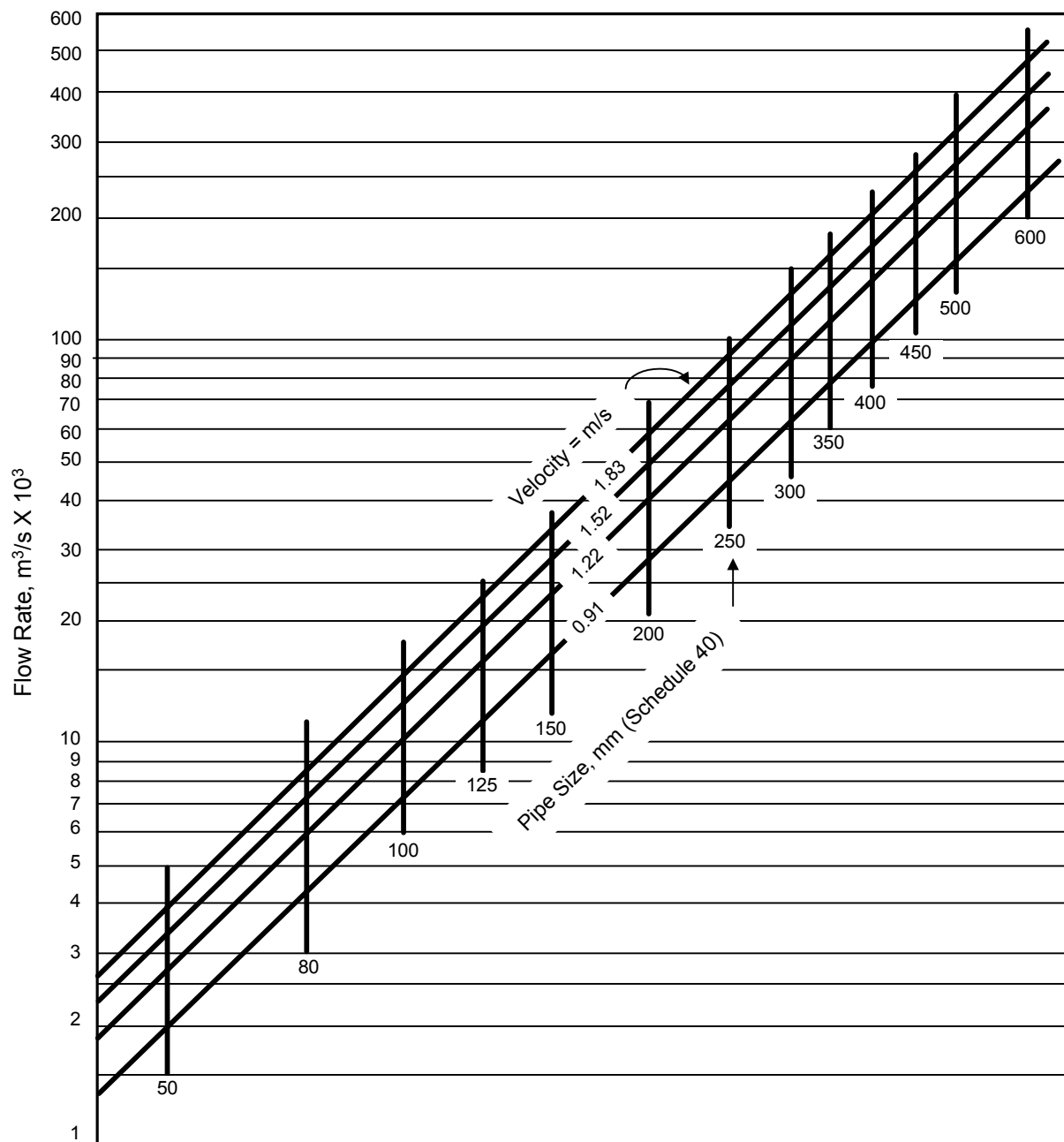


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FIGURE A18A
REBOILER LIQUID PIPE SIZE (CUSTOMARY)



DP9EIA18A

HEAT EXCHANGE EQUIPMENT
CALCULATION PROCEDURE, VAPORIZATION**FIGURE A18B**
REBOILER LIQUID PIPE SIZE (METRIC)

DP9EfA18B